

CITY UNIVERSITY OF HONG KONG

Distributed Fixed-Time Nash Equilibrium
Seeking Algorithms for Non-Cooperative
Games in Multi-Agent Systems

Submitted to

Department of Mechanical Engineering

in Partial Fulfillment of the Requirements
for the Degree of Doctor of Philosophy

by

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June 2026

Abstract

Game theory, as an approach to decision-making, has been applied widely in numerous fields such as artificial intelligence, robust control, social policymaking in political science and distributed optimization in multi-agent systems. Non-cooperative games are one of the most useful games in modeling and analyzing interactive decision-making processes, in which players are self-interested to optimize their own actions. As a key concept for non-cooperative games, Nash equilibrium, refers to an action profile in which no player can gain more payoff by unilaterally changing its own action and has been widely adopted in analysis of non-cooperative games. Based on whether or not each individual player has direct access to other players' actions, Nash equilibrium seeking approaches can be classified into two categories, that is, centralized algorithms and distributed algorithms. However, centralized algorithms, where the centralized information is to be shared by all players, would run into difficulties for large-scale network-based games. As a result, distributed algorithms, where only partial information is to be shared by neighboring players, have been receiving increasing attention. Note that, in most existing works on distributed Nash equilibrium seeking, the proposed algorithms converge to the Nash equilibrium asymptotically or exponentially. In other words, the Nash equilibrium is reached only when time goes to infinity. More recently, distributed finite-time Nash equilibrium seeking problems have been further investigated due to their faster convergence. However, the upper bound of the convergence time of the finite-time algorithms is dependent on the initial state of the underlying system and thus potentially becomes very large when the initial state deviates far away from the equilibrium. To fill this gap, distributed fixed-time Nash equilibrium seeking algorithms, where the upper bound of the convergence time is independent of the initial conditions of the systems, deserve further investigation.

Furthermore, distributed Nash equilibrium seeking for multi-agent systems in practice often encounters severe challenges, such as vulnerable communication networks under denial-of-service (DoS) attacks, which disrupt information flow and degrade

seeking performance; restricted decision spaces in constrained games, where any boundary violation may lead to system instability; and players governed by heterogeneous linear and nonlinear dynamics, making the design and analysis of the distributed fixed-time algorithms much more challenging. Addressing these challenges is crucial for real-world engineering applications, many of which explicitly call for fixed-time convergence.

Motivated by these observations, this dissertation systematically investigates distributed fixed-time Nash equilibrium seeking problems for multi-agent systems. The main contributions are outlined from four progressive aspects:

Part I: This part investigates the distributed fixed-time Nash equilibrium seeking problem for non-cooperative games. A novel distributed fixed-time Nash equilibrium seeking algorithm with a two time-scale framework is proposed based on a leader-following consensus protocol and the sliding mode theory. Different from existing works on asymptotic, exponential, or finite-time convergence, the proposed algorithm achieves fixed-time convergence with an explicitly given upper bound for the convergence time, which is independent of the initial conditions of the concerned system. Moreover, unlike the coupled passivity-based design in existing works on distributed fixed-time Nash equilibrium seeking, where verification of parameter feasibility involves solving some complex inequalities that can be challenging for large-scale systems, the proposed algorithm with a two time-scale framework can guarantee the existence of the design parameters and brings great benefits for the selection of the design parameters. Furthermore, as a significant extension, the proposed algorithm is modified to address time-varying quadratic games, ensuring fixed-time tracking of the Nash equilibrium trajectory.

Part II: This part addresses the resilient distributed fixed-time Nash equilibrium seeking problem under denial-of-service (DoS) attacks. Since communication networks of multi-agent systems are inherently vulnerable to malicious DoS attacks, which destroy network connectivity and compromise information timeliness, a resilient distributed fixed-time Nash equilibrium seeking algorithm is developed. Based on a leader-following consensus protocol, switching system theory, and sliding mode theory, the proposed algorithm ensures that the actions of the players can still converge to the Nash equilibrium in fixed time despite the presence of DoS attacks, with the upper bound of the convergence time explicitly given. Furthermore, as a notable enhancement for scenarios where DoS attack intervals are known, a time-varying gains scheme is proposed. By dynamically assigning high gains in the absence of attacks and low gains

during DoS intervals, this extended scheme enables the proposed algorithm to tolerate DoS attacks of significantly higher intensity while preserving fixed-time convergence.

Part III: This part explores the distributed Nash equilibrium seeking problem for constrained non-cooperative games. Since a player's feasible action space is often restricted by physical limitations in practice, constrained non-cooperative games have attracted extensive attention. Based on variational inequality theory, the problem of Nash equilibrium seeking for constrained non-cooperative games can be reformulated as a variational inequality problem. Building upon this reformulation, a novel distributed algorithm is developed by combining neurodynamic methods, a leader-following consensus protocol, and sliding mode theory. It is also proven that under the proposed algorithm, the actions of all players converge to the Nash equilibrium in a fixed time, which is independent of the initial conditions of the players and the upper bound of the convergence time for the proposed Nash equilibrium seeking algorithm is explicitly given.

Part IV: This part tackles the distributed fixed-time Nash equilibrium seeking problem for multi-agent systems with heterogeneous and complex physical dynamics. While the majority of existing algorithms are confined to first-order and homogeneous systems, real-world engineering applications often involve players governed by heterogeneous and complex dynamics, such as high-order integrator dynamics, general linear dynamics, and nonlinear Euler-Lagrange dynamics. To handle such heterogeneous dynamics, a novel two-layer design framework consisting of a virtual layer and a physical layer is established. In the virtual layer, distributed fixed-time Nash equilibrium seeking algorithms are utilized to generate a virtual signal that converges to the Nash equilibrium. In the physical layer, fixed-time tracking protocols are specifically designed for the heterogeneous physical dynamics so that their actual actions converge to the virtual signals. It is proven that under this decoupled framework, the physical actions of all heterogeneous players converge to the Nash equilibrium in fixed time regardless of initial conditions. Moreover, the upper bounds for their convergence times can be explicitly given.

Finally, to bridge the gap between theoretical design and practical implementation, the effectiveness and practicality of the proposed distributed algorithms are presented through three distinct engineering scenarios: a distributed energy trading game in smart microgrids, distributed computation offloading in vehicular edge computing systems, and target protection for air-ground heterogeneous multi-agent systems with conflicting goals.

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Chapter 1

Introduction

1.1 Background

In 1944, Von Neumann and Morgenstern [9] published their seminal book entitled “Theory of Games and Economic Behavior,” which was a groundbreaking book and created the interdisciplinary research field of game theory. Since then, game theory, as an approach to decision-making, has been applied widely in numerous fields such as generative adversarial networks, artificial intelligence in computer science, autonomous systems [3], artificial intelligence [3], robust/resilient control problems in control science [2, 12] and social policy making in political science [8]. Non-cooperative games are one of the most useful games in modeling and analyzing interactive decision-making processes, in which players are self-interested to optimize their own actions. Typical examples that fall into the noncooperative game setting include wind farm control [32], power flow control in communication networks [14], [34], information disclosure problems [103], energy management in smart grids [116], collaborative control and optimization of multiagent systems [31], [35], and road pricing control [102]. As a key concept for non-cooperative games, Nash equilibrium, refers to an action profile in which no player can gain more payoff by unilaterally changing its own action and has been widely adopted in analysis of non-cooperative games. Based on whether or not each individual player has direct access to other players’ actions, Nash equilibrium seeking approaches can be classified into two categories, that is, centralized algorithms and distributed algorithms. However, centralized algorithms, where the centralized information is to be shared by all players, would run into difficulties for large-scale

network-based games. As a result, distributed algorithms, where only partial information is to be shared by neighboring players, have been receiving increasing attention. The majority of current studies on distributed Nash equilibrium seeking primarily concentrate on consensus-based algorithms, though there are some other methods such as passivity-based algorithms. Since the convergence of passivity-based algorithms requires a restrictive trade-off between the network's algebraic connectivity and the inherent properties of the game mappings, the leader-following consensus-based algorithms, which forms a two time-scale structure, serves as a much more suitable and scalable approach for designing advanced distributed seeking strategies.

Note that, in most existing works on distributed Nash equilibrium seeking, the proposed algorithms converge to the Nash equilibrium asymptotically or exponentially. In other words, the Nash equilibrium is reached only when time goes to infinity. More recently, distributed finite-time converging Nash equilibrium seeking problems have been further investigated due to their faster convergence and better robustness. However, the upper bound of the convergence time of the finite-time converging algorithms is dependent on the initial state of the underlying system and thus potentially becomes very large when the initial state deviates far away from the equilibrium. To fill this gap, distributed fixed-time converging Nash equilibrium seeking algorithms, where the upper bound of the convergence times are independent of the initial conditions of the systems, deserve further investigation.

While theoretical breakthroughs in fixed-time distributed Nash equilibrium seeking have provided promising results, deploying these idealized algorithms in practical multi-agent systems often encounters several challenges that arise from cyber, decision-making and physical perspective.

First, from a cyber perspective, the distributed information exchange heavily relies on open communication networks, which are inherently exposed to malicious cyber-attacks. Such cyber-attacks can cause severe performance degradation and even the complete malfunction of the concerned multi-agent systems, which has been extensively evidenced by widely reported incidents in recent years, such as the Stuxnet virus that targeted Supervisory Control and Data Acquisition (SCADA) systems. Typically, models of cyber-attacks can be classified into two primary types: denial-of-service (DoS) attacks and deception attacks. The former primarily tends to influence the timeliness of information transfer by blocking or destroying the connectivity of networks, while the latter affects the data credibility by manipulating or damaging the integrity of the data transmitted over networks. Since distributed Nash equilibrium seeking relies

on continuous and reliable information exchange among players, the communication blockages induced by DoS attacks can easily paralyze the entire decision-making process. Therefore, designing a resilient seeking algorithm capable of withstanding DoS attacks is essential.

Second, from a decision-making perspective, practical non-cooperative games are inevitably constrained rather than ideally unconstrained. In real-world industrial applications, such as distributed economic dispatching in smart grids or cooperative collision-avoidance trajectory planning in autonomous multi-vehicle systems, players are strictly required to restrict their feasible decisions within specific closed convex sets. These restricted action spaces, which are rigidly dictated by physical actuator limitations (e.g., maximum torque or motor speed saturation), strict safety boundaries (e.g., minimum safe inter-vehicle distances), or finite battery capacities, invalidate the unconstrained seeking algorithms and even lead to system instability or optimization failure. For instance, an unconstrained algorithm might generate a theoretical optimal command that exceeds the maximum power rating of a local generator, which could trigger hardware damage or cascading failures across the grid. Consequently, to bridge the gap between idealized game-theoretic analysis and physical reality, designing Nash equilibrium seeking algorithms for constrained non-cooperative games is necessary.

Third, from a physical execution perspective, the majority of existing theoretical seeking algorithms are developed based on the restrictive assumption that the players are governed by first-order dynamics, which unrealistically implies that the agents can instantaneously achieve any commanded velocity. However, in practical engineering scenarios, the players are embodied as physical systems, such as industrial robotic manipulators, autonomous underwater vehicles, or unmanned aerial vehicles, all of which are subject to highly complex and heterogeneous mechanical dynamics. Consequently, a severe mismatch arises when directly applying Nash seeking algorithms for players of first-order dynamics to these systems. Specifically, while the Nash equilibrium seeking algorithms generate idealized, instantaneous motion commands, the actual physical agents cannot execute these sudden changes due to inherent mechanical inertia and physical resistance. Since these Nash equilibrium seeking algorithms for players of first-order dynamics fail to adapt to the varied physical realities of different machines, developing Nash equilibrium seeking for players of heterogeneous and complex dynamics is of great importance.

It is worth noting that in most existing works on these problems arising from practical multi-agent systems, the developed distributed Nash equilibrium seeking algorithms converge to the Nash equilibrium asymptotically or exponentially. However, many engineering applications in distributed Nash equilibrium seeking call for fixed-time convergence instead of exponential or asymptotical convergence. Thus, distributed fixed-time Nash equilibrium seeking problems of communication networks under denial-of-service attacks, constrained non-cooperative games and players governed by complex dynamics deserve further investigation.

1.2 Literature Review

In the past decades, a great deal of effort has been devoted to Nash equilibrium seeking of multi-agent systems. This section will briefly survey those results relevant to the focus of this thesis. First, the results on Nash Equilibrium Seeking of Multi-Agent Systems are reviewed. Second, some results on distributed Nash equilibrium seeking under denial-of-service attacks are presented. Third, some results on distributed Nash equilibrium seeking for constrained games are introduced. Then, some results on distributed Nash equilibrium seeking for players complex dynamics are reviewed. Finally, some results on finite-time/fixed-time distributed Nash equilibrium seeking are summarized.

1.2.1 Nash Equilibrium Seeking of Multi-Agent Systems

In the past decades, Nash Equilibrium Seeking of multi-agent systems have been extensively studied in different research fields due to their wide applications. Nash equilibrium seeking, as a fundamental task of non-cooperative games of multi-agent systems, aims to seek an action profile in which no player can gain more payoff by unilaterally changing its own action and has been a major research topic. To find the Nash equilibrium in non-cooperative games, there are several algorithms, including the best response strategies, fictitious play, and gradient search, to name just a few. While best response and fictitious play are well-suited for finite or discrete games, they often entail heavy computational burdens and are difficult to integrate with continuous multi-agent systems. Among these methods, gradient search stands out due to its reliance on local derivative information and its natural compatibility with continuous action spaces.

Therefore, the subsequent review primarily focuses on gradient-based distributed seeking algorithms.

Based on whether or not each individual player has direct access to other players' actions, Nash equilibrium seeking approaches can be classified into two categories, that is, centralized algorithms and distributed algorithms. However, centralized algorithms, where the centralized information is to be shared by all players, would run into difficulties for large-scale network-based games. As a result, distributed algorithms, where only partial information is to be shared by neighboring players, have been receiving increasing attention.

The majority of current studies on distributed Nash equilibrium seeking primarily concentrate on consensus-based algorithms, though there are some other methods such as passivity-based algorithms. The passivity-based algorithm reformulates the distributed seeking problem as a consensus problem, acting as a corrected gradient play augmented with an estimation term. While it possesses theoretical elegance by establishing the incremental passivity of the underlying system, its convergence requires some restrictive assumptions. Specifically, it requires that the algebraic connectivity of the communication network must be greater than a specific positive threshold determined by the inherent properties of the game mapping. This introduces a restrictive trade-off between the game properties and the graph properties. In contrast, the leader-following consensus-based algorithm provides a more flexible and robust alternative. By employing a two time-scale structure where each player maintains a local estimate of all other players' actions through a leader-following consensus protocol, this approach completely decouples the optimization process from the algebraic connectivity constraints of the network. As long as the communication graph is connected, the quasi-steady state of the consensus part guarantees accurate estimation, reducing the system to a standard gradient search algorithm. Due to this structural superiority and relaxed topological requirements, the leader-following consensus-based architecture serves as a much more suitable and scalable foundation for designing advanced distributed seeking strategies.

1.2.2 Distributed Nash Equilibrium Seeking Under Denial-of-Service Attacks

As argued before, communication networks of multi-agent systems are often exposed to DoS attacks, which implies that the communication links are intermittently and maliciously severed. Under DoS attacks, distributed Nash equilibrium seeking algorithms in [ye2017distributed, 10570469, 9209113, 10239992, HE2024111464, TIAN2025112296, 9863764, 10229248, LIU2026112603] cannot effectively work, as attacks can interrupt data transmission between players and even pose significant risks to the reliability and stability of the concerned multi-agent systems. In recent years, the problem of distributed Nash equilibrium seeking of multi-agent systems under DoS attacks has drawn much attention in the control community, see, for example, [4, 10, 11, 14, 15]. To address the challenges of DoS attacks, the authors in [1, 13] developed algorithms for solving distributed Nash equilibrium seeking problem under DoS attacks. However, these algorithms rely on a restrictive assumption, that is, the communication graph must maintain strict connectivity at every time instant. To overcome this restrictive instantaneous connectivity requirement, the authors in [4–7, 10, 11, 14, 15] relaxed the DoS attacks assumption to allow for intermittent disconnections, accommodating extreme scenarios where data transmission is completely interrupted.

1.2.3 Distributed Nash Equilibrium Seeking for Constrained Non-Cooperative Games

In practical game problems, players are often subject to individual limitations, such as the maximum power output of a generator. Therefore, it is necessary to investigate noncooperative game problems with local set constraints. To deal with local strategy set constraints, projection operator-based methods have been effectively employed in [lu_distributed_2018], where the partial gradients of cost functions are projected onto local strategy sets. Furthermore, players in practical applications frequently share limited network resources. Unlike local constraints, global constraints couple the strategy sets of all players. Consequently, distributed Nash equilibrium seeking for noncooperative game problems with global constraints, where each player's possible action is constrained by the decisions made by the other players, has garnered significant attention recently [lu2018distributed, zou2021continuous, 9343684, zeng2019generalized, 10229248, 10334027, 9345446, sun2020continuous]. To address these complex global constraints,

various techniques have been developed. For example, the projection operator was utilized to address the distributed game problems with global constraints in [lu2018distributed, zou2021continuous]. The authors in [9345446, 9343684, zeng2019generalized, 10229248, 10334027] proposed distributed Nash equilibrium seeking approaches based on the primal-dual methods for games with global constraints. The authors in [sun2020continuous] proposed a penalty-based distributed algorithm for solving Nash equilibrium seeking problems with global constraints.

1.2.4 Distributed Nash Equilibrium Seeking for Players of Complex Dynamics

Since the actions of individual players in many practical games, especially those arising from control systems—are governed by more general dynamics. To bridge the gap between game-theoretic algorithms and physical execution, it is imperative to incorporate these complex agent dynamics into the Nash equilibrium seeking framework. Consequently, recent literature has progressively extended player dynamics from basic integrators to more sophisticated physical models. For linear cases, distributed NE seeking strategies have been developed for agents characterized by second-order dynamics [...] and higher-order systems [...]. These results were subsequently generalized to accommodate players subject to general linear dynamics [...]. Furthermore, since physical systems inherently exhibit nonlinearities, NE seeking algorithms have also been investigated for games involving first-order nonlinear dynamics [...]. More recently, to accurately capture the intrinsic properties of mechanical and robotic systems, significant attention has been devoted to noncooperative games governed by nonlinear Euler-Lagrange (EL) dynamics [...]

1.2.5 Finite-Time/Fixed-Time Distributed Nash Equilibrium Seeking

In recent years, motivated by finite- and fixed-time stability theories proposed in the control field [6104367, GUAN2021105054, SONG2024105835, CHEN2018539], problems of distributed finite/fixed-time converging Nash equilibrium seeking have been investigated in [fang2021distributed, li2019distributed, poveda2022fixed, meng2025distributed] due to their faster convergence and better robustness. In particular, the authors in

[fang2021distributed, li2019distributed] proposed distributed finite-time converging Nash equilibrium seeking algorithms to non-cooperative games. Unfortunately, their convergence times, like other finite-time converging algorithms, are dependent on the initial states of the underlying systems and thus could become very large when the initial states deviate far away from the equilibrium. Distributed fixed-time converging Nash equilibrium seeking approach with its convergence time independent of initial conditions was proposed in [poveda2022fixed] for non-cooperative games under a restrictive assumption that the individual players' cost functions are available to themselves although the cost functions may depend on all players' actions. The authors in [meng2025distributed] further developed distributed fixed-time algorithm to solve the distributed fixed-time converging Nash equilibrium seeking problem for non-cooperative games without this restrictive assumption. However, as the number of players increases, the existence of feasible design parameters for the algorithm proposed in [meng2025distributed] is not always guaranteed. It should be also be pointed out that first-order integrator dynamics were considered in [fang2021distributed, li2019distributed, poveda2022fixed], while first- and second-order integrator dynamics were considered in [meng2025distributed].

1.3 Motivations for This Thesis

As discussed in the previous subsection, distributed Nash equilibrium seeking has attracted increasing attention due to its scalability advantages over centralized schemes in large-scale network-based games. Most existing distributed Nash equilibrium seeking algorithms, however, guarantee only asymptotical or exponential convergence, implying that the Nash equilibrium is reached only as time goes to infinity. Although finite-time Nash equilibrium seeking methods have been reported more recently, their convergence-time upper bounds typically depend on the initial conditions and may become excessively large when the initial state is far from the equilibrium. In many practical engineering applications, it is highly desirable to achieve fixed-time convergence, where upper bound of the convergence time is independent of initial conditions and the upper bound can be explicitly expressed.

On the other hand, distributed Nash equilibrium seeking problems in practical multi-agent systems often encounter additional challenges, including communication networks exposed to denial-of-service attacks, constrained non-cooperative games and players governed by heterogeneous complex dynamics. It is noted that most existing

results for these practical problems still focus on asymptotical or exponential convergence, and fixed-time guarantees remain largely underexplored in the Nash equilibrium seeking set-up. To the best of our knowledge, the following distributed Nash equilibrium seeking problems of multi-agent systems have not been well addressed thus far, which is the motivation for this thesis.

1. The general distributed fixed-time Nash equilibrium seeking problem of non-cooperative games without restrictive assumptions on cost function availability or network scalability.
2. The distributed fixed-time Nash equilibrium seeking problem of non-cooperative games subject to local and/or global set constraints.
3. The distributed fixed-time Nash equilibrium seeking problem of multi-agent systems governed by heterogeneous complex dynamics.
4. The resilient distributed fixed-time Nash equilibrium seeking problem of non-cooperative games over communication networks under denial-of-service attacks.

1.4 Thesis Contributions

This thesis mainly investigates two types of problems: the general distributed fixed-time Nash equilibrium seeking problem without restrictive assumptions on cost functions or network scalability, and the distributed fixed-time Nash equilibrium seeking problem of multi-agent systems subject to constrained strategy sets, heterogeneous complex dynamics, and denial-of-service attacks. The main contributions are summarized as follows:

1. A novel distributed fixed-time Nash equilibrium seeking algorithm for unconstrained non-cooperative games is proposed. Unlike those distributed Nash equilibrium seeking algorithms with asymptotical or exponential convergence for unconstrained games [9740409, liu2022distributed, shi2024distributed], our work achieves fixed-time convergence with explicitly given upper bound for the convergence time, which is independent of the initial conditions of the concerned system. In order to address the challenge of estimating the other players' actions in fixed time during the process of Nash equilibrium seeking, we develop an improved distributed fixed-time observer. Unlike existing fixed-time converging observer designs (e.g., [SONG2024105835, CHEN2018539]), which may

fail in our problem set-up since players' actions are updated via potentially unbounded cost function gradients, we utilize two additional power terms to not only accelerate the convergence of the estimators to the players' actions but also counteract the destabilizing effects induced by the unbounded gradients. The improved distributed fixed-time observer ensures that each player can estimate the others' actions in fixed time, overcoming the limitations of existing methods.

2. A highly scalable distributed fixed-time Nash equilibrium seeking algorithm is proposed for unconstrained non-cooperative games, significantly relaxing the parameter selection constraints of existing methods. Unlike the coupled passivity-based design in [meng2025distributed] where verification of parameter feasibility involves solving some complex inequalities that can be challenging for large-scale systems, we propose a novel fixed-time converging Nash equilibrium seeking approach with a two time-scale framework. Specifically, the dynamics of the distributed observers are considered as fast dynamics, while the Nash equilibrium seeking dynamics are considered as slow dynamics. It is shown that if the convergence rates of the fast dynamics is sufficiently faster than that of the slow dynamics, then the Nash equilibrium can be reached in fixed time. Since the convergence rates of the fast and slow dynamics can be individually tuned by appropriately adjusting their respective design parameters, the existence of feasible design parameters is always guaranteed. While the two-time-scale framework is widely adopted for asymptotically and exponentially converging algorithms [ye2017distributed, ye2021adaptive, ye2019distributed, DONG2021104955], to the best of our knowledge, our work is the first to apply this framework for a fixed-time converging algorithm.
3. A novel resilient distributed fixed-time Nash equilibrium seeking algorithm for constrained non-cooperative games is proposed. Unlike the distributed Nash equilibrium seeking algorithm for under DoS attacks [he2025distributed], our work achieves fixed-time convergence with an explicitly given upper bound of the convergence time, which is independent of the initial conditions of the concerned system. In particular, we develop a resilient fixed-time distributed algorithm for seeking the Nash equilibrium under DoS attacks and thereby solving the distributed Nash equilibrium seeking problem under DoS attacks. Moreover, to deal with the challenge in analyzing the fixed-time convergence of the proposed algorithm under DoS attacks, we adopt the switching system theory.

4. A novel distributed fixed-time Nash equilibrium seeking algorithm for constrained non-cooperative games is proposed. Unlike those works for constrained games [lu2018distributed, zou2021continuous, 9343684, zeng2019generalized, 10229248, 10334027, sun2020continuous, 9345446] with either asymptotic or exponential convergence, our work achieves fixed-time convergence with an explicitly given upper bound for the convergence time, which is independent of the initial conditions of the concerned system. To address the challenge arising from designing distributed fixed-time converging Nash equilibrium seeking algorithm for constrained games with coupled inequality constraints, we develop novel fixed-time converging distributed observers to estimate the other players' actions and gradients through their neighbors and fixed-time converging distributed action update dynamics based on the projected gradient approach.
5. A novel distributed fixed-time Nash equilibrium seeking framework is developed for non-cooperative games with players of heterogeneous complex dynamics. The significance of this framework lies in simultaneously addressing the limitations of existing works in both convergence performance and physical model generality. On the one hand, unlike state-of-the-art algorithms for complex physical dynamics (e.g., high-order, general linear and nonlinear systems) that achieve only asymptotic or exponential convergence [your_asymptotic_refs], the developed approach ensures fixed-time convergence to the exact Nash equilibrium. On the other hand, compared to recent finite- and fixed-time methods [fang2021distributed, li2019distributed, poveda2022fixed, 10380618, meng2025distributed] that are heavily confined to idealized first- or second-order integrators, the proposed algorithm is for heterogeneous complex dynamics and can include first- and second-order dynamics as its special cases. To deal with the challenge arising from the heterogeneous complex dynamics, we propose a two-layer design framework, which consists of a virtual layer and a physical layer, and solve the problem of distributed Nash equilibrium seeking with fixed-time convergence. Furthermore, we develop a novel tracking protocol based on sliding mode theory in the physical layer and show that the actions of the players of heterogeneous complex dynamics converge to the virtual players' actions within a fixed time, and thus solving the concerned problem

1.5 Thesis Organization

The remainder of this thesis is organized as follows. **Chapter 2** consists of the following points. Firstly, some notations in graph theory are briefly reviewed to make this thesis self-contained. Then, denial-of-service attacks are introduced, followed by some mathematical results on inequalities and functions. Next, some definitions and results on finite-time/fixed-time stabilities are presented. **Chapter 3** investigates the problem of distributed fixed-time Nash equilibrium seeking for unconstrained non-cooperative games. A novel distributed fixed-time Nash equilibrium seeking algorithm with a two time-scale framework is proposed based on the leader-following consensus protocol and the sliding mode theory. As communication networks of multi-agent systems in practice are often subject to denial-of-service attacks, **Chapter 4** studies the resilient distributed fixed-time NE seeking problem under DoS attacks. Based on switching system theory and sliding mode techniques, a resilient algorithm is developed to ensure that the actions of all players converge to the exact NE within a fixed time despite the presence of DoS attacks, providing an explicitly bounded convergence time. In practice, a player's feasible action space is often restricted by physical limitations, inevitably giving rise to constrained non-cooperative games. To address this, **Chapter 5** formulates the constrained NE seeking problem as a variational inequality problem. By integrating neurodynamic methods with sliding mode control, a novel distributed algorithm is designed. It is theoretically proven that the proposed algorithm drives the actions of all players to the exact Nash equilibrium in a fixed time, independently of the initial conditions. Furthermore, in many practical games, especially arising from the area of control systems, the actions of individual players are governed by heterogeneous complex dynamics. Therefore, **Chapter 6** studies the distributed fixed-time Nash equilibrium seeking for multi-agent systems with heterogeneous complex dynamics. A novel two-layer design framework consisting of a virtual layer and a physical layer is established. In the virtual layer, distributed fixed-time Nash equilibrium seeking algorithms are utilized to generate a virtual signal that converges to the Nash equilibrium. It is proven that, under this decoupled framework, the physical actions of all heterogeneous players converge to the exact Nash equilibrium in a fixed time from any initial condition. Moreover, the upper bounds for their convergence times can be explicitly given. While **Chapters 3-6** provide rigorous theoretical analyses accompanied by numerical examples, **Chapter 7** presents comprehensive engineering applications to validate the practical efficacy of the developed algorithms. For all applications, detailed system modeling, algorithm deployment, and simulation results are provided to

bridge the gap between theoretical game designs and physical executions. This thesis is concluded and some potential issues for future works are given in **Chapter 8**.

This thesis is typeset using **LATEX**. All the simulations are performed using **Python**.

Chapter 2

Preliminaries

2.1 Graph Notations

Let $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ denote a graph, where $\mathcal{V} = \{1, \dots, N\}$ is the vertex set and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ is the edge set. $\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}$ denotes the neighbor set of vertex i . Graph $\mathcal{G}$ is said to be connected if for any two vertices, there is a path in $\mathcal{G}$. The matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix corresponding to $\mathcal{G}$, where $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$, and $a_{ij} = 0$, otherwise. The Laplacian matrix associated with $\mathcal{A}$ is denoted as $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{N \times N}$, whose elements are defined as follows: $l_{ij} = -a_{ij}$ if $i \neq j$ and $l_{ii} = \sum_{j=1}^N a_{ij}$.

Lemma 2.1. [NI2010209] *Consider the matrix $\mathcal{H} = \mathcal{L} + \text{diag}\{a_{ij}, i, j = 1, \dots, N\}$ associated with the graph $\mathcal{G}$. If the graph $\mathcal{G}$ is undirected, the matrix $\mathcal{H}$ has the following properties: (1) the matrix $\mathcal{H}$ is positive semi-definite (i.e., $\mathcal{H} \geq 0$); and (2) the matrix $\mathcal{H}$ is strictly positive definite (i.e., $\mathcal{H} > 0$) if and only if the graph $\mathcal{G}$ is connected.*

2.2 Denial-of-Service Attacks

In this work, Adversaries are assumed to attack the edges of the communication graph $\mathcal{G}$. When an edge (j, i) is jammed by attackers, the information flow between player j and player i is blocked. Let $s_{ij}(t) = 1$ for $i, j \in \mathcal{V}$ if information between players i and j can be transmitted normally, and $s_{ij}(t) = 0$, otherwise. Then, the communication among N players by taking into consideration of DoS attacks can be described by a time-varying undirected graph $\mathcal{G}(t)$. The adjacency matrix of graph $\mathcal{G}(t)$ is defined as

$\mathcal{A}(t) = [a_{ij}(t)] \in \mathbb{R}^{N \times N}$, where $a_{ij}(t) = a_{ij} \cdot s_{ij}(t)$, for $i, j = 1, \dots, N$. If at least one edge of graph $\mathcal{G}(t)$ is jammed by the attackers, it is said that graph $\mathcal{G}(t)$ is under DoS attacks. Define the time sequences of the starting instant and ending instant of the k -th DoS attack for graph $\mathcal{G}(t)$ as t_k^a and t_k for $k \in \mathbb{N}_+$, respectively. Denote $t_0 = 0$ as the initial time. Let $[t_{k-1}, t_k^a)$ and $[t_k^a, t_k)$ be the k -th time intervals for graph $\mathcal{G}(t)$ without DoS attacks and under DoS attacks, respectively. Define

$$\Gamma_D(0, t) = \begin{cases} \bigcup_{l=1}^{k-1} [t_l^a, t_l), t \in [t_{k-1}, t_k^a), \\ \bigcup_{l=1}^{k-1} [t_l^a, t_l) \cup [t_k^a, t], t \in [t_k^a, t_k), \end{cases} \quad (2.1)$$

and

$$\Gamma_N(0, t) = [0, t] \setminus \Gamma_D(0, t), \quad (2.2)$$

as the set of time intervals for graph $\mathcal{G}(t)$ under DoS attacks and without DoS attacks, respectively. Denote

$$|\Gamma_D(0, t)| = \begin{cases} \sum_{l=1}^{k-1} (t_l - t_l^a), t \in [t_{k-1}, t_k^a), \\ \sum_{l=1}^{k-1} (t_l - t_l^a) + (t - t_k^a), t \in [t_k^a, t_k), \end{cases} \quad (2.3)$$

$$|\Gamma_N(0, t)| = t - |\Gamma_D(0, t)|. \quad (2.4)$$

2.3 Variational Inequalities and Neurodynamics

2.4 Some Mathematical Results

Lemma 2.2. *[meng2025distributed] For any $x_i \geq 0$, $i = 1, \dots, N$, and $0 < \alpha \leq 1 < \beta$, we have*

$$\left(\sum_{i=1}^N x_i \right)^\alpha \leq \sum_{i=1}^N x_i^\alpha, \quad N^{1-\beta} \left(\sum_{i=1}^N x_i \right)^\beta \leq \sum_{i=1}^N x_i^\beta.$$

Lemma 2.3. *[meng2025distributed] For any real numbers $x, y \in \mathbb{R}$, there exist positive constants p, q, ρ_2 , and continuous function $\rho_1(\cdot) > 0$ such that*

$$\rho_1(\cdot) x^p y^q \leq \rho_2 |x|^{p+q} + \frac{q}{p+q} \left(\frac{p+q}{p} \right)^{-\frac{p}{q}} \rho_1(\cdot)^{\frac{p+q}{q}} \rho_2^{-\frac{p}{q}} |y|^{p+q}.$$

Lemma 2.4. *Let p be a positive constant. For any $x, y \in \mathbb{R}$, the inequality $|\lceil x \rceil^p - \lceil y \rceil^p| \leq 2^{1-p}|x - y|^p$ holds for $0 < p \leq 1$; the inequality $|\lceil x \rceil^p - \lceil y \rceil^p| \leq p(|x - y|^p + |x - y||y|^{p-1})$ holds for $1 < p \leq 2$.*

Proof. For $0 < p \leq 1$: If $xy \geq 0$, assume $x \geq y \geq 0$ without loss of generality. It follows from Lemma 2.2 that $(x - y)^p + y^p \geq x^p$. Since $|\lceil x \rceil^p - \lceil y \rceil^p| = x^p - y^p$, one has $|\lceil x \rceil^p - \lceil y \rceil^p| \leq |x - y|^p \leq 2^{1-p}|x - y|^p$. If $xy < 0$, assuming $x > 0 > y$ without loss of generality, it follows from $\frac{x^p + (-y)^p}{2} \leq (\frac{x - y}{2})^p$ that $|\lceil x \rceil^p - \lceil y \rceil^p| = x^p + (-y)^p \leq 2^{1-p}(x - y)^p$. For $1 < p \leq 2$: Consider the continuously differentiable function $g(z) = \lceil z \rceil^p$ with derivative $g'(z) = p|z|^{p-1}$. By the Mean Value Theorem [zorich2016mathematical], there exists g_0 between x and y such that $|\lceil x \rceil^p - \lceil y \rceil^p| = p|g_0|^{p-1}|x - y|$. Since $|g_0| \leq \max(|x|, |y|) \leq |x - y| + |y|$ and $0 < p - 1 \leq 1$, it also follows from Lemma 2.2 that $|g_0|^{p-1} \leq (|x - y| + |y|)^{p-1} \leq |x - y|^{p-1} + |y|^{p-1}$. Then, it can be further inferred that $|\lceil x \rceil^p - \lceil y \rceil^p| \leq p(|x - y|^p + |x - y||y|^{p-1})$. $\square$

Lemma 2.5. [9262909] $\forall y \in \mathbb{R}^N$, suppose

$$f_1(y) = (\beta_1^2 \sum_{i=1}^N |y_i|^{2\mu} + 2\beta_1\beta_2 \sum_{i=1}^N |y_i|^{\mu+\nu} + \beta_2^2 \sum_{i=1}^N |y_i|^{2\nu}),$$

$$f_2(y) = \sum_{i=1}^N \left(\frac{\beta_1}{\mu + 1} |y_i|^{\mu+1} + \frac{\beta_2}{\nu + 1} |y_i|^{\nu+1} \right),$$

where $\beta_1, \beta_2, 1 > \mu > 0, \nu > 1$ are some positive constants. Then, one has the following inequality:

$$f_1(y) \geq \chi_1 f_2(y)^{\frac{\mu+\nu}{1+\nu}} + \chi_2 f_2(y)^{\frac{2\nu}{\nu+1}},$$

where $\chi_1 = \frac{1}{2}\beta_{\min}^2 \left(\frac{1+\mu}{\beta_{\max}}\right)^{\frac{\mu+1}{1+\nu}}$, $\chi_2 = \frac{1}{2}\beta_{\min}^2 (2N)^{\frac{1-\nu}{1+\nu}} \left(\frac{1+\mu}{\beta_{\max}}\right)^{\frac{2\nu}{\nu+1}}$, $\beta_{\min} = \min\{\beta_1, \beta_2\}$ and $\beta_{\max} = \max\{\beta_1, \beta_2\}$.

Lemma 2.6. [1098584] For a linear system

$$\dot{\xi} = A\xi + Bu, \tag{2.5}$$

where $\xi \in \mathbb{R}^m$, $u \in \mathbb{R}^p$, and A and B are the system matrices. If (A, B) is controllable and $\text{rank}(B) = p$, there exist a linear coordinate transformation $\hat{\xi} = \aleph \xi \in \mathbb{R}^m$ such that $\hat{\xi} = [\hat{\xi}_1^\top, \dots, \hat{\xi}_m^\top]$. Let ζ_i , $i = 1, \dots, p$, be the controllability index of $\hat{\xi}$. Then $\hat{\xi}$

can be described by $\hat{\xi} = [\hat{\xi}_{1,1}, \dots, \hat{\xi}_{1,\zeta_1}, \dots, \hat{\xi}_{m,1}, \dots, \hat{\xi}_{m,\zeta_m}]^\top$ and the dynamics of $\hat{\xi}_i$, $i = 1, \dots, p$, has the following form:

$$\begin{aligned}\dot{\hat{\xi}}_{i,k}(t) &= \hat{\xi}_{i,k+1}(t), \quad k = 1, \dots, \zeta_i - 1, \\ \dot{\hat{\xi}}_{i,\zeta_i}(t) &= F_i^\top A^{\zeta_i} \hat{\xi}(t) + F_i^\top A^{\zeta_i-1} B u(t),\end{aligned}$$

where $F_i^\top = i_{r_i}^\top \mathfrak{N}$, $i_k \in \mathbb{R}^m$ is a vector with the k -th element being set to 1 and all the other elements being set to 0, and $r_i = \sum_{k=0}^{i-1} (1 + \zeta_k)$ with $\zeta_0 = 0$. Moreover, define $\hat{\xi}^\zeta = [\hat{\xi}_{1,\zeta_1}, \dots, \hat{\xi}_{p,\zeta_p}]^\top \in \mathbb{R}^p$, one has $\dot{\hat{\xi}}^\zeta = G \hat{\xi} + M u$, where $G = [(F_1^\top A^{\zeta_1})^\top, \dots, (F_m^\top A^{\zeta_m})^\top]^\top$, and $M = [(F_1^\top A^{\zeta_1-1} B)^\top, \dots, (F_m^\top A^{\zeta_m-1} B)^\top]^\top$, with M being invertible.

2.5 Definitions and Lemmas on Fixed-Time Stability

Consider the following dynamic system,

$$\dot{x} = \psi(x(t)), \quad x(0) = x_0 \in \mathbb{R}^m, \quad (2.6)$$

where $\psi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is a mapping. x^* is called to be an equilibrium point of system (??) if $\psi(x^*) = 0$.

Definition 2.7. [6104367] System (2.6) is said to be fixed-time stable if it is Lyapunov stable and for any $x_0 \in \mathbb{R}^m$, there exists a constant $0 \leq T < +\infty$ such that the solution $x(t, x_0) = x^*$ for all $t > T$.

Lemma 2.8. [6104367] Let $\Omega \subseteq \mathbb{R}^m$ be a neighborhood of the equilibrium point for system (2.6). Suppose that there exists a continuously differentiable positive definite function $V : \Omega \rightarrow \mathbb{R}$ for system (2.6) such that

$$\dot{V}(x) \leq -C_1 V(x)^p - C_2 V(x)^q, \quad \forall x \in \Omega \setminus \{0\}, \quad (2.7)$$

where $C_1, C_2 > 0$, $0 < p < 1$ and $q > 1$ are four constants. Then the equilibrium point of system (2.6) is fixed-time stable with the upper bound of the convergence time given by

$$T = \frac{1}{C_1(1-p)} + \frac{1}{C_2(q-1)}. \quad (2.8)$$

Lemma 2.9. [zuo2017fixed] Consider the n -th order integrator system

$$\dot{x}_k(t) = x_{k+1}(t), \quad k = 1, \dots, n-1, \quad \dot{x}_n(t) = u(t), \quad (2.9)$$

where $x(t) = [x_1(t), x_2(t), \dots, x_n(t)]^\top \in \mathbb{R}^n$ is the state vector and $u(t) \in \mathbb{R}$ is the control input. Let the positive constants $\kappa_i, \bar{\kappa}_i > 0$, ($i = 1, 2, \dots, n$) be such that both the polynomial $s^n + \kappa_n s^{n-1} + \dots + \kappa_2 s + \kappa_1$ and $s^n + \bar{\kappa}_n s^{n-1} + \dots + \bar{\kappa}_2 s + \bar{\kappa}_1$ are Hurwitz in terms of the Laplace operator s . The integrator system can be stabilized at the origin in a fixed time under the feedback control

$$u(t) = - \sum_{i=1}^n (\kappa_i [x_i]^{\gamma_i} + \bar{\kappa}_i [x_i]^{\bar{\gamma}_i}), \quad (2.10)$$

with parameters γ_i and $\bar{\gamma}_i$ satisfying, for $i = 2, \dots, n$, $\gamma_{i-1} = \frac{\gamma_i \gamma_{i+1}}{2\gamma_{i+1} - \gamma_i}$ and $\bar{\gamma}_{i-1} = \frac{\bar{\gamma}_i \bar{\gamma}_{i+1}}{2\bar{\gamma}_{i+1} - \bar{\gamma}_i}$, where $\gamma_{n+1} = \bar{\gamma}_{n+1} = 1$, $\gamma_n = \gamma_0 \in (0, 1 - \epsilon)$ and $\bar{\gamma}_n = \bar{\gamma}_0 \in (1, 1 + \bar{\epsilon})$ for sufficiently small ϵ and $\bar{\epsilon}$. Moreover, the upper bound of the convergence time is given by

$$T = \frac{\gamma_0 \lambda_{\max}(P_1)}{(1 - \gamma_0) \lambda_{\min}(Q_1)} \lambda_{\max}^{\frac{1-\gamma_0}{\gamma_0}}(P_1) + \frac{\bar{\gamma}_0 \lambda_{\max}(P_2)}{(\bar{\gamma}_0 - 1) \lambda_{\min}(Q_2)} \lambda_{\max}^{\frac{\bar{\gamma}_0-1}{\bar{\gamma}_0}}(P_2), \quad (2.11)$$

where $P_k, Q_k > 0$ satisfy the Lyapunov equation $P_k A_k + A_k^\top P_k = -Q_k$, $k = 1, 2$, $A_1 = \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ -\kappa_1 & -\kappa_2 & \dots & -\kappa_n \end{bmatrix}$, $A_2 = \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ -\bar{\kappa}_1 & -\bar{\kappa}_2 & \dots & -\bar{\kappa}_n \end{bmatrix}$, and $\lambda_{\max}(\cdot)$, $\lambda_{\min}(\cdot)$ denote the maximum and minimum eigenvalue of the matrix $(\cdot)$, respectively.

Chapter 3

Distributed Fixed-Time Nash Equilibrium Seeking for Non-Cooperative Games

3.1 Introduction

The rapid integration of sensing, computation, and communication technologies has catalyzed the evolution of large-scale network-based systems, shifting the paradigm from centralized control to distributed decision-making architectures. Within this context, non-cooperative game theory has emerged as a powerful mathematical framework to model, analyze, and optimize the strategic interactions among selfish and rational agents. In a non-cooperative game, each player aims to minimize its own cost function, which depends not only on its own strategic action but also on the decisions made by other participants. The steady-state solution of such strategic interactions is formally characterized by the Nash equilibrium (NE), a joint action profile where no individual player can unilaterally reduce its cost by unilaterally deviating from its current strategy. Consequently, distributed NE seeking has garnered ubiquitous attention and found widespread applications in diverse engineering domains, including energy trading in smart grids, sensor coverage, and autonomous vehicle coordination [citation].

Traditional NE seeking algorithms predominantly rely on centralized paradigms, requiring a central coordinator to gather global information, compute the equilibrium,

and subsequently broadcast the optimal actions back to the individual players. However, in modern large-scale multi-agent systems, such centralized schemes suffer from severe vulnerabilities, including single points of failure, immense computational burdens, and strict privacy concerns. To circumvent these fundamental limitations, distributed NE seeking approaches have been extensively investigated. By utilizing local information exchange over a shared communication network, players can collaboratively update their strategies to asymptotically converge to the NE. Based on the gradient play and consensus protocols, a plethora of distributed asymptotic and exponential converging algorithms have been proposed in the literature [citation]. Despite their theoretical elegance, these algorithms structurally guarantee that the equilibrium is reached only as time approaches infinity. In practical engineering applications, infinite convergence time translates to prolonged transient responses and potential performance degradation, strictly limiting their applicability in time-critical tasks.

To accelerate the convergence speed, finite-time stability theory has been recently integrated into the game-theoretic framework. Distributed finite-time NE seeking algorithms have been developed in [citation], demonstrating superior transient performance and enhanced robustness against exogenous disturbances. Nevertheless, the settling-time upper bounds of these finite-time algorithms are inherently dependent on the initial states of the underlying multi-agent systems. Consequently, if the initial states deviate significantly from the equilibrium point, the convergence time can become excessively large or even practically unbounded, violating the stringent timing requirements of modern physical systems. To overcome this severe limitation, fixed-time stability theory was introduced, ensuring that the convergence time is strictly bounded by a predefined constant that is completely independent of the initial conditions.

While fixed-time NE seeking algorithms represent a significant theoretical breakthrough, their current implementations in distributed games are often hindered by restrictive structural assumptions. For instance, the authors in [citation] proposed a fixed-time converging approach under the strong premise that individual players have full access to the explicit formulations of their cost functions, which is highly unrealistic in fully distributed setups. Moreover, to relax such information requirements, recent works, such as the coupled passivity-based design in [citation], developed fixed-time algorithms. Unfortunately, the verification of parameter feasibility in these approaches heavily involves solving complex, coupled mathematical inequalities. As the number of players increases, solving these inequalities becomes computationally prohibitive,

and the existence of feasible design parameters is not always mathematically guaranteed, drastically limiting the scalability of these algorithms in large-scale networks.

Motivated by the aforementioned theoretical gaps, this chapter aims to establish a highly scalable and mathematically relaxed distributed fixed-time NE seeking framework for unconstrained non-cooperative games. To completely eliminate the restrictive parameter conditions prevalent in the existing literature, a novel two-time-scale design architecture is proposed. While the singular perturbation-like two-time-scale framework has been widely adopted in asymptotic and exponential optimization [citation], its integration into fixed-time game-theoretic algorithms remains an open and challenging problem. The main contributions of this chapter are explicitly summarized as follows:

1. **A Novel Two-Time-Scale Fixed-Time Framework:** We propose a completely new distributed fixed-time NE seeking approach explicitly partitioned into fast and slow dynamics. Specifically, the dynamics of the distributed gradient observers are designed as the fast time-scale layer, while the local action update laws are cast as the slow time-scale layer.
2. **Decoupled Parameter Feasibility:** Unlike existing passivity-based fixed-time designs where parameter selection requires solving large-scale coupled inequalities, the proposed decoupled two-time-scale framework allows the convergence rates of the fast and slow dynamics to be tuned independently. It is rigorously proven that as long as the fast layer converges sufficiently quicker than the slow layer, exact NE seeking in fixed time is achieved, thereby strictly guaranteeing parameter feasibility regardless of the network dimension.
3. **Rigorous Stability Certificates:** By seamlessly combining the leader-following consensus protocol with advanced sliding mode control theory, a rigorous Lyapunov-based stability analysis is provided. The explicit upper bound of the convergence time is analytically derived, demonstrating complete independence from the initial states of the multi-agent system.

The remainder of this chapter is structured as follows. Section 3.2 details the problem formulation and basic assumptions. The proposed two-time-scale algorithm and its fixed-time convergence analysis are presented in Section 3.3. Section 3.4 provides numerical simulations, and Section 3.5 concludes the chapter.

3.2 Problem Formulation

Consider a set of players $\mathcal{V} \triangleq \{1, \dots, N\}$ with the action of player i denoted as $x_i \in \mathbb{R}^{n_i}$ and the action vector of the other players denoted by $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$, where $n = \sum_{i=1}^N n_i$, respectively. The dynamic of player i is given as $\dot{x}_i = u_i$, where $u_i \in \mathbb{R}^{n_i}$ is the control input. Each player i aims to minimize its own objective function $f_i(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$, where $\mathbf{x} = [x_1, \dots, x_N]^T \in \mathbb{R}^n$ is the action vector of N players. If agent j is not a neighbor of agent i , which means $a_{ij} = 0$, then player i does not have direct access to player j 's action. Otherwise, player i can get the information of player j via a communication graph $\mathcal{G}$.

The unconstrained game in this article is described as

$$\min_{x_i} f_i(x_i, \mathbf{x}_{-i}). \quad (3.1)$$

First, we formalize the definition of the Nash equilibrium of the unconstrained non-cooperative game.

Definition 3.1. [ye2017distributed] A strategy profile $\mathbf{x}^* = [x_1^*, \dots, x_N^*]$ is called the Nash equilibrium of the non-cooperative game (3.1) if $\forall i \in \mathcal{V}$,

$$f_i(y_i^*, \mathbf{x}_{-i}^*) \leq f_i(x_i, \mathbf{x}_{-i}^*), \quad \forall x_i \in \mathbb{R}^{n_i}. \quad (3.2)$$

In this chapter, our objective is to design a distributed Nash equilibrium seeking algorithm for the players such that the action vector $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{x}^*$ of the game (3.1) in fixed time. Denote

$$H(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_N} \\ \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2^2} & \dots & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_1} & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_2} & \dots & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N^2} \end{bmatrix}, \quad (3.3)$$

where $\mathbf{x} = [x_1, \dots, x_N]^T \in \mathbb{R}^n$, $x_i \in \mathbb{R}^{n_i}$, $i = 1, \dots, N$. To proceed, the following mild assumptions are made. To proceed, the following mild assumptions are made.

Assumption 1. $\forall i \in \mathcal{V}$, the cost function $f_i(x_i, \mathbf{x}_{-i})$ is convex and twice continuously differentiable in $x_i \in \mathbb{R}^{n_i}$ for every fixed $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$.

Assumption 2. Let $\bar{F}(\mathbf{x}) = [\nabla_1 f_1(x_1, \mathbf{x}_{-1}), \dots, \nabla_N f_N(x_N, \mathbf{x}_{-N})]^\top$, where $\nabla_i f_i(x_i, \mathbf{x}_{-i}) = \frac{\partial f_i(x_i, \mathbf{x}_{-i})}{\partial x_i}$. $\bar{F}(\mathbf{x})$ is strongly monotone with modulus δ_m in $\mathbb{R}^n$, that is, $(\mathbf{x} - \hat{\mathbf{x}})^\top (\bar{F}(\mathbf{x}) - \bar{F}(\hat{\mathbf{x}})) \geq \delta \|\mathbf{x} - \hat{\mathbf{x}}\|^2, \forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 3. $\nabla_i f_i(\mathbf{x})$ is globally Lipschitz continuous with a positive constant θ , that is, $\|\nabla_i f_i(\mathbf{x}) - \nabla_i f_i(\hat{\mathbf{x}})\| \leq \theta_l \|\mathbf{x} - \hat{\mathbf{x}}\|, \forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 4. The communication graph $\mathcal{G}$ is undirected and connected.

Remark 3.2. Assumptions 1-3 are common in the study of Nash equilibrium seeking problems [fang2021distributed, li2019distributed, poveda2022fixed, meng2025distributed]. Under Assumptions 1-2, the Nash equilibrium $\mathbf{x}^*$ exists uniquely. Moreover, since $f_i(y_i, \mathbf{x}_{-i})$ is twice continuously differentiable, Assumption 3 guarantees that $\sup_{\mathbf{x} \in \mathbb{R}^m} \|H(\mathbf{x})\|$ is bounded [meng2025distributed], where $H(\mathbf{x})$ is defined in (3.3).

3.3 Main Results

In this section, a novel fixed-time converging distributed Nash equilibrium seeking algorithm is proposed for the unconstrained non-cooperative game (3.1).

3.3.1 Distributed Nash Equilibrium Seeking Algorithm

The distributed Nash equilibrium seeking algorithm for player i is proposed as follows:

$$\dot{x}_i = -\beta_1 [\nabla_i f_i(\mathbf{z}_i)]^\mu - \beta_2 [\nabla_i f_i(\mathbf{z}_i)]^\nu, \quad (3.4)$$

$$\begin{aligned} \dot{z}_{ij} = & -\alpha_1 [\Lambda_{ij}]^\mu - \alpha_2 [\Lambda_{ij}]^\nu - \alpha_3 [\Lambda_{ij}]^{2\mu-1} \\ & - \alpha_4 [\Lambda_{ij}]^{2\nu-1}, \end{aligned} \quad (3.5)$$

$$\Lambda_{ij} = \sum_{k=1}^N a_{ik}(z_{ij} - z_{kj}) + a_{ij}(z_{ij} - x_j), \quad (3.6)$$

where $x_i \in \mathbb{R}^{n_i}$ is the action of player i , $z_{ij} \in \mathbb{R}^{n_j}$ is the player i 's estimate on the player j 's action x_j , $\mathbf{z}_i = [z_{i1}, \dots, z_{iN}]^\top \in \mathbb{R}^n$, a_{ik} is the (i, k) -th entry of the adjacency matrix A corresponding to the communication graph $\mathcal{G}$, $\mu \in (\frac{1}{2}, 1)$, $\nu \in (1, 2)$, and $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2$ are some positive design parameters.

3.3.2 Convergence Analysis

In this section, it will be shown that the distributed Nash equilibrium seeking algorithm (3.4)-(3.6) guarantees that the players' actions $\mathbf{x}$ converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time.

Theorem 3.3. *Consider the non-cooperative game (3.1) with the distributed Nash equilibrium seeking algorithm (3.4)-(3.6) under Assumptions 1-4. Given the modulus δ and Lipschitz constant θ , the action $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time if the positive design parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2, \mu \in (\frac{1}{2}, 1), \nu \in (1, +\infty), \rho_i, i = 1, \dots, 4$, satisfy the following inequalities,*

$$\begin{aligned} \sigma_1 &= \delta_m - \frac{h_m}{2N^{\frac{1}{2}}}(c_1\rho_3 + c_2\rho_4 + 2c_3) \\ &\quad - \frac{\rho_2(c_3 + N^{\frac{1}{2}})}{2} > 0, \end{aligned} \quad (3.7)$$

$$\sigma_2 = \alpha_1 - c_1 - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2} > 0, \quad (3.8)$$

$$\sigma_3 = \alpha_2(nN)^{\frac{1-\nu}{2}} - c_2 - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2}, \quad (3.9)$$

$$\sigma_4 = \alpha_3 - \frac{c_1 h_m}{2\rho_3 N^{\frac{1}{2}}} - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2} > 0, \quad (3.10)$$

$$\sigma_5 = \alpha_4(nN)^{1-\nu} - \frac{c_2 h_m}{2\rho_4 N^{\frac{1}{2}}} - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2} > 0, \quad (3.11)$$

where $c_1 = \beta_1 N^{1-\frac{\mu}{2}} n^{\frac{1-\mu}{2}} 2^{1-\mu} \theta^\mu \xi_{\min}^{-\mu}$, $c_2 = \beta_2 N^{\frac{1}{2}} (\nu\theta^\nu + \rho_1^{1-\nu} \theta n^{1-\frac{\nu}{2}}) \xi_{\min}^{-\nu}$, $c_3 = N^{\frac{1}{2}} n^{\frac{\nu-1}{2}} \rho_1 \theta (\nu-1) n^{1-\frac{\nu}{2}}$, $h_m = \sup_\omega \|H(\omega)\|$, $H(\cdot)$ is defined in (3.3), $\xi_{\min}$ and $\xi_{\max}$ denote the minimum and maximum eigenvalue of matrix $(\mathcal{L} \otimes I_n + \mathcal{M})$, respectively, $\mathcal{M} = \text{diag}\{a_{ij} I_{n_j}, i, j = 1, \dots, N\}$ and $\mathcal{L}$ is the Laplacian matrix of the communication graph $\mathcal{G}$. Moreover, the upper bound of the convergence time can be explicitly given as

$$T_{\text{nom}} = \frac{\nu+1}{d_1(1-\mu)} + \frac{\nu+1}{d_2(\nu-1)}, \quad (3.12)$$

where $d_1 = \min\{\sigma_1 \chi_1, \sigma^*\}$, $d_2 = 2^{\frac{1-\nu}{1+\nu}} \min\{\sigma_1 \chi_2, \sigma^*\}$, $\sigma^* = \min\{\frac{\sigma_2 2^{\frac{\mu-1}{2}} \xi_{\min}^{\mu+1}}{\xi_{\max}^{\frac{\mu+1}{2}}}, \frac{\sigma_3 2^{\frac{\nu-1}{2}} \xi_{\min}^{\nu+1}}{\xi_{\max}^{\frac{\nu+1}{2}}}\} + \min\{\frac{\sigma_4 2^{\mu-1} \xi_{\min}^{2\mu}}{\xi_{\max}^\mu}, \frac{\sigma_5 2^{\nu-1} \xi_{\min}^{2\nu}}{\xi_{\max}^\nu}\}$, $\chi_1 = \frac{1}{2} \beta_{\min}^2 (\frac{1+\mu}{\beta_{\max}})^{\frac{\mu+1}{1+\nu}}$, $\chi_2 = \frac{1}{2} \beta_{\min}^2 (2n)^{\frac{1-\nu}{1+\nu}} (\frac{1+\mu}{\beta_{\max}})^{\frac{2\nu}{\nu+1}}$, $\beta_{\min} = \min\{\beta_1, \beta_2\}$ and $\beta_{\max} = \max\{\beta_1, \beta_2\}$.

Proof. Consider the Lyapunov function candidate,

$$V = V_1 + V_2, \quad (3.13)$$

where

$$V_1 = \frac{1}{2}(\mathbf{z} - \mathbf{1}_N \otimes \omega)^\top (\mathcal{L} \otimes I_n + \mathcal{M})(\mathbf{z} - \mathbf{1}_N \otimes \omega),$$

$$V_2 = \sum_{i=1}^N \left(\frac{\beta_1}{\mu+1} \|\nabla_i f_i(\omega)\|_{\mu+1}^{\mu+1} + \frac{\beta_2}{\nu+1} \|\nabla_i f_i(\omega)\|_{\nu+1}^{\nu+1} \right),$$

$\mathcal{M} = \text{diag}\{a_{ij}I_{n_j}, i, j = 1, \dots, N\}$ and $\mathcal{L}$ is the Laplacian matrix of the communication graph $\mathcal{G}$.

Since $\mathcal{G}$ is undirected and connected, then zero is a simple eigenvalue of its corresponding Laplacian matrix $\mathcal{L}$. According to Gershgorin circle theorem, $(\mathcal{L} \otimes I_n + \mathcal{M})$ is symmetric and positive definite [fang2021distributed]. It follows from the positive definiteness of $(\mathcal{L} \otimes I_m + \mathcal{M})$ that

$$\frac{1}{2}\xi_{\min}\|\mathbf{z} - \mathbf{1}_N \otimes \omega\|^2 \leq V_1 \leq \frac{1}{2}\xi_{\max}\|\mathbf{z} - \mathbf{1}_N \otimes \omega\|^2, \quad (3.14)$$

where $\xi_{\min}$ and $\xi_{\max}$ denote the minimum and maximum eigenvalue of matrix $(\mathcal{L} \otimes I_n + \mathcal{M})$, respectively. Taking the time derivative of V_1 along the trajectories of (6.9), one has

$$\begin{aligned} \dot{V}_1 &= -\Lambda^\top (\alpha_1 [\Lambda]^\mu + \alpha_2 [\Lambda]^\nu + \alpha_3 [\Lambda]^{2\mu-1} + \alpha_4 [\Lambda]^{2\nu-1} \\ &\quad + \beta_1 (1_N \otimes [F(\mathbf{z})]^\mu) + \beta_2 (1_N \otimes [F(\mathbf{z})]^\nu)) \\ &= -\alpha_1 \|\Lambda\|_{\mu+1}^{\mu+1} - \alpha_2 \|\Lambda\|_{\nu+1}^{\nu+1} - \alpha_3 \|\Lambda\|_{2\mu}^{2\mu} - \alpha_4 \|\Lambda\|_{2\nu}^{2\nu} \\ &\quad - \Lambda^\top (1_N \otimes (P(\mathbf{z}) - P(1_N \otimes \omega))) \\ &\quad - \Lambda^\top (1_N \otimes P(1_N \otimes \omega)), \end{aligned} \quad (3.15)$$

where $\Lambda = [\Lambda_{11}, \dots, \Lambda_{1N}, \Lambda_{21}, \dots, \Lambda_{NN}]^\top = (\mathcal{L} \otimes I_n + \mathcal{M})(\mathbf{z} - \mathbf{1}_N \otimes \omega)$, $F(\mathbf{z}) = [\nabla_1 f_1(\mathbf{z}_1), \dots, \nabla_N f_N(\mathbf{z}_N)]^\top$, $P(\mathbf{z}) = \beta_1 [F(\mathbf{z})]^\mu + \beta_2 [F(\mathbf{z})]^\nu$, $P(1_N \otimes \omega) = \beta_1 [F(1_N \otimes \omega)]^\mu + \beta_2 [F(1_N \otimes \omega)]^\nu$ and $F(1_N \otimes \omega) = [\nabla_1 f(\omega), \dots, \nabla_N f(\omega)]^\top$.

It follows from Lemmas 2.2 and 2.4 and Assumption 3 that

$$\begin{aligned}
& \| \lceil F(\mathbf{z}) \rceil^\mu - \lceil F(\mathbf{1}_N \otimes \omega) \rceil^\mu \| \\
& \leq \| 2^{1-\mu} \lceil \theta(\mathbf{z} - \mathbf{1}_N \otimes \omega) \rceil^\mu \| \\
& \leq 2^{1-\mu} \theta^\mu (nN)^{\frac{1-\mu}{2}} \| \mathbf{z} - \mathbf{1}_N \otimes \omega \|^\mu,
\end{aligned} \tag{3.16}$$

$$\begin{aligned}
& \| \lceil F(\mathbf{z}) \rceil^\nu - \lceil F(\mathbf{1}_N \otimes \omega) \rceil^\nu \| \\
& \leq \nu (\theta^\nu \| \mathbf{z} - \mathbf{1}_N \otimes \omega \|^\nu + \theta n^{1-\frac{\nu}{2}} \| F(\mathbf{1}_N \otimes \omega) \|^{\nu-1} \| \mathbf{z} - \mathbf{1}_N \otimes \omega \|) \\
& \leq (\nu \theta^\nu + \rho_1^{1-\nu} \theta n^{1-\frac{\nu}{2}}) \| \mathbf{z} - \mathbf{1}_N \otimes \omega \|^\nu + \rho_1 \theta (\nu - 1) n^{1-\frac{\nu}{2}} \| F(\mathbf{1}_N \otimes \omega) \|^{\nu-1},
\end{aligned} \tag{3.17}$$

$$\| P(\mathbf{1}_N \otimes \omega) \| \geq \beta_1 \| F(\mathbf{1}_N \otimes \omega) \|^{\frac{1-\nu}{2}} + \beta_2 m^{\frac{1-\nu}{2}} \| F(\mathbf{1}_N \otimes \omega) \|^{\frac{1-\nu}{2}}, \tag{3.18}$$

where ρ_1 is a positive design parameter.

From (3.14) and the definition of V_1 , one has

$$\| \mathbf{z} - \mathbf{1}_N \otimes \omega \| \leq \| \Lambda \| \xi_{min}^{-1}. \tag{3.19}$$

Combining (3.16)-(3.19) yields

$$\begin{aligned}
& \| \mathbf{1}_N \otimes (P(\mathbf{z}) - P(\mathbf{1}_N \otimes \omega)) \| \\
& \leq \beta_1 N^{\frac{1}{2}} \| \lceil F(\mathbf{z}) \rceil^\mu - \lceil F(\mathbf{1}_N \otimes \omega) \rceil^\mu \| + N^{\frac{1}{2}} \beta_2 \| \lceil F(\mathbf{z}) \rceil^\nu \\
& \quad - \lceil F(\mathbf{1}_N \otimes \omega) \rceil^\nu \| \\
& \leq c_1 \| \Lambda \|^{\frac{1-\mu}{2}} + c_2 \| \Lambda \|^{\frac{1-\nu}{2}} + c_3 \| P(\mathbf{1}_N \otimes \omega) \|,
\end{aligned} \tag{3.20}$$

where $c_1 = \beta_1 N^{1-\frac{\mu}{2}} n^{\frac{1-\mu}{2}} 2^{1-\mu} \theta^\mu \xi_{min}^{-\mu}$, $c_2 = \beta_2 N^{\frac{1}{2}} (\nu \theta^\nu + \rho_1^{1-\nu} \theta n^{1-\frac{\nu}{2}}) \xi_{min}^{-\nu}$ and $c_3 = N^{\frac{1}{2}} n^{\frac{\nu-1}{2}} \rho_1 \theta (\nu - 1) n^{1-\frac{\nu}{2}}$.

It also follows from Lemma 2.3 that

$$\begin{aligned}
& \| \Lambda \| \| P(\mathbf{1}_N \otimes \omega) \| \\
& \leq \frac{1}{2\rho_2} \| \Lambda \|^2 + \frac{\rho_2}{2} \| P(\mathbf{1}_N \otimes \omega) \|^2 \\
& \leq \frac{1}{4\rho_2} (\| \Lambda \|^{\mu+1} + \| \Lambda \|^{\nu+1} + \| \Lambda \|^{\frac{1-\mu}{2}} + \| \Lambda \|^{\frac{1-\nu}{2}}) + \frac{\rho_2}{2} \| P(\mathbf{1}_N \otimes \omega) \|^2,
\end{aligned} \tag{3.21}$$

where ρ_2 is a positive design parameter.

Under Lemma 2.2, one has $\|\Lambda\|_{\mu+1}^{\mu+1} \geq \|\Lambda\|^{\mu+1}$, $\|\Lambda\|_{\nu+1}^{\nu+1} \geq (nN)^{\frac{1-\nu}{2}} \|\Lambda\|^{\nu+1}$, $\|\Lambda\|_{2\mu}^{2\mu} \geq \|\Lambda\|^{2\mu}$ and $\|\Lambda\|_{2\nu}^{2\nu} \geq (mN)^{1-\nu} \|\Lambda\|^{2\nu}$. It then follows from (3.20)-(3.21) that (3.15) can be rewritten as

$$\begin{aligned} \dot{V}_1 \leq & -(\alpha_1 - c_1 - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2})\|\Lambda\|^{\mu+1} \\ & - (\alpha_2(mN)^{\frac{1-\nu}{2}} - c_2 - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2})\|\Lambda\|^{\nu+1} \\ & - (\alpha_3 - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2})\|\Lambda\|^{2\mu} \\ & - (\alpha_4(mN)^{1-\nu} - \frac{(c_3 + N^{\frac{1}{2}})}{4\rho_2})\|\Lambda\|^{\nu+1} \\ & + \frac{\rho_2(c_3 + N^{\frac{1}{2}})}{2}\|P(1_N \otimes \omega)\|^2, \end{aligned} \quad (3.22)$$

Taking the time derivative of V_2 along the trajectories of (3.4), one has

$$\begin{aligned} \dot{V}_2 = & -P(1_N \otimes \omega)^\top H(\omega)P(1_N \otimes \omega) \\ & - P(1_N \otimes \omega)^\top H(\omega)(P(z) - P(1_N \otimes \omega)), \end{aligned} \quad (3.23)$$

where $H(\cdot)$ is defined in (3.3).

Under Assumption 2, one has $P(\omega)^\top H(\omega)P(\omega) \geq \delta\|P(\omega)\|^2$ [ye2019distributed]. It then follows from Lemma 2.3 and (3.20) that (3.23) can be rewritten as

$$\begin{aligned} \dot{V}_2 \leq & -(\delta - \frac{h_m}{2N^{\frac{1}{2}}}(c_1\rho_3 + c_2\rho_4 + 2c_3))\|P(1_N \otimes \omega)\|^2 \\ & + \frac{c_1 h_m}{2\rho_3 N^{\frac{1}{2}}}\|\Lambda\|^{2\mu} + \frac{c_2 h_m}{2\rho_4 N^{\frac{1}{2}}}\|\Lambda\|^{2\nu}, \end{aligned} \quad (3.24)$$

where $h_m = \sup_{\omega} \|H(\omega)\|$, ρ_3 and ρ_4 are positive design parameters.

Combining (3.22) and (3.24) yields

$$\begin{aligned} \dot{V} \leq & -\sigma_1\|P(1_N \otimes \omega)\|^2 - \sigma_2\|\Lambda\|^{\mu+1} - \sigma_3\|\Lambda\|^{\nu+1} \\ & - \sigma_4\|\Lambda\|^{2\mu} - \sigma_5\|\Lambda\|^{2\nu}, \end{aligned} \quad (3.25)$$

where $\sigma_i, i = 1, \dots, 5$ are given in (3.7)-(3.11).

It follows from (3.14), (3.19) and Lemma 2.5 that

$$\|\Lambda\| \geq \frac{2^{\frac{1}{2}} \xi_{\min}}{\xi_{\max}^{\frac{1}{2}}} V_1^{\frac{1}{2}}, \quad \|P(1_N \otimes \omega)\|^2 \geq \chi_1 V_2^{\frac{\mu+\nu}{1+\nu}} + \chi_2 V_2^{\frac{2\nu}{\nu+1}}, \quad (3.26)$$

where $\chi_1 = \frac{1}{2} \beta_{\min}^2 \left(\frac{1+\mu}{\beta_{\max}} \right)^{\frac{\mu+1}{1+\nu}}$, $\chi_2 = \frac{1}{2} \beta_{\min}^2 (2n)^{\frac{1-\nu}{1+\nu}} \left(\frac{1+\mu}{\beta_{\max}} \right)^{\frac{2\nu}{\nu+1}}$, $\beta_{\min} = \min\{\beta_1, \beta_2\}$ and $\beta_{\max} = \max\{\beta_1, \beta_2\}$.

Choose $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2, \mu \in (\frac{1}{2}, 1)$, $\nu \in (1, +\infty)$, and $\rho_i, i = 1, \dots, 4$ such that the inequalities in (3.7)-(3.11) are satisfied, it then follows from (3.26) that (3.25) can be rewritten as

$$\begin{aligned} \dot{V} \leq & -\sigma_1 \chi_1 V_2^{\frac{\mu+\nu}{1+\nu}} - \sigma_1 \chi_2 V_2^{\frac{2\nu}{\nu+1}} \\ & - \frac{\sigma_2 2^{\frac{\mu+1}{2}} \xi_{\min}^{\mu+1}}{\xi_{\max}^{\frac{\mu+1}{2}}} V_1^{\frac{\mu+1}{2}} - \frac{\sigma_3 2^{\frac{\nu+1}{2}} \xi_{\min}^{\nu+1}}{\xi_{\max}^{\frac{\nu+1}{2}}} V_1^{\frac{\nu+1}{2}} \\ & - \frac{\sigma_4 2^\mu \xi_{\min}^{2\mu}}{\xi_{\max}^\mu} V_1^\mu - \frac{\sigma_5 2^\nu \xi_{\min}^{2\nu}}{\xi_{\max}^\nu} V_1^\nu. \end{aligned} \quad (3.27)$$

Since $\mu < \frac{\mu+1}{2} < \frac{\mu+\nu}{\nu+1} < 1 < \frac{2\nu}{\nu+1} < \frac{\nu+1}{2} < \nu$, one has

$$V_1^{\frac{\mu+1}{2}} + V_1^{\frac{\nu+1}{2}} > \frac{1}{2} (V_1^{\frac{\mu+\nu}{\nu+1}} + V_1^{\frac{2\nu}{\nu+1}}), \quad (3.28)$$

$$V_1^\nu + V_1^\mu > \frac{1}{2} (V_1^{\frac{\mu+\nu}{\nu+1}} + V_1^{\frac{2\nu}{\nu+1}}). \quad (3.29)$$

It further follows from (3.28)-(3.29) that (3.27) can be rewritten as

$$\begin{aligned} \dot{V} \leq & -\sigma_1 \chi_1 V_2^{\frac{\mu+\nu}{1+\nu}} - \sigma_1 \chi_2 V_2^{\frac{2\nu}{\nu+1}} - \sigma^* (V_1^{\frac{\mu+\nu}{\nu+1}} + V_1^{\frac{2\nu}{\nu+1}}) \\ \leq & -d_1 V_1^{\frac{\mu+\nu}{\nu+1}} - d_2 V_1^{\frac{2\nu}{\nu+1}}, \end{aligned} \quad (3.30)$$

where $d_1 = \min\{\sigma_1 \chi_1, \sigma^*\}$, $d_2 = 2^{\frac{1-\nu}{1+\nu}} \min\{\sigma_1 \chi_2, \sigma^*\}$ and $\sigma^* = \min\left\{ \frac{\sigma_2 2^{\frac{\mu-1}{2}} \xi_{\min}^{\mu+1}}{\xi_{\max}^{\frac{\mu+1}{2}}}, \frac{\sigma_3 2^{\frac{\nu-1}{2}} \xi_{\min}^{\nu+1}}{\xi_{\max}^{\frac{\nu+1}{2}}} \right\} + \min\left\{ \frac{\sigma_4 2^{\mu-1} \xi_{\min}^{2\mu}}{\xi_{\max}^\mu}, \frac{\sigma_5 2^{\nu-1} \xi_{\min}^{2\nu}}{\xi_{\max}^\nu} \right\}$.

Via Lemma 2.8, one has that z_{ij} converges to ω_j and $\nabla_i f_i(\omega)$ converges to $\mathbf{0}_{m_i}$ in fixed time T_1 , where T_1 is explicitly given in (3.12).

Under Assumptions 1-2, it follows from [meng2025distributed] that the Nash equilibrium $\mathbf{x}^*$ is reached if and only if $\nabla_i f_i(\mathbf{x}^*) = \mathbf{0}_{m_i}$, $i = 1, \dots, N$. One thus has that ω converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time T_1 .

The proof of Theorem 3.3 is thus completed. $\square$

Remark 3.4. It should be pointed out that the proposed fixed-time converging Nash equilibrium seeking algorithm forms a two time-scale framework. Specifically, the dynamics of the distributed observers in (6.9)-(3.6) are considered as fast dynamics, while the Nash equilibrium seeking dynamics in (3.4) are considered as slow dynamics. Furthermore, the convergence rates of the fast and slow dynamics can be individually tuned by adjusting the design parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and β_1, β_2 , respectively. It can be observed from the inequalities (3.7)-(3.11) that given the network parameters and other design parameters μ, ν , these inequalities can be satisfied by choosing sufficiently large design parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, which, in turn, result in faster convergence of the fast dynamics, or by choosing smaller enough design parameters β_1 and β_2 , which result in slower convergence of the slow dynamics. This “separation” characteristic of dynamics can guarantee the existence of the design parameters and bring great benefits for the analysis of the proposed algorithm (3.4)-(3.6) compared with the passivity-based Nash equilibrium seeking algorithm proposed in [meng2025distributed].

Remark 3.5. According to the proof of Theorem 3.3, the design parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2$ and $\rho_i, i = 1, \dots, 4$ can be determined as follows: 1) Choose β_1, β_2 arbitrarily; 2) Choose $\rho_i, i = 1, \dots, 4$ according to (3.7); 3) Choose $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ according to (3.8)-(3.11).

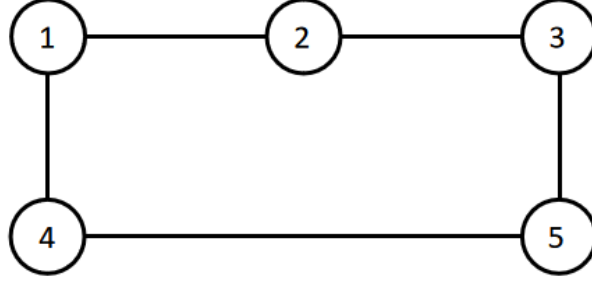


Figure 3.1. The communication graph.

3.3.3 Extension to Time-Varying Quadratic Games

3.4 Numerical Examples

In this section, a simulation example involving five players is presented to illustrate the result established in Theorem 3.3. The players' cost functions are given as follows:

$$\begin{aligned}
 f_1(x) &= 0.75x_1^2 + 0.25x_1x_2 - 0.15x_1x_3 - 0.04x_1x_4 + 0.15x_1x_5 - x_1 \\
 f_2(x) &= -0.25x_1x_2 + 0.80x_2^2 + 0.25x_2x_3 - 0.12x_2x_4 - 0.05x_2x_5 - x_2 \\
 f_3(x) &= 0.12x_1x_3 - 0.05x_2x_3 + 0.80x_3^2 + 0.26x_3x_4 - 0.18x_3x_5 - 3x_3 \\
 f_4(x) &= -0.07x_1x_4 + 0.07x_2x_4 - 0.34x_3x_4 + 0.90x_4^2 + 0.20x_4x_5 - 4x_4 \\
 f_5(x) &= -0.20x_1x_5 - 0.05x_2x_5 + 0.04x_3x_5 - 0.30x_4x_5 + 0.90x_5^2 - 5x_5
 \end{aligned}$$

The communication graph is given in Figure 3.1 and the game is given as follows:

$$\min_{x_i \in \mathbb{R}^{n_i}} f_i(x_i, \mathbf{x}_{-i}). \quad (3.31)$$

It can be verified that Assumptions 1-3 are satisfied. Besides, it can be calculated that the unique Nash equilibrium point of the game (3.31) is given as $\mathbf{x}^* = [-1, 0.167, 0.333]^\top$.

Table 3.1. Design parameters for the distributed Nash equilibrium seeking algorithm (3.4)–(3.6).

	μ	ν	α_1	α_2	α_3	α_4	β_1	β_2	ρ_1	ρ_2	ρ_3	ρ_4
Value	$\frac{3}{5}$	$\frac{9}{7}$	200	250	320	1200	3.5	0.5	0.006	0.004	0.008	0.01

Table 3.2. Parameters for Verification of (3.7)-(3.11) .

Parameter	Value	Parameter	Value	Parameter	Value
δ	0.1000	h_m	0.2000	ξ_{min}	0.2434
ξ_{max}	4.4382	c_1	19.1105	c_2	30.8592
c_3	0.0057	σ_1	0.0761	σ_2	27.4405
σ_3	20.4873	σ_4	69.0278	σ_5	35.4908

Figure 3.2. Convergence time for different initial values..

The distributed Nash equilibrium seeking algorithm (3.4)-(3.6) is applied to seek the Nash equilibrium of the game with the design parameters shown in Tables ?? . Moreover, it can be verified from the parameters in Table 3.2 that the inequalities in (3.7)-(3.11) are all satisfied. Then the initial states of players are given by $\mathbf{x}(0) = [0, 0, 0, 0, 0]^\top$.

The simulation results are presented in Figure 5.3. It can be observed that all players' collected action $\mathbf{x}$ converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time.

In addition, we conduct a comparison between the fixed-time converging Nash equilibrium seeking algorithm (3.4)-(3.6) and the finite-time converging Nash equilibrium seeking algorithm proposed in [fang2021distributed]. The design parameters of the fixed-time algorithm are given in Table 3.1 and the design parameters for the finite-time algorithm in [fang2021distributed] are chosen as $\alpha_i = 5$, $\eta_i = 1000$. Figure 3.2 shows the respective convergence times to the same residual set $\{\mathbf{x}(t) | e_{\mathbf{x}}(t) \leq 0.05\}$ across various initial states, where $e_{\mathbf{x}} = \|\mathbf{x} - \mathbf{x}^*\|$. It can be observed from Figure 3.2 that the convergence time based on the fixed-time algorithm remains small and well within the fixed upper bound across a range of initial conditions, in contrast to that for the finite-time algorithm where significant variations exist in response to different initial conditions.

3.5 Conclusions

In this chapter, the distributed fixed-time Nash equilibrium seeking problem for unconstrained non- In this section, a novel distributed fixed-time converging Nash equilibrium seeking algorithm for unconstrained non-cooperative games is proposed. A

key feature of the proposed Nash equilibrium seeking algorithm is its two time-scale framework, which brings great benefits for the selection of the design parameters and the analysis of the proposed algorithm. The fixed-time convergence can be proven by utilizing Lyapunov techniques. Furthermore, numerical simulation is then presented to illustrate the effectiveness of the proposed algorithm.

Chapter 4

Resilient Distributed Fixed-Time Nash Equilibrium Seeking Against Denial-of-Service Attacks

4.1 Introduction

In the preceding chapter, a generalized distributed fixed-time Nash equilibrium seeking algorithm was established under the fundamental assumption of an ideal communication environment. However, the deployment of multi-agent systems in real-world engineering scenarios inherently transforms them into complex Cyber-Physical Systems (CPS). In such systems, distributed players heavily rely on open, shared, and wireless communication networks to iteratively exchange gradient estimators and strategy information. While wireless networks offer exceptional flexibility and scalability, they inevitably expose the game-theoretic algorithms to severe cyber vulnerabilities and malicious intrusions. Among various cyber threats, Denial-of-Service (DoS) attacks pose one of the most critical challenges to distributed control and optimization paradigms. By maliciously overwhelming communication channels, DoS attacks induce intermittent data packet losses, severe communication delays, and consecutive topology disconnections, fundamentally disrupting the cooperative information exchange required for reaching a global equilibrium.

The resilience of distributed multi-agent systems against cyber-attacks has attracted considerable attention within the control community. Substantial research efforts have

been dedicated to analyzing and synthesizing resilient protocols for basic consensus and distributed optimization problems under DoS attacks [citation]. For example, resilient consensus protocols over intermittent switching topologies were investigated in [citation], where time-varying communication links induced by adversaries were mitigated using dwell-time methodologies. In parallel, the impact of DoS attacks on distributed optimization and game theory has been initially explored. The authors in [citation] proposed an asymptotically convergent NE seeking algorithm subject to periodic DoS attacks. Furthermore, randomized or probabilistic communication models have been utilized to capture the intermittent nature of DoS attacks, ensuring convergence in expectation or with probability one [citation].

Despite these remarkable advancements in secure control, existing resilient distributed NE seeking algorithms predominantly yield only asymptotic or exponential convergence properties. In the presence of unpredictable and potentially prolonged DoS attacks, the convergence time of these asymptotic methods can be drastically extended, leading to unacceptable delays in safety-critical applications such as autonomous military swarms or power grid frequency regulation. Bridging the gap between strict cybersecurity requirements and explicit convergence time guarantees necessitates the development of resilient fixed-time NE seeking frameworks. However, extending fixed-time stability theories to cyber-attacked environments is not a trivial task. The intermittent disconnections caused by DoS attacks explicitly destroy the continuous flow of information, severely complicating the construction of fixed-time Lyapunov functions and making the derivation of an exact settling-time upper bound extremely challenging. Furthermore, ensuring that the tracking errors do not diverge uncontrollably during the attack intervals while recovering rapidly during the active communication intervals remains a critical open problem in fixed-time game theory.

Motivated by these pressing theoretical and practical challenges, this chapter systematically investigates the resilient distributed fixed-time NE seeking problem for multi-agent systems operating over vulnerable communication networks subjected to DoS attacks. By synthesizing switching system theory with non-smooth control techniques, a robust countermeasure is developed. To the best of our knowledge, this work represents one of the first rigorous attempts to simultaneously address DoS attacks and fixed-time convergence within a distributed non-cooperative game framework. The primary contributions of this chapter are highlighted as follows:

1. **Resilient Fixed-Time Convergence Framework:** A novel resilient distributed algorithm is synthesized to achieve exact NE seeking within a mathematically guaranteed fixed time, even when the communication network undergoes frequent and unpredictable topology breakdowns induced by DoS attacks.
2. **Integration of Switching and Sliding Mode Theories:** To effectively tackle the discontinuous information flow, the cyber-physical interaction is modeled as a switched dynamic system. By elegantly combining leader-following consensus, switching system theory, and sliding mode control, the proposed protocol rigorously ensures that the cumulative tracking convergence during the communication-active intervals strictly dominates the potential state divergence during the DoS-active intervals.
3. **Explicit Settling-Time Upper Bounds:** Unlike existing resilient asymptotic methods, this chapter provides a definitive and explicit analytical formula for the upper bound of the convergence time. This bound is proven to be completely independent of the players' initial conditions and explicitly incorporates the characteristics of the DoS attacks (e.g., attack duration and frequency), offering vital guidance for resilient network design.

The subsequent sections of this chapter are organized as follows. Section 4.2 mathematically models the DoS attacks and formulates the resilient game problem. Section 4.3 details the proposed resilient algorithm and provides a rigorous mathematical proof of the fixed-time convergence. Simulation verifications under varying attack scenarios are given in Section 4.4, followed by concluding remarks in Section 4.5.

4.2 Problem Formulation

Consider a set of players $\mathcal{V} \triangleq \{1, \dots, N\}$ with the action of player i denoted as $x_i \in \mathbb{R}^{n_i}$ and the action vector of the other players denoted by $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$, where $n = \sum_{i=1}^N n_i$, respectively. The dynamic of player i is given as $\dot{x}_i = u_i$, where $u_i \in \mathbb{R}^{n_i}$ is the control input. Each player i aims to minimize its own objective function $f_i(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$, where $\mathbf{x} = [x_1, \dots, x_N]^T \in \mathbb{R}^n$ is the action vector of N players. If agent j is not a neighbor of agent i , which means $a_{ij} = 0$, then player i does not have direct access to player j 's action. Otherwise, player i can get the information of player j via the communication graph $\mathcal{G}(t)$.

The game in this chapter is described as

$$\min_{x_i} f_i(x_i, \mathbf{x}_{-i}). \quad (4.1)$$

First, we formalize the definition of the Nash equilibrium of the unconstrained non-cooperative game.

Definition 4.1. [ye2017distributed] A strategy profile $\mathbf{x}^* = [y_1^*, \dots, y_N^*]$ is called the Nash equilibrium of the non-cooperative game (4.1) if $\forall i \in \mathcal{V}$,

$$f_i(y_i^*, \mathbf{x}_{-i}^*) \leq f_i(y_i, \mathbf{x}_{-i}^*), \quad \forall y_i \in \mathbb{R}^{n_i}. \quad (4.2)$$

In this chapter, our objective is to design a resilient distributed Nash equilibrium seeking algorithm for the players such that the action vector $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{x}^*$ of the game (4.1) in fixed time. Denote

$$H(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_N} \\ \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_1} & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_2} & \cdots & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N^2} \end{bmatrix}, \quad (4.3)$$

where $\mathbf{x} = [x_1, \dots, x_N]^\top \in \mathbb{R}^n$, $x_i \in \mathbb{R}^{n_i}$, $i = 1, \dots, N$. To proceed, the following mild assumptions are made. To proceed, the following mild assumptions are made.

Assumption 5. $\forall i \in \mathcal{V}$, the cost function $f_i(x_i, \mathbf{x}_{-i})$ is convex and twice continuously differentiable in $x_i \in \mathbb{R}^{n_i}$ for every fixed $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$.

Assumption 6. $\nabla_i f_i(\mathbf{x})$ is globally Lipschitz continuous with a positive constant θ , that is, $\|\nabla_i f_i(\mathbf{x}) - \nabla_i f_i(\hat{\mathbf{x}})\| \leq \theta_l \|\mathbf{x} - \hat{\mathbf{x}}\|$, $\forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 7. $H(\mathbf{y})$ is strictly block diagonally dominant with positive definite diagonal blocks, that is, there exists a constant $\varphi > 0$ such that $\lambda_{\min} \left(\frac{\partial^2 f_i(\mathbf{x})}{\partial x_i^2} \right) - \sum_{j \neq i}^N \left\| \frac{\partial^2 f_i(\mathbf{x})}{\partial x_i \partial x_j} \right\| > \varphi$, $i = 1, \dots, N$.

Assumption 8. The undirected communication graph $\mathcal{G}$ is connected.

Assumption 9. (DoS Duration) There exist constants $1 > \varrho_d \geq 0$ and $\vartheta_d \geq 0$ such that $|\Gamma_D(0, t)| \leq \varrho_d t + \vartheta_d$, $\forall t \geq 0$.

Remark 4.2. Assumptions 5-6 are common in the study of Nash equilibrium seeking problems [poveda2022fixed, meng2025distributed]. Since $f_i(x_i, \mathbf{x}_{-i})$ is twice continuously differentiable, Assumption 2 guarantees that $\sup_{\mathbf{x} \in \mathbb{R}^n} \|H(\mathbf{x})\|$ is bounded [meng2025distributed], where $H(\mathbf{x})$ is defined in (4.3). Moreover, under Assumption 7, it can be inferred that $H(\mathbf{x})^\top + H(\mathbf{y}) \succ 2\varphi I_n$, which implies $F(\mathbf{x}) = [\nabla_1 f_1(x_1, \mathbf{x}_{-1}), \dots, \nabla_N f_N(x_N, \mathbf{x}_{-N})]^\top$ is strongly monotone with modulus φ in $\mathbb{R}^n$ [ye2019distributed]. Although Assumption 3 is somewhat more restrictive than strong monotonicity, this block-diagonal structural property is necessary to dominate the cross terms induced by the nonlinear fixed-time protocols, thereby ensuring the fixed-time convergence of the proposed algorithm under DoS attacks. Note that this assumption is adopted from [10181129] and also holds for many practical game models, such as the energy consumption games and mobile sensor networks studied in [10229248, he2025distributed]. It is noted that the Nash equilibrium $\mathbf{y}^*$ exists uniquely under Assumptions 1-3.

Remark 4.3. Assumption 8 is common for networked Nash equilibrium seeking problems [ye2017distributed, YE2018266, ye2019distributed, 9209113, 10570469]. Assumption 9 is adopted from [de2015input, CAO2025112099], which guarantees that DoS duration is bounded.

4.3 Main Results

In this section, a novel resilient fixed-time converging distributed Nash equilibrium seeking algorithm is proposed for the non-cooperative game (4.1).

4.3.1 Resilient Distributed Nash Equilibrium Seeking Algorithm

The resilient fixed-time distributed Nash equilibrium seeking algorithm for player i is proposed as follows:

$$\dot{x}_i = -\beta_1 [\nabla_i f_i(\mathbf{z}_i)]^\mu - \beta_2 [\nabla_i f_i(\mathbf{z}_i)]^\nu, \quad (4.4)$$

$$\dot{z}_{ij} = -\alpha_1 \Lambda_{ij}(\mu) - \alpha_2 \Lambda_{ij}(\nu), \quad (4.5)$$

$$\Lambda_{ij}(p) = \sum_{k=1}^N a_{ik}(t) [z_{ij} - z_{kj}]^p + a_{ij}(t) [z_{ij} - x_j]^p, \quad (4.6)$$

where $x_i \in \mathbb{R}^{n_i}$ is the virtual signal of player i , $z_{ij} \in \mathbb{R}^{n_j}$ is the player i 's estimate on the player j 's virtual signal x_j , $\mathbf{z}_i = [z_{i1}, \dots, z_{iN}]^\top \in \mathbb{R}^n$, $a_{ik}(t)$ is the (i, k) -th entry of the adjacency matrix $\mathcal{A}(t)$ corresponding to the communication graph $\mathcal{G}(t)$, and $\alpha_1, \alpha_2, \beta_1, \beta_2, \mu \in (0, 1), \nu \in (1, 2)$ are some positive design parameters.

4.3.2 Convergence Analysis

Before presenting the main result, we introduce the following definitions to streamline the subsequent analysis. Let $r = [r_1, \dots, r_5]^\top$. Define

$$\sigma_1(r) = c_1 + \frac{N^{\frac{1}{2}}}{\mu + 1} r_2^{-\mu} + \frac{h_m c_1 \mu}{N^{\frac{1}{2}}(\mu + 1)} r_4^{-\frac{1}{\mu}}, \quad (4.7)$$

$$\begin{aligned} \sigma_2(r) = & c_2 + c_3 r_1^{1-\nu} + \frac{c_4}{\nu + 1} r_1 r_3^{-\nu} + \frac{N^{\frac{1}{2}}}{\nu + 1} r_3^{-\nu} \\ & + \frac{h_m c_2 \nu}{N^{\frac{1}{2}}(\nu + 1)} r_5^{-\frac{1}{\nu}} + \frac{h_m c_3 \nu}{N^{\frac{1}{2}}(\nu + 1)} r_1^{1-\nu} r_5^{-\frac{1}{\nu}}, \end{aligned} \quad (4.8)$$

$$\sigma_3(r) = \varphi - \frac{N^{\frac{1}{2}} n^{\frac{1-\mu}{2}} \mu}{\mu + 1} r_2 - \frac{h_m c_1}{N^{\frac{1}{2}}(\mu + 1)} r_4, \quad (4.9)$$

$$\begin{aligned} \sigma_4(r) = & \varphi - \frac{c_4 \nu}{\nu + 1} r_1 r_3 - \frac{N^{\frac{1}{2}} \nu}{\nu + 1} r_3 - \frac{h_m c_4}{N^{\frac{1}{2}}} r_1 \\ & - \frac{h_m c_2}{N^{\frac{1}{2}}(\nu + 1)} r_5 - \frac{h_m c_3}{N^{\frac{1}{2}}(\nu + 1)} r_1^{1-\nu} r_5, \end{aligned} \quad (4.10)$$

where $c_1 = N^{1-\frac{\mu}{2}} n^{1-\mu} 2^{1-\mu} \theta^\mu$, $c_2 = N^{\frac{1}{2}} \nu \theta^\nu$, $c_3 = N^{\frac{1}{2}} \theta n^{1-\frac{\nu}{2}}$, $c_4 = N^{\frac{1}{2}} \theta n^{1-\frac{\nu}{2}} (\nu - 1)$ and $h_m = \sup_{\mathbf{y}} \|H(\mathbf{y})\|$.

We are now ready to present the main result.

Theorem 4.4. *Consider the non-cooperative game (4.1) with the resilient distributed Nash equilibrium seeking algorithm (4.4)-(4.6) under Assumptions 1-6. Given the positive constant φ , Lipschitz constant θ and the DoS attacks parameters ϱ_d and ϑ_d , the*

virtual signal $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{y}^*$ in fixed time if there exist positive design parameters $\alpha_1, \alpha_2, \beta_1, \beta_2, \mu \in (0, 1), \nu \in (1, 2)$ and scaling parameter $\phi \in (0, 1)$ such that the following conditions are satisfied:

$$\frac{\alpha_1(2\lambda_1)^{\frac{\mu+1}{2}}}{2} - \beta_1\sigma_1(\varepsilon) > \beta_1\phi\varphi, \quad (4.11)$$

$$\frac{\alpha_2 b(2\lambda_2)^{\frac{\nu+1}{2}}}{2} - \beta_2\sigma_2(\varepsilon) > \beta_2\phi\varphi, \quad (4.12)$$

$$\varrho_d < \frac{\phi\varphi}{\phi\varphi + \max\{2^{\frac{1-\mu}{2}}\sigma_1(\delta), 2^{\frac{\nu-1}{2}}\sigma_2(\delta)\}}, \quad (4.13)$$

where λ_1 and λ_2 are the minimum eigenvalues of $\mathcal{H}(\frac{2}{\mu+1})$ and $\mathcal{H}(\frac{2}{\nu+1})$, respectively, $b = (N^2(n+1))^{\frac{1-\nu}{2}} 2^{\frac{2(1-\nu)}{(\nu+1)^2}}$, ε and δ denote the solutions to the equations $\sigma_3(\varepsilon) = \sigma_4(\varepsilon) = \phi\varphi$ and $\sigma_1(\delta) = |\sigma_3(\delta)|$, $\sigma_2(\delta) = |\sigma_4(\delta)|$, respectively. The matrix $\mathcal{H}(p)$ is defined as $\mathcal{H}(p) = \mathcal{L}(p) \otimes I_n + \text{diag}\{a_{ij}^p I_{n_j}, i, j = 1, \dots, N\}$, with the weighted Laplacian $\mathcal{L}(p) = [l_{ij}(p)]$ given by $l_{ij}(p) = -a_{ij}^p$, if $i \neq j$ and $l_{ii} = \sum_{k=1}^N a_{ik}^p$. Moreover, the upper bound of the convergence time is explicitly given by

$$T_{dos} = \frac{(d^* + 1)\vartheta_d + \frac{2}{d_1(1-\mu)} + \frac{2}{d_2(\nu-1)}}{1 - \varrho_d(d^* + 1)}, \quad (4.14)$$

where $d^* = (\phi\varphi)^{-1} \max\{2^{\frac{1-\mu}{2}}\sigma_1(\delta), 2^{\frac{\nu-1}{2}}\sigma_2(\delta)\}$, $d_1 = \beta_1\phi\varphi$ and $d_2 = 2^{\frac{1-\nu}{2}}\beta_2\phi\varphi$.

Proof: Define $e_{ij} = z_{ij} - x_j$ and a direct calculation yields $e_{ij} - e_{kj} = z_{ij} - z_{kj}$. Consider the Lyapunov function candidate,

$$V = V_1 + V_2, \quad (4.15)$$

where $V_1 = \frac{1}{2} \sum_{i,j=1}^N e_{ij}^\top e_{ij}$, $V_2 = \frac{1}{2} \sum_{i=1}^N \|\nabla_i f_i(\mathbf{x})\|^2$.

Taking the time derivative of V_1 along the trajectories of (4.5), one has

$$\begin{aligned} \dot{V}_1 = & - \sum_{i,j=1}^N e_{ij}^\top (\alpha_1 \Lambda_{ij}(\mu) + \alpha_2 \Lambda_{ij}(\nu)) - \sum_{i,j=1}^N e_{ij}^\top (\beta_1 [\nabla_j f_j(\mathbf{z}_j)]^\mu \\ & + \beta_2 [\nabla_j f_j(\mathbf{z}_j)]^\nu). \end{aligned} \quad (4.16)$$

Since the communication graph $\mathcal{G}(t)$ is undirected, for any $p > 0$, one has

$$a_{ij}(t) = a_{ji}(t) \text{ and}$$

$$\sum_{i,j=1}^N e_{ij}^\top \Lambda_{ij}(p) = \frac{1}{2} \left(\sum_{i,j=1}^N \left(\sum_{k=1}^N a_{ik}(t) \|e_{ij} - e_{kj}\|_{p+1}^{p+1} + 2a_{ij}(t) \|e_{ij}\|_{p+1}^{p+1} \right) \right). \quad (4.17)$$

Define the matrices $\tilde{\mathcal{L}}(p) = [\tilde{l}_{ij}(p)] \in \mathbb{R}^{N \times N}$ as follows: $\tilde{l}_{ii}(p) = -\sum_{j=1}^N (a_{ij}(t))^p$ and $\tilde{l}_{ij}(p) = (a_{ij}(t))^p$ if $i \neq j$. Denote $\tilde{\mathcal{H}}(p) = \tilde{\mathcal{L}}(p) \otimes I_n + \text{diag}\{(a_{ij}(t))^p I_{n_j}, i, j = 1, \dots, N\}$.

It then follows from Lemmas 2.2 that

$$\alpha_1 \sum_{i,j=1}^N e_{ij}^\top \Lambda_{ij}(\mu) \geq \frac{\alpha_1}{2} (2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\mu+1}) \mathbf{e})^{\frac{\mu+1}{2}}, \quad (4.18)$$

$$\alpha_2 \sum_{i,j=1}^N e_{ij}^\top \Lambda_{ij}(\nu) \geq \frac{\alpha_2 b}{2} (2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\nu+1}) \mathbf{e})^{\frac{\nu+1}{2}}, \quad (4.19)$$

where $\mathbf{e} = [e_{11}, \dots, e_{1N}, e_{21}, \dots, e_{NN}]^\top$, $b = (N^2(n+1))^{\frac{1-\nu}{2}} 2^{\frac{2(1-\nu)}{(\nu+1)^2}}$.

Denote $F(\mathbf{z}) = [\nabla_1 f_1(\mathbf{z}_1), \dots, \nabla_N f_N(\mathbf{z}_N)]^\top$, $F(1_N \otimes \mathbf{x}) = [\nabla_1 f(\mathbf{x}), \dots, \nabla_N f(\mathbf{x})]^\top$, $P(\mathbf{z}) = \beta_1 [F(\mathbf{z})]^\mu + \beta_2 [F(\mathbf{z})]^\nu$ and $P(1_N \otimes \mathbf{x}) = \beta_1 [F(1_N \otimes \mathbf{x})]^\mu + \beta_2 [F(1_N \otimes \mathbf{x})]^\nu$. Then, one has $\sum_{i,j=1}^N e_{ij}^\top (-\beta_1 [\nabla_j f_j(\mathbf{z}_j)]^\mu - \beta_2 [\nabla_j f_j(\mathbf{z}_j)]^\nu) = \mathbf{e}^\top (1_N \otimes (-P(\mathbf{z}) + P(1_N \otimes \mathbf{x}) - P(1_N \otimes x)))$.

To facilitate the subsequent inequality analysis, we introduce a time-varying switching scaling parameter $\rho(t) : [0, \infty) \rightarrow \mathbb{R}_+^5$, defined as

$$\rho(t) = \begin{cases} \varepsilon, & t \in [t_{k-1}, t_k^a), \\ \delta, & t \in [t_k^a, t_k). \end{cases} \quad (4.20)$$

Then, it follows from Lemmas 2.2 and 2.4 and Assumption 7 that

$$\begin{aligned} \|[F(\mathbf{z})]^\mu - [F(1_N \otimes \mathbf{x})]^\mu\| &\leq 2^{1-\mu} (nN)^{\frac{1-\mu}{2}} \|F(\mathbf{z}) - F(1_N \otimes \mathbf{x})\|^\mu \\ &\leq 2^{1-\mu} \theta^\mu (nN)^{\frac{1-\mu}{2}} \|\mathbf{e}\|^\mu, \end{aligned} \quad (4.21)$$

$$\begin{aligned}
\| [F(\mathbf{z})]^\nu - [F(1_N \otimes \mathbf{x})]^\nu \| &\leq \nu(\theta^\nu \|\mathbf{e}\|^\nu + \theta n^{1-\frac{\nu}{2}} \|F(1_N \otimes \mathbf{x})\|^{\nu-1} \|\mathbf{e}\|) \\
&\leq (\nu\theta^\nu + \theta n^{1-\frac{\nu}{2}} \rho_1(t)^{1-\nu}) \|\mathbf{e}\|^\nu + \theta n^{1-\frac{\nu}{2}} (\nu \\
&\quad - 1) \rho_1(t) \|F(1_N \otimes \mathbf{x})\|^\nu.
\end{aligned} \tag{4.22}$$

Combining (4.21)-(4.22) yields

$$\begin{aligned}
&\|1_N \otimes (P(\mathbf{z}) - P(1_N \otimes \mathbf{x}))\| \\
&\leq \beta_1 c_1 \|\mathbf{e}\|^\mu + \beta_2 (c_2 + c_3 \rho_1(t)^{1-\nu}) \|\mathbf{e}\|^\nu \\
&\quad + \beta_2 c_4 \rho_1(t) \|F(1_N \otimes \mathbf{x})\|^\nu,
\end{aligned} \tag{4.23}$$

where $c_1 = N^{1-\frac{\mu}{2}} n^{1-\mu} 2^{1-\mu} \theta^\mu$, $c_2 = N^{\frac{1}{2}} \nu \theta^\nu$, $c_3 = N^{\frac{1}{2}} \theta n^{1-\frac{\nu}{2}}$ and $c_4 = N^{\frac{1}{2}} \theta n^{1-\frac{\nu}{2}} (\nu - 1)$. It follows from Lemma 2.3 that

$$\begin{aligned}
&\|\mathbf{e}\| \|P(1_N \otimes x)\| \\
&\leq \frac{\beta_1 \rho_2(t)^{-\mu}}{\mu+1} \|\mathbf{e}\|^{\mu+1} + \frac{\beta_1 \rho_2(t) n^{\frac{1-\mu}{2}} \mu}{\mu+1} \|F(1_N \otimes \mathbf{x})\|^{\mu+1} \\
&\quad + \frac{\beta_2 \rho_3(t)^{-\nu}}{\nu+1} \|\mathbf{e}\|^{\nu+1} + \frac{\beta_2 \rho_3(t) \nu}{\nu+1} \|F(1_N \otimes \mathbf{x})\|^{\nu+1}.
\end{aligned} \tag{4.24}$$

It also follows from (4.23)-(4.24) and Lemma ?? that

$$\begin{aligned}
&\|\mathbf{e}\| \|1_N \otimes (P(z) - P(1_N \otimes x))\| \\
&\leq c_1 \beta_1 \|\mathbf{e}\|^{\mu+1} + \beta_2 (c_2 + c_3 \rho_1(t)^{1-\nu} + \frac{c_4 \rho_1(t) \rho_3(t)^{-\nu}}{\nu+1}) \|\mathbf{e}\|^{\nu+1} \\
&\quad + \frac{\beta_2 \nu c_4 \rho_1(t) \rho_3(t)}{\nu+1} \|F(1_N \otimes \mathbf{x})\|^{\nu+1}.
\end{aligned} \tag{4.25}$$

It then follows from (4.18)-(4.25) that (4.16) can be rewritten as

$$\begin{aligned}
\dot{V}_1 \leq & -\frac{\alpha_1}{2}(2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\mu+1})\mathbf{e})^{\frac{\mu+1}{2}} - \frac{\alpha_2 b}{2}(2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\nu+1})\mathbf{e})^{\frac{\nu+1}{2}} \\
& + \beta_1(c_1 + \frac{N^{\frac{1}{2}}\rho_2(t)^{-\mu}}{\mu+1})\|\mathbf{e}\|^{\mu+1} \\
& + \beta_2(c_2 + c_3\rho_1(t)^{1-\nu} + \frac{c_4\rho_1(t)\rho_3(t)^{-\nu}}{\nu+1} + \frac{N^{\frac{1}{2}}\rho_3(t)^{-\nu}}{\nu+1})\|\mathbf{e}\|^{\nu+1} \\
& + \frac{\beta_1 N^{\frac{1}{2}}\rho_2(t)n^{\frac{1-\mu}{2}}\mu}{\mu+1}\|F(1_N \otimes \mathbf{x})\|^{\mu+1} \\
& + \frac{\beta_2(c_4\rho_1(t) + N^{\frac{1}{2}})\rho_3(t)\nu}{\nu+1}\|F(1_N \otimes \mathbf{x})\|^{\nu+1}.
\end{aligned} \tag{4.26}$$

Taking the time derivative of V_2 along the trajectories of (4.4), one has

$$\begin{aligned}
\dot{V}_2 = & -\frac{1}{2}F(1_N \otimes \mathbf{x})^\top (H(\mathbf{x}) + H(\mathbf{x})^\top)P(1_N \otimes \mathbf{x}) \\
& - \frac{1}{2}F(1_N \otimes \mathbf{x})^\top (H(\mathbf{x}) + H(\mathbf{x})^\top)(P(z) - P(1_N \otimes x)),
\end{aligned} \tag{4.27}$$

where $H(\cdot)$ is defined in (4.3). Under Assumption 6, one has $F(1_N \otimes \mathbf{x})^\top (H(\mathbf{x}) + H(\mathbf{x})^\top)P(1_N \otimes \mathbf{x}) > 2\varphi(\beta_1\|F(1_N \otimes \mathbf{x})\|_{\mu+1}^{\mu+1} + \beta_2\|F(1_N \otimes \mathbf{x})\|_{\nu+1}^{\nu+1})$. It then follows from Lemma 2.3 and (4.23) that (4.27) can be rewritten as

$$\begin{aligned}
\dot{V}_2 \leq & -\varphi(\beta_1\|F(1_N \otimes \mathbf{x})\|_{\mu+1}^{\mu+1} + \beta_2\|F(1_N \otimes \mathbf{x})\|_{\nu+1}^{\nu+1}) \\
& + \frac{h_m c_1 \rho_4(t)^{-\frac{1}{\mu}}\mu}{N^{\frac{1}{2}}(\mu+1)}\|\mathbf{e}\|^{\mu+1} + \frac{\beta_1 h_m c_1 \rho_4(t)}{N^{\frac{1}{2}}(\mu+1)}\|F(1_N \otimes \mathbf{x})\|^{\mu+1} \\
& + \frac{\beta_2 h_m (c_2 + c_3\rho_1(t)^{1-\nu})\rho_5(t)^{-\frac{1}{\nu}}\nu}{N^{\frac{1}{2}}(\nu+1)}\|\mathbf{e}\|^{\nu+1} + \beta_2(\frac{h_m c_4 \rho_1(t)}{N^{\frac{1}{2}}}) \\
& + \frac{h_m (c_2 + c_3\rho_1(t)^{1-\nu})\rho_5(t)}{N^{\frac{1}{2}}(\nu+1)}\|F(1_N \otimes \mathbf{x})\|^{\nu+1}.
\end{aligned} \tag{4.28}$$

Since $0 < \mu < 1 < \nu$, it follows from Lemma 2.2 that $\|F(1_N \otimes \mathbf{x})\|_{\mu+1}^{\mu+1} \geq \|F(1_N \otimes \mathbf{x})\|^{\mu+1}$ and $\|F(1_N \otimes \mathbf{x})\|_{\nu+1}^{\nu+1} \geq n^{\frac{1-\nu}{2}} \|F(1_N \otimes \mathbf{x})\|^{\nu+1}$. Combining (4.26) and (4.28) yields

$$\begin{aligned} \dot{V} \leq & -\frac{\alpha_1}{2} (2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\mu+1})\mathbf{e})^{\frac{\mu+1}{2}} - \beta_1 \sigma_3(\rho(t)) \|F(1_N \otimes \mathbf{x})\|^{\mu+1} \\ & - \frac{\alpha_2 b}{2} (2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\nu+1})\mathbf{e})^{\frac{\nu+1}{2}} - \beta_2 \sigma_4(\rho(t)) \|F(1_N \otimes \mathbf{x})\|^{\nu+1} \\ & + \beta_1 \sigma_1(\rho) \|\mathbf{e}\|^{\mu+1} + \beta_2 \sigma_2(\rho(t)) \|\mathbf{e}\|^{\nu+1}, \end{aligned} \quad (4.29)$$

where $\sigma_i(\cdot)$, $i = 1, \dots, 5$, are given in (4.7)-(4.10).

Owing to the presence of DoS attacks for graph $\mathcal{G}(t)$, the following two cases are considered to show the fixed-time convergence of the proposed Nash equilibrium seeking algorithm.

Case 1: For $t \in [t_{k-1}, t_k^a)$.

It follows from (4.20) that $\rho(t) = \varepsilon$. Under Assumption 8, by Lemma 2.1, one has $\tilde{\mathcal{H}}(\frac{2}{\mu+1}) = \mathcal{H}(\frac{2}{\mu+1}) > 0$ and $\tilde{\mathcal{H}}(\frac{2}{\nu+1}) = \mathcal{H}(\frac{2}{\nu+1}) > 0$. It can be further inferred that $(2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\mu+1})\mathbf{e})^{\frac{\mu+1}{2}} \geq (2\lambda_1)^{\frac{\mu+1}{2}} \|\mathbf{e}\|^{\mu+1}$ and $(2\mathbf{e}^\top \tilde{\mathcal{H}}(\frac{2}{\nu+1})\mathbf{e})^{\frac{\nu+1}{2}} \geq (2\lambda_2)^{\frac{\nu+1}{2}} \|\mathbf{e}\|^{\nu+1}$, where λ_1 and λ_2 denote the minimum eigenvalues of the matrix $\mathcal{H}(\frac{2}{\mu+1})$ and $\mathcal{H}(\frac{2}{\nu+1})$, respectively.

Since the design parameters $\alpha_1, \alpha_2, \beta_1, \beta_2$ satisfy the inequalities (4.11)-(4.12) and $\sigma_3(\varepsilon) = \sigma_4(\varepsilon) = \phi\varphi$, it then follows from the definition of V_1 and V_2 that (4.29) can be rewritten as

$$\dot{V} \leq -d_1 V^{\frac{\mu+1}{2}} - d_2 V^{\frac{\nu+1}{2}}, \quad (4.30)$$

where $d_1 = \beta_1 \phi\varphi$ and $d_2 = 2^{\frac{1-\nu}{2}} \beta_2 \phi\varphi$.

Case 2: For $t \in [t_k^a, t_k)$.

It follows from (4.20) that $\rho(t) = \delta$. During DoS attacks intervals, graph $\mathcal{G}(t)$ might not be connected. By Lemma 2.1, one can infer that $\tilde{\mathcal{H}}(\frac{2}{\mu+1}), \tilde{\mathcal{H}}(\frac{2}{\nu+1}) \geq 0$. It then follows from $\sigma_1(\delta) = |\sigma_3(\delta)|$ and $\sigma_2(\delta) = |\sigma_4(\delta)|$ that (4.29) can be rewritten as

$$\begin{aligned} \dot{V} \leq & \beta_1 \sigma_1(\delta) V_1^{\frac{\mu+1}{2}} + \beta_1 |\sigma_3(\delta)| V_2^{\frac{\mu+1}{2}} + \beta_2 \sigma_2(\delta) V_1^{\frac{\nu+1}{2}} + \beta_2 |\sigma_4(\delta)| V_2^{\frac{\nu+1}{2}} \\ \leq & d_1 V^{\frac{\mu+1}{2}} + d_2 V^{\frac{\nu+1}{2}}, \end{aligned} \quad (4.31)$$

where $d_3 = 2^{\frac{1-\mu}{2}} \beta_1 \sigma_1(\delta)$ and $d_4 = \beta_2 \sigma_2(\delta)$.

Combining (4.30) and (4.31) yields

$$\begin{cases} \dot{V}(t) \leq -d_1 V^{\frac{\mu+1}{2}} - d_2 V^{\frac{\nu+1}{2}}, & t \in [t_{k-1}, t_k^a), \\ \dot{V}(t) \leq d_3 V^{\frac{\mu+1}{2}} + d_4 V^{\frac{\nu+1}{2}}, & t \in [t_k^a, t_k). \end{cases} \quad (4.32)$$

Based on the switching behavior described in (4.32), the proof of fixed-time convergence is detailed below.

When $V(t_0) = 0$, it follows from (4.32) that for any $t \geq t_0 = 0$, $V(t) = 0$, which also implies $V(T_1) = 0$. When $V(t_0) > 0$, it follows from (4.32) that

$$\begin{cases} \int_{V(t_{k-1})}^{V(t_k^a)} \frac{dV}{d_1 V^{\frac{\mu+1}{2}} + d_2 V^{\frac{\nu+1}{2}}} \leq -(t_k^a - t_{k-1}), & t \in [t_{k-1}, t_k^a) \\ \int_{V(t_k^a)}^{V(t_k)} \frac{dV}{d_1 V^{\frac{\mu+1}{2}} + d_2 V^{\frac{\nu+1}{2}}} \leq d^*(t_k - t_k^a), & t \in [t_k^a, t_k), \end{cases} \quad (4.33)$$

where $d^* = \max\{\frac{d_3}{d_1}, \frac{d_4}{d_2}\} = (\phi\varphi)^{-1} \max\{2^{\frac{1-\mu}{2}} \sigma_1(\delta), 2^{\frac{\nu-1}{2}} \sigma_2(\delta)\}$.

Integrating both sides of the (4.33), one has

$$\int_{V(t_0)}^{V(t)} \frac{dV}{d_1 V^{\frac{\mu+1}{2}} + d_2 V^{\frac{\nu+1}{2}}} \leq (d^* + 1)\vartheta_d - (1 - \varrho_d(d^* + 1))t, \quad (4.34)$$

Given $\varrho_d < \frac{1}{1+d^*} = \frac{\phi\varphi}{\phi\varphi + \max\{2^{\frac{1-\mu}{2}} \sigma_1(\delta), 2^{\frac{\nu-1}{2}} \sigma_2(\delta)\}}$, there exists a positive constant t_f , such that $V(t_f) = 0$. Since $\int_0^{+\infty} \frac{dV}{d_1 V^{\frac{\mu+1}{2}} + d_2 V^{\frac{\nu+1}{2}}} < \frac{2}{d_1(1-\mu)} + \frac{2}{d_2(\nu-1)}$ [6104367], it follows from (4.34) that

$$t_f < \frac{(d^* + 1)\vartheta_d + \frac{2}{d_1(1-\mu)} + \frac{2}{d_2(\nu-1)}}{1 - \varrho_d(d^* + 1)} = T_1. \quad (4.35)$$

In summary, for any $V(t_0) \geq 0$, there exists a positive constant T_1 such that $V(t) = 0, \forall t \geq T_1$, where T_1 is given in (4.14). Furthermore, it follows from the definition of $V(t)$ given in (4.15) that $z_{ij}(t) = x_j(t)$ and $\nabla_i f_i(\mathbf{x}) = \mathbf{0}_{n_i}, \forall t > T_1, i, j = 1, \dots, N$. Under Assumptions 1-3, it follows from [meng2025distributed] that the Nash equilibrium $\mathbf{y}^*$ is achieved if and only if $\nabla_i f_i(\mathbf{y}^*) = \mathbf{0}_{n_i}, i = 1, \dots, N$. One thus has that x converges to the Nash equilibrium $\mathbf{y}^*$ in fixed time T_1 . The proof of Theorem 4.4 is thus completed.

Remark 4.5. The existence of parameter vectors $\varepsilon, \delta \in \mathbb{R}_+^5$ satisfying $\sigma_3(\varepsilon) = \sigma_4(\varepsilon) = \phi\varphi$, $\sigma_1(\delta) = |\sigma_3(\delta)|$, and $\sigma_2(\delta) = |\sigma_4(\delta)|$ can be verified by the Intermediate Value Theorem [zorich2016mathematical]. It follows from (4.7)-(4.10) that $\sigma_3(\varepsilon) - \phi\varphi$ and $\sigma_4(\varepsilon) - \phi\varphi$ monotonically vary from a positive value $(1 - \phi)\varphi$ to $-\infty$ as the components of ε increase from 0^+ to $+\infty$. Similarly, $\sigma_1(\delta) - |\sigma_3(\delta)|$ and $\sigma_2(\delta) - |\sigma_4(\delta)|$ vary from $+\infty$ to $-\infty$ as δ increase from 0^+ to $+\infty$. Given the continuity of $\sigma_i(\cdot)$ ($i = 1, \dots, 4$) on $\mathbb{R}_+^5$, the solutions to these equations are thus ensured to exist.

Remark 4.6. The terms $\beta_1[\nabla_i f_i(\mathbf{z}_i)]^\mu$, $\beta_2[\nabla_i f_i(\mathbf{z}_i)]^\nu$, $\alpha_1\Lambda_{ij}(\mu)$ and $\alpha_2\Lambda_{ij}(\nu)$ in (4.4)-(4.6) play a crucial role in the convergence of $\mathbf{x}$ to the Nash equilibrium $\mathbf{y}^*$ in fixed time. Furthermore, our result can include the cases of finite-time convergence and exponential convergence as its special cases. If $\beta_2 = \alpha_2 = 0$, then our algorithm reduces to the case of finite-time convergence. If $\mu = \nu = 1$, then our algorithm reduces to the case of exponential convergence. It is worth noting that, due to the consideration of DoS attacks, even the reduced finite-time algorithm represents new contributions compared to existing literature.

Remark 4.7. It follows from (4.7)-(4.13) that an allowable upper bound of ϱ_d for the DoS attacks duration is given by $\bar{\varrho}_d = \varphi/(\varphi + \max\{2^{\frac{1-\mu}{2}}\sigma_1(\delta), 2^{\frac{\nu-1}{2}}\sigma_2(\delta)\})$. Thus, given $\varrho_d < \bar{\varrho}_d$, one of the practical guidelines for selecting the design parameters is summarized as follows. First, we choose $\phi \in (0, 1)$ such that $\varrho_d < \phi\varphi/(\phi\varphi + \max\{2^{\frac{1-\mu}{2}}\sigma_1(\delta), 2^{\frac{\nu-1}{2}}\sigma_2(\delta)\})$. Then, we can choose small enough design parameters β_1 and β_2 or large enough design parameters α_1 and α_2 such that the inequalities (4.11)-(4.12) are satisfied.

Remark 4.8. It is worth noting that the algorithm presented in (4.4)-(4.6) is designed for the case that DoS attacks intervals are unknown to players and constant gains are thus employed, which leads to a conservative upper bound on ϱ_d . To guarantee fixed-time convergence, the algorithm employs nonlinear dynamics with power terms. While these terms accelerate convergence to the Nash equilibrium in the absence of DoS attacks, they amplify the estimation errors accumulated during DoS intervals. Furthermore, since the control gains are time-invariant, the system states may diverge at a speed comparable to the convergence rate. Therefore, a conservative upper bound on ϱ_d is required to guarantee the fixed-time convergence of the proposed algorithm.

4.3.3 Extension to Known DoS Attack Intervals

Theorem 4.4 provides a convergence time estimate for the proposed algorithm (4.4)-(4.6) under the case that the DoS attacks intervals are unknown to the players, which leads to a conservative assumption on the maximum value of ϱ_d . However, if the DoS attacks intervals are known by the players, a less conservative result can be obtained by replacing the fixed gains β_1 and β_2 in (4.4) with time-varying gains. Choose $\beta_1(t)$ and $\beta_2(t)$ as follows:

$$\beta_1(t) = \tau_1, \beta_2(t) = \tau_2, t \in [t_{k-1}, t_k^a), \quad (4.36)$$

$$\beta_1(t) = \tau_3, \beta_2(t) = \tau_4, t \in [t_k^a, t_k), \quad (4.37)$$

where τ_1, τ_2 are positive design parameters, and τ_3 and τ_4 are non-negative design parameters. Then, a corollary with a less conservative condition on the maximum value of ϱ_d can be directly obtained from Theorem 4.4.

Corollary 4.9. *Consider the non-cooperative game (4.1) with the resilient distributed Nash equilibrium seeking algorithm (4.4)-(4.6) with the time-varying gains (4.36)-(4.37) under Assumptions 1-6. Given the positive constant φ , Lipschitz constant θ and the DoS attacks parameters ϱ_d and ϑ_d , the virtual signal $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{y}^*$ in fixed time if there exist design parameters $\alpha_1, \alpha_2, \tau_1, \tau_2 > 0, \tau_3, \tau_4 \geq 0, \mu \in (0, 1), \nu \in (1, 2)$ and scaling parameter $\phi \in (0, 1)$ such that the following conditions are satisfied:*

$$\frac{\alpha_1(2\lambda_1)^{\frac{\mu+1}{2}}}{2} - \tau_1\sigma_1(\varepsilon) > \tau_1\phi\varphi, \quad (4.38)$$

$$\frac{\alpha_2 b(2\lambda_2)^{\frac{\nu+1}{2}}}{2} - \tau_2\sigma_2(\varepsilon) > \tau_2\phi\varphi, \quad (4.39)$$

$$\varrho_d < \frac{\phi\varphi}{\phi\varphi + \max\{\tau_1^{-1}\tau_3 2^{\frac{1-\mu}{2}}\sigma_1(\delta), \tau_2^{-1}\tau_4 2^{\frac{\nu-1}{2}}\sigma_2(\delta)\}}, \quad (4.40)$$

where $\lambda_1, \lambda_2, b, \varepsilon$ and δ are as defined in Theorem 4.4. Moreover, the upper bound of the convergence time is explicitly given by

$$\bar{T}_1 = \frac{(\bar{d}^* + 1)\vartheta_d + \frac{2}{d_1(1-\mu)} + \frac{2}{d_2(\nu-1)}}{1 - \varrho_d(\bar{d}^* + 1)}, \quad (4.41)$$

where $\bar{d}^* = (\phi\varphi)^{-1} \max\{\tau_1^{-1}\tau_3 2^{\frac{1-\mu}{2}} \sigma_1(\delta), \tau_2^{-1}\tau_4 2^{\frac{\nu-1}{2}} \sigma_2(\delta)\}$, $\bar{d}_1 = \tau_1\phi\varphi$ and $\bar{d}_2 = 2^{\frac{1-\nu}{2}} \tau_2\phi\varphi$.

Remark 4.10. It is worth noting that increasing the design parameters τ_1 and τ_2 can accelerate the speed of convergence, whereas minimizing the design parameters τ_3 and τ_4 (potentially to zero) can suppress the divergence of the players' actions during DoS intervals. Moreover, it follows from (4.40) that larger τ_1, τ_2 or smaller τ_3, τ_4 lead to a less conservative condition on the maximum value of ϱ_d , thereby improving the tolerance to DoS attacks. Furthermore, it implies that one can always choose a proper set of design parameters τ_1, τ_2, τ_3 and τ_4 such that the inequality in (4.40) is satisfied for any $0 \leq \varrho_d < 1$, which improves the practical applicability of the proposed algorithm.

4.4 Numerical Examples

In this section, an example will be presented to verify our main results. In electric markets, the competition among distributed energy resources can be modeled as a Nash–Cournot game (referring to other studies [7,33] and [34]). Here, we consider the Cournot Competition game with six generation systems whose communication graphs are depicted in Figure 1. The left graph shows that the communication topology is connected when attacks are not happening. And the others mean that the network suffers from two different types of attacks and the communication graphs are unconnected.

4.5 Conclusions

This chapter has addressed the resilient distributed fixed-time Nash equilibrium seeking problem for multi-agent systems operating over vulnerable communication networks subjected to Denial-of-Service (DoS) attacks. A robust control protocol has been developed by synthesizing switching system theory, the leader-following consensus protocol, and non-smooth sliding mode techniques. The proposed resilient algorithm effectively mitigates the adverse impacts of intermittent communication disconnections by ensuring that the cumulative tracking convergence during the active communication intervals strictly dominates the potential state divergence during the attack intervals. A rigorous mathematical proof has been provided to establish that the exact Nash equilibrium is achieved within an explicitly bounded fixed time. Notably, the derived upper

bound of the convergence time directly incorporates the characteristics of the DoS attacks (such as attack frequency and duration) but remains completely independent of the players' initial conditions, thereby offering explicit guidance for resilient cyber-physical system design.

Chapter 5

Distributed Fixed-Time Nash Equilibrium Seeking for Constrained Non-Cooperative Games

5.1 Introduction

The methodologies developed in Chapters 3 and 4 successfully address the fundamental convergence speed and cyber-resilience of distributed NE seeking algorithms. However, these frameworks implicitly rely on the assumption that the strategy space of each player is unbounded, constituting an unconstrained game. In realistic physical and engineering systems, players are perpetually subjected to strict local and operational limitations. For instance, in a distributed energy market, a generator's power output is strictly bounded by its physical generation capacity; in autonomous driving, an agent's acceleration and steering angles are strictly confined by mechanical bounds and safety regulations. These unavoidable physical limitations mathematically translate into constrained non-cooperative games, where ensuring the exact feasibility of players' actions throughout the strategic interactions becomes as crucial as reaching the equilibrium itself.

In the past decade, handling constrained strategy sets in distributed game theory has been a focal point of intense research. The most classical and widely adopted mathematical tool for dealing with constrained action spaces is the projection operator. By projecting an infeasible intermediate state onto the boundary of the strictly

convex constraint set, projection-based distributed NE seeking algorithms iteratively guide the players toward a feasible equilibrium [citation]. While conceptually straightforward, projection operators are inherently non-smooth and highly nonlinear. The continuous-time implementation of projection-based dynamics often involves differential inclusions and discontinuous right-hand sides, severely complicating the rigorous stability analysis and leading to potential chattering effects in physical implementations. Alternatively, penalty function methods and primal-dual dynamics based on the Karush-Kuhn-Tucker (KKT) multipliers have been extensively proposed to convert the constrained optimization problem into an unconstrained one [citation]. However, these approaches typically suffer from tuning difficulties associated with penalty parameters or demonstrate sluggish oscillatory behaviors in the dual variables.

More crucially, the vast majority of existing algorithms addressing constrained non-cooperative games guarantee only asymptotic convergence. While a few recent studies have ventured into finite-time constrained NE seeking [citation], their convergence boundaries remain stubbornly dependent on the initial states, rendering them unsuitable for dynamically changing environments. To ensure strict real-time feasibility and rapid responses, it is highly desirable to develop fixed-time convergent algorithms for constrained games. Yet, integrating fixed-time stability with strict set constraints introduces unprecedented mathematical hurdles. The inherent non-smoothness of the constraint-handling mechanisms (such as projections) intrinsically conflicts with the fractional-power terms heavily utilized in standard fixed-time protocol designs. Consequently, the development of a smooth, exact, and globally fixed-time convergent framework for constrained games remains a largely unresolved challenge in the current literature.

To bridge this substantial gap, this chapter investigates the distributed fixed-time NE seeking problem for constrained non-cooperative games. Instead of relying on non-smooth projection operators, we leverage the advanced mathematical machinery of Variational Inequality (VI) theory and neurodynamic optimization to systematically handle the physical limitations. The main contributions of this chapter are distinctively summarized as follows:

1. Variational Inequality Reformulation: We formally map the constrained non-cooperative game problem into a distributed variational inequality problem. This rigorous mathematical translation allows for the seamless handling of coupled constraints and optimization objectives within a unified geometric framework.

2. **Neurodynamic-Based Fixed-Time Design:** To elegantly circumvent the non-smoothness of traditional projection operators, a novel distributed algorithm is developed by integrating recurrent neurodynamic methods with the leader-following consensus protocol and sliding mode control theory. This smooth formulation inherently guarantees the strict feasibility of all intermediate actions without generating undesirable chattering.
3. **Exact Fixed-Time Constraint Satisfaction:** It is rigorously proven that the proposed neurodynamic algorithm drives the actions of all players into the constrained feasible set and subsequently to the exact Nash equilibrium within a strictly bounded fixed time. The derived upper bound of the convergence time is totally decoupled from the initial conditions, verifying the robust fixed-time performance for constrained environments.

The remainder of this chapter is structured as follows. Section 5.2 introduces the variational inequality formulation for constrained games. Section 5.3 presents the neurodynamic-based fixed-time algorithm and establishes its theoretical proofs. Section 5.4 demonstrates the superiority of the proposed method via numerical simulations, and Section 5.5 draws the chapter's conclusions.

5.2 Problem Formulation

Consider a set of players $\mathcal{V} \triangleq \{1, \dots, N\}$ with the action of player i denoted as $x_i \in \mathbb{R}^{n_i}$ and the action vector of the other players denoted by $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$, where $n = \sum_{i=1}^N n_i$, respectively. The dynamic of player i is given as $\dot{x}_i = u_i$, where $u_i \in \mathbb{R}^{n_i}$ is the control input.

Each player i aims to minimize its own cost function $f_i(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} = [x_1, \dots, x_N]^T \in \mathbb{R}^n$ denotes the action vector of N players. If player j is not a neighbor of Euler-Lagrange i , which means $a_{ij} = 0$, then player i has no access to player j 's action directly. Otherwise, player i can get the information of player j via an undirected communication graph $\mathcal{G}$.

The game in this article is described as

$$\begin{aligned} & \min_{x_i} f_i(x_i, \mathbf{x}_{-i}), \\ & \text{subject to } x_i \in Q_i(\mathbf{x}_{-i}), \end{aligned} \tag{5.1}$$

where $Q_i(\mathbf{x}_{-i}) = \{x_i \in \Omega_i | g(x_i, \mathbf{x}_{-i}) \leq 0\}$ is a closed convex set, and $\Omega_i = [\omega_{i1}^l, \omega_{i1}^u] \times [\omega_{i2}^l, \omega_{i2}^u] \times \cdots \times [\omega_{in_i}^l, \omega_{in_i}^u] \subset \mathbb{R}^{n_i}$, $i \in \mathcal{V}$. It is noted that the constraint function $g(\mathbf{x})$ is shared by all players.

Denote $\Omega = \Omega_1 \times \Omega_2 \times \cdots \times \Omega_N$ and $\chi = \{\mathbf{x} \in \mathbb{R}^n | g(\mathbf{x}) \leq 0\}$. Then the feasible strategy set of this game is $B = \Omega \cap \chi$.

First, we formalize the definition of the Nash equilibrium of the constrained non-cooperative game.

Definition 5.1. [zou2021continuous] A strategy profile $\mathbf{x}^* = [x_1^*, \dots, x_N^*]$ is called a Nash equilibrium of the constrained non-cooperative game (5.1) if $\forall i \in \mathcal{V}$,

$$f_i(x_i^*, \mathbf{x}_{-i}^*) \leq f_i(x_i, \mathbf{x}_{-i}^*), x_i \in Q_i(\mathbf{x}_{-i}^*). \quad (5.2)$$

Our objective is to design a distributed Nash equilibrium seeking algorithm for the players such that their actions $\mathbf{x}$ converge to the Nash equilibrium of the constrained game (5.1) in fixed time.

The following mild assumptions are made.

Assumption 10. For any $i \in \mathcal{V}$, the cost function $f_i(x_i, \mathbf{x}_{-i})$ is convex and twice continuously differentiable in $x_i \in \mathbb{R}^{n_i}$ for every fixed $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$.

Assumption 11. Let $F(\mathbf{x}) = [\nabla_1 f_1(x_1, \mathbf{x}_{-1}), \dots, \nabla_N f_N(x_N, \mathbf{x}_{-N})]^\top$, where $\nabla_i f_i(x_i, \mathbf{x}_{-i}) = \frac{\partial f_i(x_i, \mathbf{x}_{-i})}{\partial x_i}$. F is strongly monotone with modulus δ_m in $\mathbb{R}^n$ and globally Lipschitz continuous with a positive constant θ_l , that is, $(\mathbf{x} - \hat{\mathbf{x}})^\top (F(\mathbf{x}) - F(\hat{\mathbf{x}})) \geq \delta_m \|\mathbf{x} - \hat{\mathbf{x}}\|^2$, $\|F(\mathbf{x}) - F(\hat{\mathbf{x}})\| \leq \theta_l \|\mathbf{x} - \hat{\mathbf{x}}\|$, $\forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 12. The constraint function $g(\mathbf{x})$ is convex and the feasible strategy set B is non-empty, that is, $B \neq \emptyset$. Furthermore, the game (5.1) satisfies the Slater's condition, that is, there exists $\mathbf{x} \in B$ such that $\omega_{ik}^l < x_{ik} < \omega_{ik}^u$, $g(\mathbf{x}) < 0$, $i = 1, \dots, N, j = 1, \dots, n_i$.

Assumption 13. The communication graph $\mathcal{G}$ is undirected and connected.

5.3 Main Results

5.3.1 Distributed Nash Equilibrium Seeking Algorithm

In this section, we propose a fixed-time converging distributed Nash equilibrium seeking algorithm for the constrained non-cooperative game (5.1).

Inspired by the distributed approaches in [li2019distributed, poveda2022fixed] and the variational inequality solving approach proposed in [9919333], we propose the following distributed fixed-time converging algorithm to the constrained game problem (5.1). For any $i, j \in \mathcal{V}$,

$$\dot{x}_i = -\eta_{i,1} \frac{x_i - \zeta_{ii}}{d_i^{1-\mu}} - \eta_{i,2} \frac{x_i - \zeta_{ii}}{d_i^{1-\nu}}, \quad (5.3)$$

$$\dot{z}_{ij} = -(\alpha_{i,1} [\Lambda_{ij}]^\mu + \alpha_{i,2} [\Lambda_{ij}]^\nu) - \gamma \text{sign}(\Lambda_{ij}), \quad (5.4)$$

$$\dot{v}_{ij} = -(\beta_{i,1} [\rho_{ij}]^\mu + \beta_{i,2} [\rho_{ij}]^\nu) - \varepsilon \text{sign}(\rho_{ij}), \quad (5.5)$$

$$\Lambda_{ij} = \sum_{k=1}^N a_{ik} (z_{ij} - z_{kj}) + a_{ij} (z_{ij} - x_j), \quad (5.6)$$

$$\rho_{ij} = \sum_{k=1}^N a_{ik} (v_{ij} - v_{kj}) + a_{ij} (v_{ij} - \nabla_j f_j(\mathbf{z}_j)), \quad (5.7)$$

$$\zeta_i = \mathbf{P}_B(\mathbf{z}_i - k\mathbf{v}_i), \quad (5.8)$$

$$d_i = \min\{\|\psi_i - \zeta_i\|, s\}, \quad (5.9)$$

where $x_i = [x_{i1}, \dots, x_{in_i}]^\top$ is the action of the player i , $z_{ij} = [z_{ij1}, \dots, z_{ijn_j}]^\top$ is the player i 's estimate on the player j 's action, $v_{ij} = [v_{ij1}, \dots, v_{ijn_j}]^\top$ is the player i 's estimate on $\nabla_j f_j(\mathbf{z}_j)$, $\mathbf{z}_i = [z_{i1}, \dots, z_{iN}]^\top$ is the player i 's estimates on all players' actions, $\mathbf{v}_i = [v_{i1}, \dots, v_{iN}]^\top$, $\zeta_{ii} = [\zeta_{ii1}, \dots, \zeta_{iin_i}]^\top$ is the i -th element of $\zeta_i = [\zeta_{i1}, \dots, \zeta_{ii}, \dots, \zeta_{iN}]$, $\psi_i = [z_{i1}, \dots, z_{i(i-1)}, x_i, z_{i(i+1)}, \dots, z_{iN}]^\top$, $\mathbf{P}_B(\cdot)$ is the projection operator defined in the notation part, a_{ik} is the (i, k) -th entry of the adjacency matrix A corresponding to the communication graph $\mathcal{G}$, $\alpha_{i,1}, \alpha_{i,2}, \beta_{i,1}, \beta_{i,2}, \eta_{i,1}, \eta_{i,2}, s, k, \gamma, \varepsilon, \mu \in (0, 1), \nu \in (1, +\infty)$ are some positive design parameters.

The proposed algorithm consists of two parts: Nash equilibrium seeking and distributed estimation. For player i , the distributed estimators in (5.4)-(5.7) with their

states denoted as z_{ij} and v_{ij} , respectively, are designed based on leader-following consensus protocols to estimate player j 's action and the gradient of player j 's cost function, respectively. In the Nash equilibrium seeking part (5.3) with (5.8) and (5.9), the projected gradients for action updates are constructed using the estimates of all players' actions and cost function gradients. Furthermore, a design parameter s is introduced to prevent unbounded action updates.

Remark 5.2. In this work, the projection operator is utilized to address distributed game problems with constraints. Specifically, we construct the convex feasible set B by intersecting the inequality and set constraints and use a projection-based gradient update to maintain feasibility and ensure convergence to the constrained Nash equilibrium. A similar method is also adopted in [9919333]. However, unlike [9919333], our algorithm is distributed and addresses a multi-player optimization problem. Compared with the primal-dual methods [9343684, zeng2019generalized, 10229248, 10334027] that necessitate each player to compute Lagrange multipliers for handling constraints and subsequently achieve consensus on these Lagrange multipliers across the communication graph, the proposed approach does not require estimation of each player's Lagrange multiplier and thus reduces the number of variables to be communicated for distributed implementation. Compared with the penalty-based Nash equilibrium seeking algorithm [sun2020continuous], our approach guarantees that player i 's action x_i is bounded during the Nash equilibrium seeking process.

It is worth noting that the proposed algorithm guarantees that the virtual player i 's action $x_i = [x_{i1}, \dots, x_{in_i}]^\top$ is bounded during the Nash equilibrium seeking process. The following lemma is presented to describe the bounds of $x_{ij}(t)$.

Lemma 5.3. *Consider the constrained non-cooperative game (5.1) with the Nash equilibrium seeking algorithm (5.3)-(5.9) under Assumptions 1-4. There exists a positive constant $\tau_i = \max\{|x_{ij}(0)|, |\omega_{ij}^l|, |\omega_{ij}^u|, j = 1, \dots, n_i\}$ such that $|x_{ij}(t)| \leq \tau_i, \forall t \geq 0$, where $\omega_{ij}^l, \omega_{ij}^u$ are some constants related to the set constraints described in Assumption 12.*

Proof. We can prove $|x_{ij}(t)|$ is bounded by τ_i by contradiction. Suppose that this is not true. In such a case, based on the continuity of $x_{ij}(t)$, there exists a time $t_1 \geq 0$ such that $x_{ij}(t_1) = -\tau_i$ with $\dot{x}_{ij}(t_1) < 0$ or $x_{ij}(t_1) = \tau_i$ with $\dot{x}_{ij}(t_1) > 0$. It follows from the definition of the projection operator that $\omega_{ij}^l \leq \zeta_{ij} \leq \omega_{ij}^u$, one has $\dot{y}_{ij}(t_1) = -\eta_{i,1} \frac{-\tau_i - \zeta_{ij}}{d_i^{1-\mu}} - \eta_{i,2} \frac{-\tau_i - \zeta_{ij}}{d_i^{1-\nu}} \geq 0$, or $\dot{y}_{ij}(t_1) = -\eta_{i,1} \frac{\tau_i - \zeta_{ij}}{d_i^{1-\mu}} - \eta_{i,2} \frac{\tau_i - \zeta_{ij}}{d_i^{1-\nu}} \leq 0$. This brings

in contradiction. Thus, given any $y_{ij}(0) \in \mathbb{R}$, it follows that $|x_{ij}(t)| \leq \tau_i, \forall t \geq 0$. The proof is completed. $\square$

It follows from Lemma 5.3 that $|x_{ij}(t)|$ is bounded by $\tau_i = \max\{|x_{ij}(0)|, |\omega_{ij}^l|, |\omega_{ij}^u|, j = 1, \dots, n_i\}$, $i \in \mathcal{V}$. If $s > 2\sqrt{n}\tau_{\max}$, where $\tau_{\max} = \max\{\tau_i, i = 1, \dots, N\}$, then

$$d_i = \min\{\|\psi_i - \zeta_i\|, s\} \geq \min\{\|x_i - \zeta_{ii}\|, s\} \geq \|x_i - \zeta_{ii}\|. \quad (5.10)$$

The last inequality in (5.10) follows from the fact that

$$\|x_i - \zeta_{ii}\| \leq \|x_i\| + \|\zeta_{ii}\| \leq 2\sqrt{n}\tau_{\max}. \quad (5.11)$$

Thus, it can be further inferred that

$$\left\| \frac{x_i - \zeta_{ii}}{d_i^{1-\mu}} \right\| \leq \frac{\|x_i - \zeta_{ii}\|}{\|x_i - \zeta_{ii}\|^{1-\mu}} \leq (2\sqrt{n}\tau_{\max})^\mu, \quad (5.12)$$

$$\left\| \frac{x_i - \zeta_{ii}}{d_i^{1-\nu}} \right\| \leq \frac{\|x_i - \zeta_{ii}\|}{s^{1-\nu}} \leq 2\sqrt{n}\tau_{\max}s^{\nu-1}. \quad (5.13)$$

5.3.2 Convergence Analysis

In this section, it is ready to present the following result on the fixed-time convergence of the distributed Nash equilibrium seeking algorithm (5.3)-(5.9).

Theorem 5.4. *Consider the constrained non-cooperative game (5.1) with the Nash equilibrium seeking algorithm (5.3)-(5.9) for the virtual players of first-order dynamics under Assumptions 1-4. For any given $k \in (0, \frac{2\sigma_m}{\theta_l^2})$, let $c = \sqrt{1 - 2k\sigma_m + k^2\theta_l^2}$, $r = \max\{\eta_{i,m}/\eta_{j,m}, i, j = 1, \dots, N, m = 1, \dots, 2\}$ and $\eta_{\max,m} = \max\{\eta_{i,m}, i = 1, \dots, N\}$, $m=1, \dots, 2$. Then the Nash equilibrium x^* can be reached in fixed time with the proposed algorithm if the following conditions are satisfied:*

$$\gamma > \eta_{\max,1}(2\sqrt{n}\tau_{\max})^\mu + 2\eta_{\max,2}\sqrt{n}\tau_{\max}s^{\nu-1}, \quad (5.14)$$

$$\varepsilon > L\sqrt{N}\gamma, \quad r < \frac{1}{c}, \quad s > 2\sqrt{n}\tau_{\max}. \quad (5.15)$$

And the upper bound of its convergence time T is given by

$$T_{cons} = \frac{1+\nu}{G_1(1-\mu)} + \frac{1+\nu}{G_2(\nu-1)} + \frac{1+\nu}{G_3(1-\mu)} + \frac{1+\nu}{G_4(\nu-1)} + \frac{2}{G_5(1-\mu)} + \frac{2}{G_6(\nu-1)}, \quad (5.16)$$

where $C_1, C_2, C_3, C_4, C_5, C_6$ are some positive constants.

Proof. We will present the proof in three parts. The first part is to show that the player i 's estimate z_{ij} converges to the player j 's action x_j in fixed time; the second part is to show that v_{ij} converges to $\nabla_j f_j(\mathbf{x})$ in fixed time; and the third part is to show that the players' actions $\mathbf{x}$ converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time.

Part I: Proof of the convergence of z_{ij} to x_j in fixed time.

Consider the Lyapunov function candidate,

$$V_1 = \frac{1}{2}(\mathbf{z} - \mathbf{1}_N \otimes \mathbf{x})^\top (\mathcal{L} \otimes I_n + \mathcal{M})(\mathbf{z} - \mathbf{1}_N \otimes \mathbf{x}), \quad (5.17)$$

where $\mathbf{z} = [z_{11}, \dots, z_{1N}, z_{21}, \dots, z_{NN}]^\top$, $\mathbf{x} = [x_1, \dots, x_N]^\top$, $\mathcal{M} = \text{diag}\{a_{ij}I_{n_j}, i, j = 1, \dots, N\}$ and $\mathcal{L}$ is the Laplacian matrix of the communication graph $\mathcal{G}$.

Since $\mathcal{G}$ is undirected and connected, then zero is a simple eigenvalue of its corresponding Laplacian matrix $\mathcal{L}$. According to Gershgorin's disc theorem, $(\mathcal{L} \otimes I_n + \mathcal{M})$ is symmetric and positive definite [fang2021distributed]. It follows from the positive definiteness of $(\mathcal{L} \otimes I_n + \mathcal{M})$ that

$$\frac{1}{2}\lambda_{\min}\|\mathbf{z} - \mathbf{1}_N \otimes \mathbf{x}\|^2 \leq V_1 \leq \frac{1}{2}\lambda_{\max}\|\mathbf{z} - \mathbf{1}_N \otimes \mathbf{x}\|^2, \quad (5.18)$$

where $\lambda_{\min}$ and $\lambda_{\max}$ denote the minimum and maximum eigenvalue of matrix $(\mathcal{L} \otimes I_n + \mathcal{M})$, respectively.

Taking the time derivative of V_1 along the trajectories of (5.4), one has

$$\begin{aligned} \dot{V}_1 = & -\Lambda^\top (\tilde{\alpha}_1 [\Lambda]^\mu + \tilde{\alpha}_2 [\Lambda]^\nu) \\ & - \Lambda^\top (\gamma \text{sign}(\Lambda) + \mathbf{1}_N \otimes (\hat{\eta}_1 [\frac{x_i - \zeta_{ii}}{d_i^{1-\mu}}]_{vec} + \hat{\eta}_2 [\frac{x_i - \zeta_{ii}}{d_i^{1-\nu}}]_{vec})), \end{aligned} \quad (5.19)$$

where $\Lambda = [\Lambda_{11}, \dots, \Lambda_{1N}, \Lambda_{21}, \dots, \Lambda_{NN}]^\top = (\mathcal{L} \otimes I_n + \mathcal{M})(z - 1_N \otimes \mathbf{x})$, $\tilde{\alpha}_m = \text{diag}\{\alpha_{i,m} I_n, i = 1, \dots, N\}$, $\hat{\eta}_m = \text{diag}\{\eta_{i,m} I_n, i = 1, \dots, N\}$, $m = 1, 2$, $[\frac{x_i - \zeta_{ii}}{d_i^{1-\mu}}]_{\text{vec}} = [\frac{x_{11} - \zeta_{11}}{d_1^{1-\mu}}, \dots, \frac{x_{NN} - \zeta_{NN}}{d_N^{1-\mu}}]^\top$, $[\frac{x_i - \zeta_{ii}}{d_i^{1-\nu}}]_{\text{vec}} = [\frac{x_{11} - \zeta_{11}}{d_1^{1-\nu}}, \dots, \frac{x_{NN} - \zeta_{NN}}{d_N^{1-\nu}}]^\top$.

For any $i \in \mathcal{V}$, it follows from the inequalities (5.12) and (5.13) that

$$\eta_{i,1} \left\| \frac{x_i - \zeta_{ii}}{d_i^{1-\mu}} \right\| + \eta_{i,2} \left\| \frac{x_i - \zeta_{ii}}{d_i^{1-\nu}} \right\| \leq \eta_{\max,1} (2\sqrt{n}\tau_{\max})^\mu + 2\eta_{\max,2} \sqrt{n}\tau_{\max} s^{\nu-1}. \quad (5.20)$$

Given $\gamma > \eta_{\max,1} (2\sqrt{n}\tau_{\max})^\mu + 2\eta_{\max,2} \sqrt{n}\tau_{\max} s^{\nu-1}$, it follows from (5.20) that

$$\Lambda^\top (\gamma \text{sign}(\Lambda) + 1_N \otimes (\hat{\eta}_1 [\frac{x_i - \zeta_{ii}}{d_i^{1-\mu}}]_{\text{vec}} + \hat{\eta}_2 [\frac{x_i - \zeta_{ii}}{d_i^{1-\nu}}]_{\text{vec}})) > 0. \quad (5.21)$$

It then follows from (5.19) and (5.21) that

$$\dot{V}_1 \leq -\alpha_{\min,1} \|\Lambda\|_{\mu+1}^{\mu+1} - \alpha_{\min,2} \|\Lambda\|_{\nu+1}^{\nu+1}, \quad (5.22)$$

where $\alpha_{\min,m} = \min\{\alpha_{i,m}, i = 1, \dots, N\}$, $m = 1, 2$, $\|\cdot\|_{\mu+1}$ is the $(\mu+1)$ -norm, and $\|\cdot\|_{\nu+1}$ is the $(\nu+1)$ -norm.

It follows from (5.17) and (5.18) that

$$\|\Lambda\|^2 = \|(\mathcal{L} \otimes I_n + \mathcal{M})(z - 1_N \otimes \mathbf{x})\|^2 \geq \lambda_{\min}^2 \|z - 1_N \otimes \mathbf{x}\|^2 \geq \frac{2\lambda_{\min}^2}{\lambda_{\max}} V_1. \quad (5.23)$$

It further follows from (5.22), (5.23) and Lemma 2.5 given in Appendix that

$$\dot{V}_1 \leq -G_1 V_1^{\frac{\mu+1}{2}} - G_2 V_1^{\frac{\nu+1}{2}}, \quad (5.24)$$

where $G_1 = \alpha_{\min,1} \varpi^{\frac{\mu+1}{2}}$, $G_2 = \alpha_{\min,2} (nN)^{\frac{1-\nu}{2}} \varpi^{\frac{\nu+1}{2}}$, $\varpi = \frac{2\lambda_{\min}^2}{\lambda_{\max}}$.

Via Lemma 2.8, one thus has that the estimate z_{ij} converges to x_j in fixed time with the upper bound of the convergence time given by

$$T_1 = \frac{2}{G_1(1-\mu)} + \frac{2}{G_2(\nu-1)}. \quad (5.25)$$

After $t = T_1$, one has $z_{ij} = x_j$ and $\rho_{ij} = \sum_{k=1}^N a_{ik}(v_{ij} - v_{kj}) + a_{ij}(v_{ij} - \nabla_j f_j(\mathbf{x}))$.

Part II: Proof of the convergence of v_{ij} to $\nabla_j f_j(\mathbf{x})$ in fixed time.

For $t \geq T_1$, consider the Lyapunov function candidate,

$$V_2 = \frac{1}{2}(\mathbf{v} - 1_N \otimes F(\mathbf{x}))^\top (\mathcal{L} \otimes I_n + \mathcal{M})(\mathbf{v} - 1_N \otimes F(\mathbf{x})), \quad (5.26)$$

where $F(\mathbf{x}) = [\nabla_1 f_1(\mathbf{x}), \dots, \nabla_N f_N(\mathbf{x})]^\top$, and $\mathbf{v} = [v_{11}, \dots, v_{1N}, v_{21}, \dots, v_{NN}]^\top$.

Taking the time derivative of V_2 along the trajectories of (5.5), one has

$$\dot{V}_2 = -\rho^\top (\tilde{\beta}_1 [\rho]^\mu + \tilde{\beta}_2 [\rho]^\nu) - \rho^\top (\varepsilon \text{sign}(\rho) + 1_N \otimes \frac{d(F(\mathbf{x}))}{dt}), \quad (5.27)$$

where $\rho = [\rho_{11}, \rho_{12}, \dots, \rho_{1N}, \rho_{21}, \dots, \rho_{NN}]^\top = (\mathcal{L} \otimes I_n + \mathcal{M})(v - 1_N \otimes F(\mathbf{x}))$, $\tilde{\beta}_m = \text{diag}\{\beta_{i,m} I_n, i = 1, \dots, N\}$, $m = 1, 2$.

It follows from Assumption 2 and (5.20) that

$$\left\| \frac{dF(\mathbf{x})}{dt} \right\| \leq \theta_l \lim_{\Delta t \rightarrow 0} \frac{\|\mathbf{x}(t + \Delta t) - \mathbf{x}(t)\|}{\Delta t} \leq \theta_l \sqrt{N} \gamma. \quad (5.28)$$

Given $\varepsilon > \theta_l \sqrt{N} \gamma$, similar to the arguments of the first part in (5.21)-(5.24), one has

$$\dot{V}_2 \leq -G_3 V_2^{\frac{\mu+1}{2}} - G_4 V_2^{\frac{\nu+1}{2}}, \quad (5.29)$$

where $G_3 = \beta_{\min,1} \varpi^{\frac{\mu+1}{2}}$, $G_4 = \beta_{\min,2} (nN)^{\frac{1-\nu}{2}} \varpi^{\frac{\nu+1}{2}}$, and $\beta_{\min,m} = \min\{\beta_{i,m}, i = 1, \dots, N\}$, $m = 1, 2$.

Via Lemma 2.8, one has that v_{ij} converges to $\nabla_j f_j(\mathbf{x})$ in fixed time with the upper bound of the convergence time given by

$$T_2 = \frac{2}{G_3(1-\mu)} + \frac{2}{G_4(\nu-1)}. \quad (5.30)$$

Since $\|\mathbf{x}\|$ is bounded from Lemma 5.3, the proof of this part is to be completed when the boundedness of $\|v\|$ is shown within the time interval $[0, T_1)$. For $0 \leq t < T_1$, consider the Lyapunov function candidate,

$$W_1 = \frac{1}{2}(\mathbf{v} - 1_N \otimes \hat{F}(\mathbf{z}))^\top (\mathcal{L} \otimes I_n + \mathcal{M})(\mathbf{v} - 1_N \otimes \hat{F}(\mathbf{z})), \quad (5.31)$$

where $\rho = [\rho_{11}, \rho_{12}, \dots, \rho_{1N}, \rho_{21}, \dots, \rho_{NN}]^\top = (\mathcal{L} \otimes I_n + \mathcal{M})(v - 1_N \otimes \hat{F}(\mathbf{z}))$, and $\hat{F}(\mathbf{z}) = [\nabla_1 f_1(\mathbf{z}_1), \dots, \nabla_N f_N(\mathbf{z}_N)]^\top$.

Taking the time derivative of W_1 along the trajectories of (5.5), one has

$$\begin{aligned}\dot{W}_1 &= -\rho^\top (\tilde{\beta}_1 [\rho]^\mu + \tilde{\beta}_2 [\rho]^\nu) - \rho^\top (\varepsilon \text{sign}(\rho) + 1_N \otimes \frac{d(\hat{F}(\mathbf{z}))}{dt}) \\ &\leq -\beta_{\min} \|\rho\|^2 + \sqrt{N} \|\rho\| \left\| \frac{d(\hat{F}(\mathbf{z}))}{dt} \right\|,\end{aligned}\quad (5.32)$$

where $\beta_{\min} = \min\{\beta_{i,m}, i = 1, \dots, N, m = 1, 2\}$.

For any $i \in \mathcal{V}$, one has $\left\| \frac{d(\nabla_i f_i(\mathbf{z}_i))}{dt} \right\| \leq \left\| \frac{d(F(\mathbf{z}_i))}{dt} \right\|$. It further follows from Assumption 2 and (5.4) that

$$\begin{aligned}\left\| \frac{d(F(\mathbf{z}_i))}{dt} \right\| &\leq \theta_l \lim_{\Delta t \rightarrow 0} \frac{\|\mathbf{z}_i(t + \Delta t) - \mathbf{z}_i(t)\|}{\Delta t} \\ &\leq \theta_l (\alpha_{i,1} \|\Lambda_i\|^\mu + \alpha_{i,2} \|\Lambda_i\|^\nu + \gamma),\end{aligned}\quad (5.33)$$

where $F(\mathbf{z}_i) = [\nabla_1 f_1(\mathbf{z}_i), \dots, \nabla_N f_N(\mathbf{z}_i)]^\top$, and $\Lambda_i = [\Lambda_{i1}, \dots, \Lambda_{iN}]$.

It follows from (5.17) and (5.24) that $\|\Lambda\| = \|(\mathcal{L} \otimes I_n + \mathcal{M})(z - 1_N \otimes \mathbf{x})\|$ is bounded within the time interval $[0, T_1)$. Then, one has $\left\| \frac{d(\hat{F}(\mathbf{z}))}{dt} \right\|$ is also bounded within the time interval $[0, T_1)$, that is, there exists a positive constant D_1 such that $\left\| \frac{d(\hat{F}(\mathbf{z}))}{dt} \right\| \leq D_1$.

It further follows from Lemma 2.3 that $\|\rho\| \|1_N \otimes \frac{d(\hat{F}(\mathbf{z}))}{dt}\| \leq \frac{k_1}{2} \|\rho\|^2 + \frac{1}{2k_1} D_1$, where k_1 is a positive constant.

Similar to the argument in (5.23), one has

$$\|\rho\|^2 = \|(\mathcal{L} \otimes I_n + \mathcal{M})(v - 1_N \otimes \hat{F}(\mathbf{z}))\|^2 \geq \frac{2\lambda_{\min}^2}{\lambda_{\max}} W_1. \quad (5.34)$$

It then follows from (5.31)-(5.34) that

$$\dot{W}_1 \leq -\frac{(2\beta_{\min,3} - k_1 \sqrt{N})\lambda_{\min}^2}{\lambda_{\max}} W_1 + \frac{\sqrt{N}}{2k_1} D_1. \quad (5.35)$$

Choosing suitable parameter k_1 such that $2\beta_{\min,3} - k_1 \sqrt{N} > 0$, then one has $\|\mathbf{v} - 1_N \otimes \hat{F}(\mathbf{z})\|$ is bounded within the time interval $[0, T_1)$.

It follows from Lemma 5.3, (5.17) and (5.24) that both $\|\mathbf{x}\|$ and $\|\mathbf{z} - 1_N \otimes \mathbf{x}\|$ are bounded within the time interval $[0, T_1)$, then it can be further inferred that $\|\mathbf{z}\|$ is bounded within the time interval $[0, T_1)$. Under Assumption 1-2, there exists $\bar{\mathbf{x}} \in \mathbb{R}^n$ such that for any $i \in \mathcal{V}$, $\nabla_i f_i(\bar{\mathbf{x}}) = 0$ [ye2017distributed]. It also follows from

Assumption 2 that $\|\nabla_i f(\mathbf{z}_i)\| = \|\nabla_i f(\mathbf{z}_i) - \nabla_i f(\bar{\mathbf{x}})\| \leq \theta_l \|\mathbf{z}_i - \bar{\mathbf{x}}\|$, which implies $\|\nabla_i f(\mathbf{z}_i)\|$ is bounded within the time interval $[0, T_1)$. Since both $\|\mathbf{v} - \mathbf{1}_N \otimes \hat{F}(\mathbf{z})\|$ and $\|\hat{F}(\mathbf{z})\|$ are bounded within the time interval $[0, T_1)$, one has $\|v\|$ is also bounded within the time interval $[0, T_1)$, that is, no finite-time escape occurs.

After $t = T_1 + T_2$, $z_{ij} = y_j$, $v_{ij} = \nabla_j f_j(\mathbf{x})$. It follows from Lemma 5.3 that

$$\|\psi_i - \zeta_i\| = \|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\| \leq 2\sqrt{n}\tau_{max}, \quad (5.36)$$

where $\psi_i = [z_{i1}, \dots, z_{i(i-1)}, x_i, z_{i(i+1)}, \dots, z_{iN}]^\top$.

Furthermore, given $s > 2\sqrt{n}\tau_{max}$, it follows from (5.9) and (5.36) that $d_i = \|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|$ and the dynamics of the virtual players' actions can be expressed as:

$$\dot{\mathbf{x}} = -(\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))) \left(\frac{\hat{\eta}_1}{\|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|^{1-\mu}} + \frac{\hat{\eta}_2}{\|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|^{1-\nu}} \right), \quad (5.37)$$

where $F(\mathbf{x}) = [\nabla_1 f_1(\mathbf{x}), \dots, \nabla_N f_N(\mathbf{x})]$, and $\hat{\eta}_m = \text{diag}\{\eta_{i,m} I_{n_i}, i = 1, \dots, N\}$, $m = 1, 2, 3$.

Part III: Proof of the convergence of $\mathbf{x}$ to $\mathbf{x}^*$ in fixed time.

Before presenting the main proof, a lemma is introduced to bridge the variational inequality and the Nash equilibrium.

Lemma 5.5. [10229248] *Given a convex set $B \subset \mathbb{R}^N$, and a mapping $F : B \rightarrow \mathbb{R}^N$, solving the variational inequality $\mathbf{VI}(B, F)$ is to find a vector $\mathbf{x}^* \in B$ such that*

$$(\mathbf{x} - \mathbf{x}^*)^\top F(\mathbf{x}^*) \geq 0, \forall \mathbf{x} \in B. \quad (5.38)$$

Then $\mathbf{x}^$ is the solution to $\mathbf{VI}(B, F)$ if and only if $\mathbf{x}^* = \mathbf{P}_B(\mathbf{x}^* - F(\mathbf{x}^*))$. Under Assumptions 10-??, every solution to $\mathbf{VI}(B, F)$ is also a Nash equilibrium of the constrained non-cooperative game (5.1), where $F(\mathbf{x}) = [\nabla_1 f_1(x_1, \mathbf{x}_{-1}), \dots, \nabla_N f_N(x_N, \mathbf{x}_{-N})]^\top$.*

It follows from Lemma 5.5 that if $\mathbf{x}^*$ is the solution to the variational inequality $\mathbf{VI}(B, F)$, then $\mathbf{x}^*$ is also the Nash equilibrium of game (5.1). Thus we only need to prove that $\mathbf{x}$ converges to the solution to $\mathbf{VI}(B, F)$ in fixed time.

For $t \geq T_1 + T_2$, consider the Lyapunov function candidate,

$$V_3 = \frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|^2, \quad (5.39)$$

where $\hat{\mathbf{x}}$ is the solution to the variational inequality $\mathbf{VI}(B, F)$ described in Lemma 5.5 given in Appendix.

Taking the time derivative of V_3 along the trajectories of (5.37), one has

$$\dot{V}_3 = -\frac{(\mathbf{x} - \hat{\mathbf{x}})^\top \hat{\eta}_1(\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x})))}{\|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|^{1-\mu}} - \frac{(\mathbf{x} - \hat{\mathbf{x}})^\top \hat{\eta}_2(\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x})))}{\|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|^{1-\nu}}. \quad (5.40)$$

It also follows from Lemma 5.5 that if $\hat{\mathbf{x}}$ is the solution to $\mathbf{VI}(B, F)$, then $(\mathbf{x} - \hat{\mathbf{x}})^\top F(\hat{\mathbf{x}}) \geq 0, \forall \mathbf{x} \in B$, and $\hat{\mathbf{x}} = \mathbf{P}_B(\hat{\mathbf{x}} - kF(\hat{\mathbf{x}}))$. It can be further inferred that given $k > 0$, $(\mathbf{x} - \hat{\mathbf{x}})^\top kF(\hat{\mathbf{x}}) \geq 0, \forall \mathbf{x} \in B$. Thus, $\hat{\mathbf{x}}$ is also the solution to $\mathbf{VI}(B, kF)$ and

$$\hat{\mathbf{x}} = \mathbf{P}_B(\hat{\mathbf{x}} - kF(\hat{\mathbf{x}})). \quad (5.41)$$

It follows from Assumption 2 and (5.41) that

$$\begin{aligned} \|\hat{\mathbf{x}} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\|^2 &\leq \|\mathbf{x} - \hat{\mathbf{x}}\|^2 - 2k(F(\mathbf{x}) - F(\hat{\mathbf{x}}))^\top (\mathbf{x} - \hat{\mathbf{x}}) \\ &\quad + k^2\|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|^2 \\ &\leq c^2\|\mathbf{x} - \hat{\mathbf{x}}\|^2, \end{aligned} \quad (5.42)$$

$$\begin{aligned} \|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\| &\geq (1 - c)\|\mathbf{x} - \hat{\mathbf{x}}\|, \\ \|\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))\| &\leq (1 + c)\|\mathbf{x} - \hat{\mathbf{x}}\|, \end{aligned} \quad (5.43)$$

where $c = \sqrt{1 - 2k\sigma_m + k^2\theta_l^2}$.

Since $(\mathbf{x} - \hat{\mathbf{x}})^\top (\mathbf{x} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x}))) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 - (\mathbf{x} - \hat{\mathbf{x}})^\top (\hat{\mathbf{x}} - \mathbf{P}_B(\mathbf{x} - kF(\mathbf{x})))$, it then follows from (5.40)-(5.43) that

$$\begin{aligned} \dot{V}_3 &\leq -(\eta_{\min,1} - c\eta_{\max,1}) \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|^{\mu+1}}{(1+c)^{1-\mu}} - (\eta_{\min,2} - c\eta_{\max,2}) \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|^{\nu+1}}{(1-c)^{1-\nu}} \\ &= -G_5 V_3^{\frac{\mu+1}{2}} - G_6 V_3^{\frac{\nu+1}{2}}, \end{aligned} \quad (5.44)$$

where $G_5 = 2^{(\mu+1)/2}(\eta_{\min,1} - c\eta_{\max,1})/(1+c)^{1-\mu}$, $G_6 = 2^{(\nu+1)/2}(\eta_{\min,2} - c\eta_{\max,2})/(1-c)^{1-\nu}$. Given $r = \max\{\frac{\eta_{i,m}}{\eta_{j,m}}, i, j = 1, \dots, N, m = 1, \dots, 2\} < \frac{1}{c}$, it can be obtained that G_5 and $G_6 > 0$.

Via Lemma 2.8, one has that the virtual players' actions $\mathbf{x}$ converge to $\hat{\mathbf{x}}$ in fixed time and the upper bound of the convergence time is given by

$$T_3 = \frac{2}{G_5(1-\mu)} + \frac{2}{G_6(\nu-1)}. \quad (5.45)$$

It follows from Lemma 5.5 given in Appendix that if $\hat{\mathbf{x}}$ is the solution to the variational inequality $\mathbf{VI}(B, F)$, then $\hat{\mathbf{x}}$ is also the Nash equilibrium of the constrained game (5.1), that is, $\hat{\mathbf{x}} = \mathbf{x}^*$.

It follows from Lemma (5.3) that $|x_{ij}(t)|$ is bounded for all $t \geq 0$, that is, no finite-time escape occurs.

Combining the three parts, it can be inferred that the distributed Nash equilibrium seeking algorithm (5.3)-(5.9) for the virtual players of first-order dynamics converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time with the upper bound of the convergence time given by

$$T = T_1 + T_2 + T_3, \quad (5.46)$$

where T_1 , T_2 , and T_3 are positive constants given in (5.25), (5.30), and (5.45), respectively. The proof is thus completed. $\square$

Remark 5.6. There are several design parameters in the distributed Nash equilibrium seeking algorithm (5.3)-(5.9) and they can be selected as follows. Given the modulus σ_m , the Lipschitz constant θ_l , we can choose $0 < k < \frac{2\sigma_m}{\theta_l^2}$ to ensure that $c = \sqrt{1 - 2k\sigma_m + k^2\theta_l^2} < 1$. Then, we select the design parameters $\eta_{i,m}$, $m = 1, 2$, such that $r = \max\{\frac{\eta_{i,m}}{\eta_{j,m}}, i, j = 1, \dots, N, m = 1, \dots, 3\} < \frac{1}{c}$. Next, we evaluate the value of $\tau_{max} = \max\{\tau_i, i = 1, \dots, N\}$, where $\tau_i = \max\{|y_{ij}(0)|, |\omega_{ij}^l|, |\omega_{ij}^u|, j = 1, \dots, n_i\}$, and select γ, ε , and s such that $\gamma > \eta_{max,1}(2\sqrt{n}\tau_{max})^\mu + 2\eta_{max,2}\sqrt{n}\tau_{max}s^{\nu-1}$, $\varepsilon > \theta_l\sqrt{N}\gamma$ and $s > 2\sqrt{n}\tau_{max}$, respectively. The positive design parameters $\alpha_{i,m}$ and $\beta_{i,m}$, $m = 1, 2$, can be selected arbitrarily. By following the parameter selection guidance outlined above, it can be guaranteed that we can obtain a set of feasible design parameters and solve the Nash equilibrium seeking problem.

Remark 5.7. It is noted from Theorem 5.4 that the upper bound of the convergence time $T_E = T_1 + T_2 + T_3$ presented in (5.4) depends on the design parameters $\alpha_{i,m}, \beta_{i,m}, \eta_{i,m}, i = 1, \dots, N, m = 1, 2$. In particular, T_1 can be reduced by increasing $\alpha_{i,m}$, T_2 can be reduced by increasing $\beta_{i,m}$, T_3 can be reduced by increasing $\eta_{i,m}$.

Remark 5.8. The terms $\eta_{i,1}(y_i - \zeta_{ii})/d_i^{1-\mu}$, $\eta_{i,2}(y_i - \zeta_{ii})/d_i^{1-\nu}$, $\alpha_{i,1}[\Lambda_{ij}]^\mu$, $\alpha_{i,2}[\Lambda_{ij}]^\nu$, $\beta_{i,1}[\rho_{ij}]^\mu$, and $\beta_{i,2}[\rho_{ij}]^\nu$ in (5.3)-(5.5) play a crucial role in the convergence of $\mathbf{x}$ to $\mathbf{x}^*$

in fixed time. Furthermore, our result can include the cases of finite-time convergence and asymptotic convergence as its special cases. If $\nu = 1, i = 1, \dots, N$, then our algorithm reduces to the case of finite-time convergence. If $\mu = \nu = 1, i = 1, \dots, N$, then our algorithm reduces to case of asymptotic convergence. It should be pointed out that the reduced algorithm of finite-time convergence is also novel in comparison to existing works.

5.4 Numerical Examples

In this section, we consider a practical example of an energy consumption game[hu2022distributed] of N players. The cost of each player mainly consists of two parts: the load curtailment cost and the cost incurred to purchase the energy:

$$J_i(x) = l_i(x_i - x_i^r)^2 + x_i P(x), \quad (5.47)$$

where x_i (kWh) is the energy spent by player i , x_i^r (kWh) is the nominal value of energy consumption for player i , l_i (SGD/kWh) is a positive constant coefficient, and $P(\mathbf{x})$ is the pricing function modeled by $P(\mathbf{x}) = a \sum_{j=1}^N x_j + b$, with $a > 0$ and b being constant coefficients.

For the constrained game, consider $x_i^l \leq x_i \leq x_i^u$ and a shared inequality constraint $\sum_{j=1}^N x_j \leq h$, where x_i^l and x_i^u are the lower and upper bounds of x_i respectively, and $h > 0$ is a constant. The energy consumption game is given by

$$\min_{x_i \in [x_i^l, x_i^u]} J_i(x), \quad \text{s.t.} \quad \sum_{j=1}^N x_j \leq h, \quad \forall i \in \mathcal{V}. \quad (5.48)$$

In this example, we consider five players ($N = 5$). The game parameters are set as $x_1^r = 5, x_2^r = 5.5, x_3^r = 6, x_4^r = 6.5, x_5^r = 7, x_i^u = 5.5, x_i^l = 0, l_i = 0.5, a = 0.1, b = 1.25, h = 18, i = 1, \dots, N$. It can be easily computed that the constrained game admits a unique Nash Equilibrium at $x^* = [2.06, 2.51, 2.97, 3.42, 3.88]$ kWh. The communication digraph is shown in Figure 5.1, which is strongly connected.

Algorithm (??)-(??) is applied to seek the Nash Equilibrium of the constrained game with the design parameters shown in Table 5.1. We conducted simulation involving 100 trials with random initial states to verify the fixed-time convergence property of the proposed algorithm. The initial conditions are selected with their norms ranging

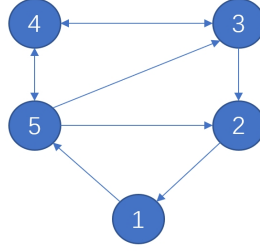


Figure 5.1. The communication digraph.

Table 5.1. Design parameters for the distributed Nash equilibrium seeking algorithm (??)–(??). Parameters: $\mu = 0.85$, $\nu = 1.15$, $k = 0.42$, $s = 2.24 \times 10^4$, $\gamma = 3.81 \times 10^5$, $\varepsilon = 1.37 \times 10^6$..

	α_i	β_i	φ_i	σ_i	δ_i	η_i
Player 1	50.3	51.4	52.2	53.5	8.0	8.1
Player 2	51.2	50.3	53.5	50.3	8.1	8.2
Player 3	52.3	51.6	51.1	51.1	8.2	8.0
Player 4	53.5	53.2	50.4	52.1	8.2	8.2
Player 5	50.5	53.5	50.3	51.3	8.0	8.1

Table 5.2. Comparison of different distributed Nash Equilibrium seeking algorithms.

Algorithms	Design parameters
Asymptotic algorithm [lu2018distributed]	$\alpha(t) = \frac{4}{t+1}$.
Asymptotic algorithm [zou2021continuous]	$\eta = 4$.
Asymptotic algorithm [sun2020continuous]	$\alpha = 10, \bar{k}_i = 4, \omega_{ij} = 8$
Fixed-Time algorithm (5.3)-(5.9)	$\alpha_i = \beta_i = \varphi_i = \sigma_i = 0.5, \delta_i = \eta_i = 2, \mu = 0.85, \nu =$

widely from 10^1 to 10^4 . Given the Lipschitz constant $L = 1.6$ and modulus $m = 1.1$, it can be verified that the design parameters shown in Table 5.1 meet the conditions stipulated in Theorem ??.

The simulation results are shown in Figure 5.2. In each subplot of Figure 5.2, the black dashed vertical line represents the theoretical upper bound of the convergence time for the corresponding state, which is calculated based on our stability analysis. It can be observed from Figure 5.2 that all error trajectories rapidly converge to zero well within the theoretical fixed time bound, regardless of the magnitude of the initial states. This behavior aligns with the results observed in fixed-time stable systems [**poveda2022fixed**, **tang2025lyapunov**], which verifies the fixed-time convergence property of the proposed algorithm (5.3)-(5.9).

To illustrate the transient behavior, we present a specific trial selected from the 100 simulations, where the initial state is $\mathbf{x}(0) = [4.5, 3.5, 2.5, 1.5, 1]^\top$. The corresponding simulation results are shown in Figures 5.3 and 5.4. It can be observed from Figure 5.3 that the players' actions converge to the Nash equilibrium in fixed time, which also verifies the effectiveness of the proposed algorithm.

In this part, we choose the asymptotic convergence algorithms [**lu2018distributed**, **zou2021continuous**, **sun2020continuous**], and the fixed-time convergence algorithm (??)-(??) for comparison with initial state $\mathbf{x}(0) = [0, 0, 0, 0, 0]^\top$. The same communication graph is adopted as depicted in Figure ??, with the edges being undirected. The design parameters of those algorithms are shown in Table 5.2, where the convergence time is defined with an error tolerance of 5×10^{-4} . It can be observed from Figure 5.5 and Table 5.2 that our algorithm (??)-(??) converges to the Nash equilibrium faster than those asymptotic algorithms.

In order to illustrate the impact of design parameters on the transient performance of the proposed algorithm. Four different sets of the design parameters $\alpha_i, \beta_i, \varphi_i, \sigma_i, \eta_i, \delta_i$

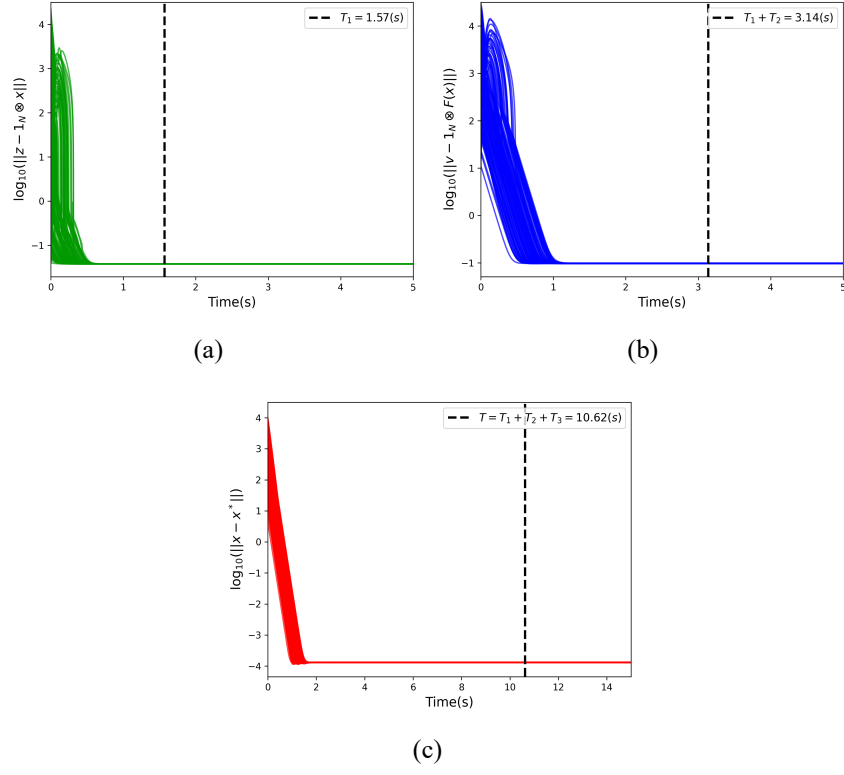


Figure 5.2. Trajectories of logarithmic error responses of the players' states under 100 different initial conditions. (a) $\log_{10}(\|z - \mathbf{1}_N \otimes \mathbf{x}\|)$, (b) $\log_{10}(\|v - \mathbf{1}_N \otimes F(\mathbf{x})\|)$, (c) $\log_{10}(\|x - x^*\|)$.

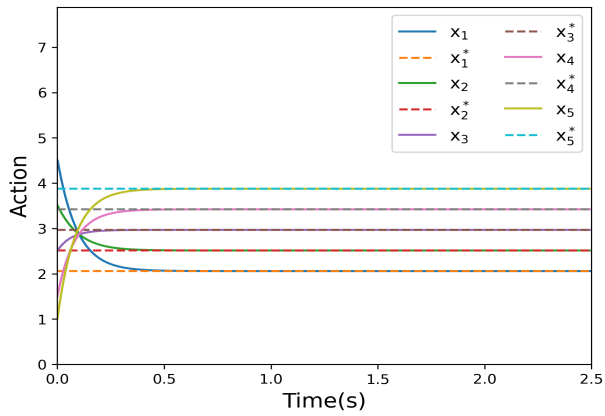


Figure 5.3. Trajectories of players' actions..

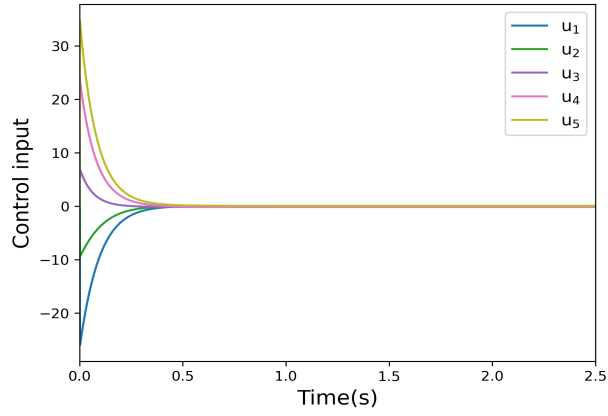


Figure 5.4. Trajectories of players' control inputs..

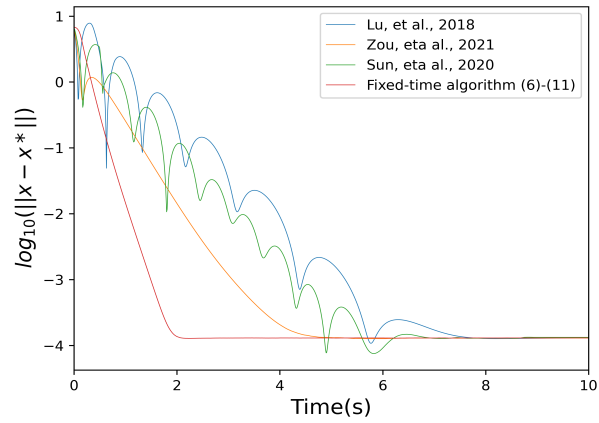


Figure 5.5. Comparison of logarithmic error responses among the asymptotic convergence algorithms [lu2018distributed, zou2021continuous, sun2020continuous], and the fixed-time convergence algorithm (6)-(11)..

in algorithm (??)-(??) are selected. Set 1 adopts $\alpha_i = \beta_i = \varphi_i = \sigma_i = \delta_i = \eta_i = 2$, $i = 1, \dots, 5$, $\mu = 0.85$, $\nu = 1.15$, $k = 0.42$, $s = 40$, $\gamma = 80$, $\varepsilon = 290$. Set 2 adopts 20 times of parameters α_i and β_i while keeping others the same as those in Set 1. Set 3 adopts 20 times of parameters φ_i and σ_i while keeping others the same as those in Set 1, Set 4 adopts 2 times of parameters η_i and δ_i while keeping others the same as those in Set 1.

The simulation results are shown in Figure 5.6 and Table 5.3 with initial state $\mathbf{x}(0) = [4.5, 3.5, 2.5, 1.5, 1]^\top$. The convergence time in Table 5.3 is defined with an error tolerance of 5×10^{-4} . It can be observed from Figure 5.6 that the larger the value of the design parameters $\alpha_i, \beta_i, \varphi_i, \sigma_i, \eta_i, \delta_i$, the faster x_i converges to the Nash equilibrium x_i^* , $i = 1, \dots, N$. These results are consistent with the discussions in Remark 5.

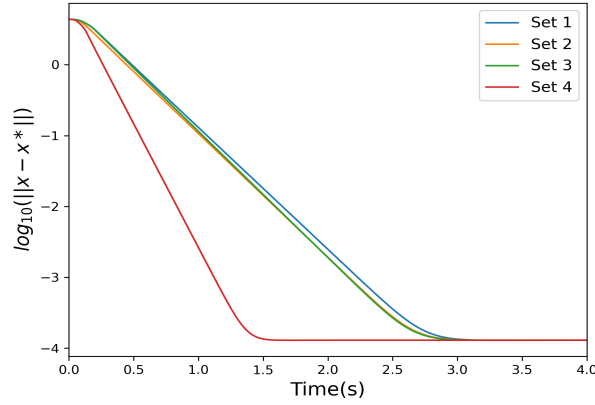


Figure 5.6. Comparison of logarithmic error responses among the different sets of the design parameters..

Table 5.3. Convergence times under different sets of the design parameters.

	Set 1	Set 2	Set 3	Set 4
Convergence time(s)	2.412	2.375	2.332	1.209

5.5 Conclusions

In this chapter, the distributed Nash equilibrium seeking problem for constrained non-cooperative games has been thoroughly investigated. To effectively handle local and

global set constraints without invoking non-smooth projection operators that severely complicate fixed-time stability analysis, the constrained game problem has been elegantly reformulated as a distributed variational inequality problem. Subsequently, a novel distributed fixed-time algorithm has been developed by seamlessly integrating recurrent neurodynamic optimization methods with sliding mode control theory. Theoretical analysis has rigorously demonstrated that the proposed smooth framework strictly enforces the feasibility of all players' actions throughout the transient process and guarantees exact convergence to the Nash equilibrium within a predefined fixed time. The derived upper bound of the convergence time is totally decoupled from the initial conditions, thereby offering a smooth, robust, and constraint-aware optimization mechanism for physical multi-agent systems with strict operational limitations.

Chapter 6

Distributed Fixed-Time Nash Equilibrium Seeking for Multi-Agent Systems with Heterogeneous Complex Dynamics

6.1 Introduction

The distributed game-theoretic frameworks developed thus far have provided robust, fast, and constraint-aware solutions for Nash equilibrium seeking. However, an implicit structural assumption common to these frameworks—and indeed prevalent throughout the vast majority of the distributed optimization literature—is that the decision-making players are modeled as idealized, kinematic entities, typically governed by simple single-integrator dynamics. In sharp contrast, modern multi-agent systems arising from advanced control engineering (such as robotic manipulators, autonomous aerial vehicles, and mobile rovers) are physical systems strictly governed by sophisticated, high-order, and profoundly nonlinear dynamics. Consequently, there exists a critical gap between theoretical game-theoretic algorithms formulated in the kinematic domain and their direct execution on physical platforms with intrinsic momentum, friction, and complex mechanical constraints.

Recognizing the necessity to bridge theoretical game designs and physical executions, a growing body of literature has dedicated efforts to integrating specific agent

dynamics into NE seeking frameworks. Progressively advancing from simple kinematics, distributed NE seeking strategies have been developed for agents characterized by second-order dynamics and general linear systems [citation]. For linear systems, internal model principles and observer-based architectures are routinely utilized to decouple the game algorithms from the plant dynamics. Nevertheless, practical physical agents frequently exhibit inherent, severe nonlinearities. Most notably, a broad class of mechanical and robotic systems is mathematically characterized by the nonlinear Euler-Lagrange (EL) equations, which inherently encompass complex mass-inertia matrices, Coriolis/centripetal forces, and gravitational vectors. While recent efforts have explored distributed NE seeking for non-cooperative games governed by EL dynamics [citation], these pioneering works predominantly yield asymptotic or exponential convergence properties.

Synthesizing fixed-time convergence properties with heterogeneous complex nonlinear dynamics (such as EL systems) poses profound control-theoretic challenges. Current finite- and fixed-time methods in game theory are heavily confined to idealized first- or second-order integrators because their mathematical proofs strictly rely on the continuous injection of fractional-power gradient signals. Directly applying these fractional-power signals as physical control inputs (torques or forces) to EL systems inevitably induces severe actuation chattering and unmodeled dynamic excitation. Furthermore, handling external disturbances and unmodeled dynamics inherently present in physical EL agents, while simultaneously maintaining strict fixed-time convergence to the exact Nash equilibrium, remains a highly restrictive bottleneck.

To systematically overcome these critical limitations, this chapter proposes a generalized, highly inclusive distributed fixed-time NE seeking framework for non-cooperative games played by multi-agent systems governed by heterogeneous complex dynamics, encompassing general nonlinear models and specific Euler-Lagrange equations. By explicitly decoupling the cyber-layer optimization from the physical-layer execution, this chapter significantly advances both the physical model generality and the convergence performance of existing literature. The main contributions are summarized as follows:

1. A Novel Virtual-Physical Two-Layer Architecture: To gracefully isolate the game-theoretic optimization complexities from the severe mechanical nonlinearities, a novel two-layer design framework is constructed. The cyber/virtual layer computes the theoretical fixed-time NE trajectories, while the physical layer governs the actual actuators. This separation principle inherently prevents algebraic loops and chattering in the physical control inputs.

2. **Fixed-Time Sliding Mode Tracking for EL Systems:** A novel fixed-time tracking protocol based on advanced sliding mode theory is synthesized specifically in the physical layer. It is theoretically guaranteed that despite the presence of heterogeneous unmodeled dynamics, Coriolis matrices, and external disturbances, the physical states of the EL players strictly synchronize with the virtual players' optimal actions within a fixed time.
3. **Complete Fixed-Time Generality:** Unlike existing algorithms strictly restricted to single or double integrators, the proposed decoupled framework rigorously accommodates severe nonlinearities while naturally encompassing first-order, second-order, and general linear dynamics as its special cases. The exact convergence of the physical agents to the Nash equilibrium is achieved in a globally bounded fixed time independent of any initial states.

6.2 Problem Formulation

Consider a set of players $\mathcal{V} \triangleq \{1, \dots, N\}$ with the action of player i denoted as $x_i \in \mathbb{R}^{n_i}$ and the action vector of the other players denoted by $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$, where $n = \sum_{i=1}^N n_i$, respectively. The dynamics of the players are heterogeneous. In this chapter, we specifically consider three representative cases of dynamics commonly investigated in multi-agent systems, that is, high-order integrator dynamics, general linear dynamics and Euler-Lagrange dynamics. To characterize this heterogeneity, the player set is partitioned into three mutually disjoint subsets, denoted by $\mathcal{V}_H$, $\mathcal{V}_L$, and $\mathcal{V}_E$, such that $\mathcal{V} = \mathcal{V}_H \cup \mathcal{V}_L \cup \mathcal{V}_E$.

- **High-order integrator dynamics:** For each player $i \in \mathcal{V}_H$, the dynamics of the i -th player modeled by an r_i -th order integrator are described as follows,

$$\begin{cases} \dot{x}_{i,1} &= x_{i,2}, \\ \dot{x}_{i,2} &= x_{i,3}, \\ &\vdots \\ \dot{x}_{i,r_i} &= u_i, \end{cases} \quad (6.1)$$

where $x_{i,1} = x_i \in \mathbb{R}^{n_i}$ is the action of player i , $x_{i,k}(t) \in \mathbb{R}^{n_i}$ for $k \in \{1, 2, \dots, r_i\}$ represents its higher-order states, and $u_i(t) \in \mathbb{R}^{n_i}$ is the control input.

- **General linear dynamics:** For each player $i \in \mathcal{V}_L$, the dynamics of the i -th player are described as follows,

$$\dot{y}_i = A_i y_i + B_i u_i, \quad x_i = C_i y_i, \quad (6.2)$$

where $y_i \in \mathbb{R}^{m_i}$, $A_i \in \mathbb{R}^{m_i \times m_i}$, $B_i \in \mathbb{R}^{m_i \times p_i}$ and $C_i \in \mathbb{R}^{n_i \times m_i}$ are some constant matrices, $u_i \in \mathbb{R}^{p_i}$ is the control input.

- **Euler-Lagrange dynamics:** For each player $i \in \mathcal{V}_E$, the dynamics of the i -th player are described as follows,

$$M_i(x_i) \ddot{x}_i + C_i(x_i, \dot{x}_i) \dot{x}_i + G_i(x_i) = u_i + \bar{b}_i, \quad (6.3)$$

where $x_i, \dot{x}_i, \ddot{x}_i \in \mathbb{R}^{n_i}$ represent the generalized position/action, velocity, and acceleration vectors, respectively, $u_i \in \mathbb{R}^{n_i}$ is the control torque vector, $M_i(x_i) \in \mathbb{R}^{n_i \times n_i}$ is the symmetric positive-definite inertia, $C_i(x_i, \dot{x}_i) \in \mathbb{R}^{n_i \times n_i}$ is the Coriolis and centrifugal torque matrix, $G_i(x_i) \in \mathbb{R}^{n_i}$ is the gravitational torque vector, and $\bar{b}_i \in \mathbb{R}^{n_i}$ is a bounded external disturbance.

In this chapter, each player i aims to minimize its own cost function $f_i(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$. If player j is not a neighbor of player i , which means $a_{ij} = 0$, then player i has no access to player j 's action directly. Otherwise, player i can get the information of player j via the communication graph $\mathcal{G}$.

The game in this paper is described as

$$\min_{y_i \in \mathbb{R}^{n_i}} f_i(y_i, \mathbf{y}_{-i}). \quad (6.4)$$

First, we formalize the definition of the Nash equilibrium of the unconstrained non-cooperative game.

Definition 6.1. [ye2017distributed] A strategy profile $\mathbf{x}^* = [x_1^*, \dots, x_N^*]$ is called the Nash equilibrium of the non-cooperative game (6.4) if $\forall i \in \mathcal{V}$,

$$f_i(y_i^*, \mathbf{x}_{-i}^*) \leq f_i(x_i, \mathbf{x}_{-i}^*), \quad \forall x_i \in \mathbb{R}^{n_i}. \quad (6.5)$$

In this chapter, our objective is to design a distributed Nash equilibrium seeking algorithm for the players of the heterogeneous dynamics given in such that the action vector $\mathbf{x}$ converges to the Nash equilibrium $\mathbf{x}^*$ of the game (6.4) in fixed time. Denote

$$H(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f_1(\mathbf{x})}{\partial x_1 \partial x_N} \\ \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2^2} & \dots & \frac{\partial^2 f_2(\mathbf{x})}{\partial x_2 \partial x_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_1} & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N \partial x_2} & \dots & \frac{\partial^2 f_N(\mathbf{x})}{\partial x_N^2} \end{bmatrix}, \quad (6.6)$$

where $\mathbf{x} = [x_1, \dots, x_N]^\top \in \mathbb{R}^n$, $x_i \in \mathbb{R}^{n_i}$, $i = 1, \dots, N$.

To proceed, the following mild assumptions are made.

Assumption 14. $\forall i \in \mathcal{V}$, the cost function $f_i(x_i, \mathbf{x}_{-i})$ is convex and twice continuously differentiable in $x_i \in \mathbb{R}^{n_i}$ for every fixed $\mathbf{x}_{-i} \in \mathbb{R}^{n-n_i}$.

Assumption 15. Let $\bar{F}(\mathbf{x}) = [\nabla_1 f_1(x_1, \mathbf{x}_{-1}), \dots, \nabla_N f_N(x_N, \mathbf{x}_{-N})]^\top$, where $\nabla_i f_i(x_i, \mathbf{x}_{-i}) = \frac{\partial f_i(x_i, \mathbf{x}_{-i})}{\partial x_i}$. $\bar{F}(\mathbf{x})$ is strongly monotone with modulus δ_m in $\mathbb{R}^n$, that is, $(\mathbf{x} - \hat{\mathbf{x}})^\top (\bar{F}(\mathbf{x}) - \bar{F}(\hat{\mathbf{x}})) \geq \delta \|\mathbf{x} - \hat{\mathbf{x}}\|^2$, $\forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 16. $\nabla_i f_i(\mathbf{x})$ is globally Lipschitz continuous with a positive constant θ , that is, $\|\nabla_i f_i(\mathbf{x}) - \nabla_i f_i(\hat{\mathbf{x}})\| \leq \theta_l \|\mathbf{x} - \hat{\mathbf{x}}\|$, $\forall \mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^n$.

Assumption 17. The communication graph $\mathcal{G}$ is undirected and connected.

Assumption 18. The external disturbance $\bar{b}_i$ given in (6.3) is continuous and bounded, that is, $\|\bar{b}_i\| \leq k_b^i$, $i = 1, \dots, N$, where k_b^i is a positive constant.

Assumption 19. For player $i \in \mathcal{V}_L$, (A_i, B_i) is controllable and $\text{rank}(B_i) = p_i$. In addition, there exist matrices $K_{1i} \in \mathbb{R}^{p_i \times n_i}$ and $K_{2i} \in \mathbb{R}^{p_i \times n_i}$ such that

$$\begin{bmatrix} C_i B_i & \mathbf{0} \\ -A_i B_i & B_i \end{bmatrix} \begin{bmatrix} K_{1i} \\ K_{2i} \end{bmatrix} = \begin{bmatrix} I_{n_i} \\ \mathbf{0} \end{bmatrix}. \quad (6.7)$$

Remark 6.2. Assumptions 1-3 are common in the study of Nash equilibrium seeking problems [fang2021distributed, li2019distributed, poveda2022fixed, meng2025distributed]. Under Assumptions 1-2, the Nash equilibrium $\mathbf{x}^*$ exists uniquely. Moreover, since $f_i(y_i, \mathbf{x}_{-i})$ is twice continuously differentiable, Assumption 3 guarantees that $\sup_{\mathbf{x} \in \mathbb{R}^m} \|H(\mathbf{x})\|$ is bounded [meng2025distributed], where $H(\mathbf{x})$ is defined in (6.6).

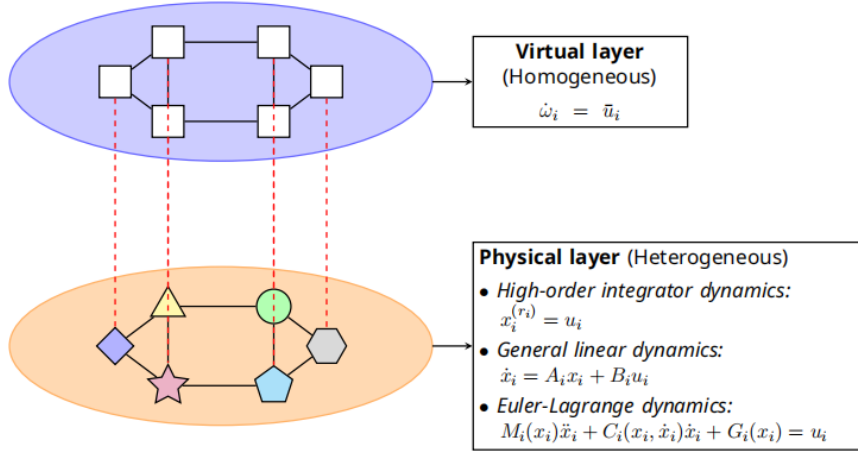


Figure 6.1. Two-layer design framework.

Remark 6.3. Assumption 6 is adopted from [10229248]. The solvability of the matrix equations (6.7) is a sufficient condition to guarantee the solvability of the Nash equilibrium seeking problem of heterogeneous linear dynamics described in (6.2). In fact, the existence of K_{1i} and K_{2i} , $i = 1, \dots, N$, implies the existence of the optimal state $y_i^* = B_i K_{1i} x_i^*$ such that $C_i y_i^* = x_i^*$.

6.3 Main Results

Inspired by the hierarchical designs in [GIRARD2009566], we propose a two-layer design framework, which is shown in Figure 6.1, to solve the Nash equilibrium seeking problem for the game (6.4) with the players governed by heterogeneous dynamics described in (6.1)-(6.3). In the virtual layer, we design a distributed Nash equilibrium seeking algorithm with fixed-time convergence for the constrained game with the players governed by the dynamics $\dot{\omega}_i = \hat{u}_i$, where $\omega_i, \hat{u}_i \in \mathbb{R}^{n_i}$ are the action and the control input of the virtual player i , respectively. In the physical layer, a tracking protocol is proposed for the actual players governed by heterogeneous dynamics described in (6.1)-(6.3) such that their actions $\mathbf{x}$ converge to the actions of the virtual players ω in fixed time.

6.3.1 Virtual Layer Design for Nash Equilibrium Seeking

In this subsection, the distributed Nash equilibrium seeking algorithm 3.4-3.6 is applied for the virtual players modeled by the dynamics $\dot{\omega}_i = \hat{u}_i$, where $\omega_i, \hat{u}_i \in \mathbb{R}^{n_i}$, $i \in \mathcal{V}$. In particular, for any $i, j \in \mathcal{V}$, the distributed algorithm is as follows:

$$\dot{\omega}_i = -\beta_1 [\nabla_i f_i(\mathbf{z}_i)]^\mu - \beta_2 [\nabla_i f_i(\mathbf{z}_i)]^\nu, \quad (6.8)$$

$$\begin{aligned} \dot{z}_{ij} = & -\alpha_1 [\Lambda_{ij}]^\mu - \alpha_2 [\Lambda_{ij}]^\nu - \alpha_3 [\Lambda_{ij}]^{2\mu-1} \\ & - \alpha_4 [\Lambda_{ij}]^{2\nu-1}, \end{aligned} \quad (6.9)$$

$$\Lambda_{ij} = \sum_{k=1}^N a_{ik}(z_{ij} - z_{kj}) + a_{ij}(z_{ij} - \omega_j), \quad (6.10)$$

where $\omega_i \in \mathbb{R}^{n_i}$ is the virtual signal of player i , $z_{ij} \in \mathbb{R}^{n_j}$ is the player i 's estimate on the player j 's virtual signal ω_j , $\mathbf{z}_i = [z_{i1}, \dots, z_{iN}]^\top \in \mathbb{R}^n$, a_{ik} is the (i, k) -th entry of the adjacency matrix A corresponding to the communication graph $\mathcal{G}$, $\mu \in (\frac{1}{2}, 1)$, $\nu \in (1, 2)$, and $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2$ are some positive design parameters.

6.3.2 Physical Layer Control for Heterogeneous Dynamics

In subsection 6.3.1, the distributed Nash equilibrium seeking algorithm (6.8)-(6.10) was proposed to ensure that the actions of the virtual players converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time. In this subsection, novel tracking protocols are proposed such that the actions $\mathbf{x}$ of the actual players governed by heterogeneous dynamics described in (6.1)-(6.3) converge to the virtual players' actions ω in fixed time. Consequently, $\mathbf{x}$ can also converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time.

- **High-order integrator dynamics:** For $i \in \mathcal{V}_H$, it follows from (6.1) that x_i converges to ω_i if and only if $x_{i,1} = \omega_i$ and $x_{i,k} = 0$, $k = 2, \dots, r_i$. Then, the tracking error the player i can be defined by $e_i = [e_{i,1}, \dots, e_{i,r_i}]^\top \in \mathbb{R}^{p_i}$, where $p_i = n_i r_i$,

$$\begin{aligned} e_{i,1} &= x_{i,1} - \omega_i, \\ e_{i,k} &= x_{i,k}, \quad k = 2, \dots, r_i. \end{aligned} \quad (6.11)$$

Then, it follows from (6.11) that the time derivative of $e_{i,k}$ can be expressed as follows:

$$\dot{e}_{i,1} = e_{i,2} - \dot{\omega}_i,$$

$$\begin{aligned}\dot{e}_{i,k} &= e_{i,k+1}, \quad k = 2, \dots, r_i - 1, \\ \dot{e}_{i,r_i} &= u_i,\end{aligned}\tag{6.12}$$

where u_i is the control input of the player i .

To drive the tracking error (6.11) to zero in fixed time, we propose the following sliding surface for the player i :

$$s_i = e_{i,r_i} + \sum_{k=1}^{r_i} \int_0^t (\kappa_k^i [e_{i,k}(\tau)]^{\gamma_k^i} + \bar{\kappa}_k^i [e_{i,k}(\tau)]^{\bar{\gamma}_k^i}) d\tau, \tag{6.13}$$

where the parameters κ_k^i , $\bar{\kappa}_k^i$, γ_k^i and $\bar{\gamma}_k^i$ are chosen as described in Lemma 2.9.

The control input u_i , for the player $i \in \mathcal{V}_H$ is proposed as follows:

$$u_{ij} = - \sum_{k=1}^{r_i} \left(\kappa_k^i [e_{i,k}]^{\gamma_k^i} + \bar{\kappa}_k^i [e_{i,k}]^{\bar{\gamma}_k^i} \right) - (\eta_i [s_i]^{\mu_1} + \zeta_i [s_i]^{\nu_1}), \tag{6.14}$$

where $\eta_i, \zeta_i, \mu_1 \in (0, 1)$, $\nu_1 \in (1, +\infty)$ are some positive constants.

- **General linear dynamics:** For each player $i \in \mathcal{V}_E$, the tracking error of player i be defined by

$$e_i = x_i - B_i K_{1i} \omega_i. \tag{6.15}$$

It follows from Assumption 19 that if $e_i = 0$, then $y_i - \omega_i = 0$.

The following controller is proposed for player i ,

$$u_i = v_i - K_{2i} \omega_i + K_{1i} \dot{\omega}_i, \tag{6.16}$$

where v_i will be determined later for achieving fixed-time tracking.

It follows from Assumption 19 and (6.15)-(6.16) that

$$\begin{aligned}\dot{e}_i &= A_i x_i + B_i (-K_{2i} \omega_i + K_{1i} \dot{\omega}_i) + B_i v_i - B_i K_{1i} \dot{\omega}_i \\ &= A_i e_i + B_i v_i.\end{aligned}\tag{6.17}$$

Since (A_i, B_i) is controllable and $\text{rank}(B_i) = p_i$, via Lemma 2.6, there exists a linear coordinate transformation $\hat{e}_i =_i e_i = [\hat{e}_{i1,1}, \dots, \hat{e}_{i1,\zeta_1}, \dots, \hat{e}_{ip_i,1}, \dots,$

$\hat{e}_{ip_i, \zeta_{p_i}}]^\top$, where $\zeta_j, j = 1, \dots, p_i$, denotes the controllability index of $\hat{e}_{ij}$. Let $\hat{e}_i^\zeta = [\hat{e}_{i1, \zeta_1}, \dots, \hat{e}_{ip_i, \zeta_{p_i}}]^\top$. Then the v_i is designed as follows:

$$v_i = M_i^{-1}(\Omega_i - G_i \hat{e}_i), \quad (6.18)$$

where G_i and M_i are given as in Lemma 2.6, $\Omega_i = [\Omega_{i1}, \dots, \Omega_{ip_i}]^\top$ with

$$\Omega_{ij} = - \sum_{l=1}^{\zeta_j} (\kappa_{ij,l} [\hat{e}_{ij,l}]^{\gamma_{ij,l}} + \bar{\kappa}_{ij,l} [\hat{e}_{ij,l}]^{\bar{\gamma}_{ij,l}}) \quad (6.19)$$

, and $\kappa_{ij,l}, \bar{\kappa}_{ij,l}$, and $\gamma_{ij,l}, \bar{\gamma}_{ij,l}, j = 1, \dots, p_i, l = 1, \dots, \zeta_j$, are the positive gains that can be designed as in Lemma 2.6.

Thus, the controller for player i is given as

$$u_i = M_i^{-1}(\Omega_i - G_i \hat{e}_i) - K_{2i} \omega_i + K_{1i} \dot{\omega}_i. \quad (6.20)$$

- **Euler-Lagrange dynamics:** For each player $i \in \mathcal{V}_E$, the tracking error e_i and its time derivative for player i are first defined as follows:

$$e_i = x_i - \omega_i, \quad \dot{e}_i = \dot{x}_i - \dot{\omega}_i. \quad (6.21)$$

Based on the tracking error, the sliding surface s_i of the player i is defined as follows:

$$s_i = \dot{e}_i + q_i([\dot{e}_i]^{\mu_2} + [\dot{e}_i]^{\nu_2}), \quad (6.22)$$

where $q_i, \mu_2 \in (0, 1), \nu_2 \in (1, +\infty)$ are some positive constants.

Then the control torque vector u_i for the player $i \in \mathcal{V}_E$ is proposed as follows:

$$\begin{aligned} u_i = & G_i(x_i) + C_i(x_i, \dot{x}_i) \dot{x}_i - p_i M_i(x_i) ([s_i]^{\mu_2} + [s_i]^{\nu_2} + s_i + \theta_i \text{sign}(s_i)) \\ & - q_i M_i(x_i) (\mu_2 ([\dot{e}_i]^{\mu_2-1} \odot \dot{e}_i) + \nu_2 ([\dot{e}_i]^{\nu_2-1} \odot \dot{e}_i)), \end{aligned} \quad (6.23)$$

where $p_i, \theta_i \geq p_i^{-1} k_b^i \sup_{x_i} \|M_i^{-1}(x_i)\|$ are some positive constants, k_b^i is the upper bound of the external disturbance $\bar{b}_i$, and $\odot$ is the Hadamard product defined in the notation part.

6.3.3 Convergence Analysis

In this subsection, it will be shown that the distributed Nash equilibrium seeking algorithm (6.8)-(6.10) and the tracking protocols (6.14)-(6.23) guarantee that the players' actions $\mathbf{x}$ converge to the Nash equilibrium $\mathbf{x}^*$ in fixed time.

Theorem 6.4. *Consider the non-cooperative game (6.4) with the distributed Nash equilibrium seeking algorithm (6.8)-(6.10) and the tracking protocols (6.14)-(6.23) for players of heterogeneous dynamics described in (6.1)-(6.3) under Assumptions 14-19. The action $\mathbf{x}$ of the players converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time. Furthermore, the upper bound of the convergence time T can be explicitly given by*

$$T_{total} = T_{vir} + T_{phy}, \quad (6.24)$$

where T_1 and T_2 are given in (6.29) and (6.32), respectively.

Proof: First, it follows from Theorem 3.3 that ω converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time $T_{vir} = T_{nom}$, where T_{nom} is given in (3.12), that is, $\omega = \mathbf{x}^*$, for $t \geq T_{vir}$.

After $t = T_{vir}$, the proof will be presented in three parts, which show that the actions of players of high-order dynamics (6.1), general linear dynamics (6.2) and (6.3) converge to the Nash equilibrium in fixed time respectively.

Part I: Proof that the actions of players with high-order dynamics (6.1) converge to x_i^* in fixed time.

Consider the Lyapunov function candidate,

$$V_1 = \frac{1}{2} \sum_{i \in \mathcal{V}_H} s_i^2. \quad (6.25)$$

It follows from (6.12)-(6.14) that

$$\dot{s}_i = -(\eta_i \lceil s_i \rceil^{\mu_1} + \zeta_i \lceil s_i \rceil^{\nu_1}). \quad (6.26)$$

Taking the time derivative of V_3 along the trajectories of (??), one has

$$\dot{V}_1 = - \sum_{i \in \mathcal{V}_H} (\eta_i |s_i|^{\mu_1+1} + \zeta_i |s_i|^{\nu_1+1}). \quad (6.27)$$

It follows from Lemma 2.2 given in Appendix that

$$\dot{V}_1 \leq -G_1 V_1^{\frac{\mu_1+1}{2}} - G_2 V_1^{\frac{\nu_1+1}{2}}, \quad (6.28)$$

where $G_1 = 2^{\frac{\mu_1+1}{2}} \eta_{\min}$, $G_2 = 2^{\frac{\nu_1+1}{2}} m^{\frac{1-\nu_1}{2}} \zeta_{\min}$, $\eta_{\min} = \min\{\eta_1, \dots, \eta_N\}$ and $\zeta_{\min} = \min\{\zeta_1, \dots, \zeta_N\}$.

Via Lemma 2.8, one has that s_i converges to $\mathbf{0}_{n_i}$ in fixed time. Moreover, the upper bound of the convergence time T_1 is given by

$$T_1 = \frac{1}{G_1(1-\mu_1)} + \frac{1}{G_2(\nu_1-1)}. \quad (6.29)$$

It can be further obtained that after $t = T_{vir} + T_1$, $s_i = \mathbf{0}_{n_i}$, in addition to $\omega_i = x_i^*$, $\dot{\omega}_i = 0$. It follows from (6.12)-(6.14) that

$$\begin{aligned} \dot{e}_{ij,1} &= e_{ij,2} - \dot{\omega}_{ij}, \\ \dot{e}_{ij,k} &= e_{ij,k+1}, \quad k = 2, \dots, n_i - 1, \\ \dot{e}_{ij,n_i} &= u_{ij}, \end{aligned} \quad (6.30)$$

Via Lemma 2.8, one has that the tracking error e_i converges to $\mathbf{0}_{n_i}$ in fixed time, which implies that x_i also converges to ω_i in fixed time, $i \in \mathcal{V}_H$. Furthermore, the upper bound of the convergence time given by

$$T_2^i = \frac{\gamma_0^i \lambda_{\max}(P_1^i)}{(1-\gamma_0^i) \lambda_{\min}(Q_1^i)} \lambda_{\max}^{\frac{1-\gamma_0^i}{\gamma_0^i}}(P_1^i) + \frac{\bar{\gamma}_0^i \lambda_{\max}(P_2^i)}{(\bar{\gamma}_0^i - 1) \lambda_{\min}(Q_2^i)} \lambda_{\max}^{\frac{\bar{\gamma}_0^i - 1}{\bar{\gamma}_0^i}}(P_2^i), \quad (6.31)$$

where $P_1^i, P_2^i, Q_1^i, Q_2^i$ are given as in Lemma 2.9, and $\lambda_{\max}(\cdot)$, $\lambda_{\min}(\cdot)$ denote the maximum and minimum eigenvalue of the matrix $(\cdot)$, respectively. Denote

$$T_2 = \max\{T_2^i, i = 1, \dots, N\}, \quad (6.32)$$

where T_2^i is given in (6.31). In summary, the action $\mathbf{y}$ of the players of the high-order integrator dynamics (6.1) converges to the Nash equilibrium $\mathbf{x}^*$ in fixed time with the upper bound of the convergence time given by

$$T_H = T_{vir} + T_1 + T_2. \quad (6.33)$$

However, the proof of this part is not completed until the boundedness of x_i , $i = 1, \dots, N$ is shown within the time interval $[0, T_{\text{virtual}} + T_1)$. It follows from (6.26) and (6.28) that s_i is bounded for any $t \geq 0$. Furthermore, it also follows from the proof of Theorem 3.3 that both ω_i and $\dot{\omega}_i$ are both bounded for any $t \geq 0$, $i = 1, \dots, N$. From (6.12) and (6.13) and the boundedness of $\dot{\omega}_i$, the unboundedness of $e_{i,(k+1)}$ implies that $e_{i,k}$ will escape to infinity in finite time and have the same sign with $e_{i,(k+1)}$ ultimately, $k = 1, \dots, r_i - 1$, which leads to the unboundedness of s_i . This leads to a contradiction. Therefore, we can conclude that the tracking error e_i and x_i are both bounded on any $t \geq 0$, that is, no finite-time escape occurs. The proof of this part is thus completed.

Part II: Proof that the actions of players with general linear dynamics (6.1) converge to x_i^* in fixed time.

Via Lemma 2.6 given in Appendix, one can obtain that the dynamics of $\hat{e}_{ij}$, $i \in \mathcal{V}_L$, $j = 1, \dots, p_i$, have the following form:

$$\begin{cases} \dot{\hat{e}}_{ij,k}(t) = \hat{e}_{ij,k+1}, & k = 1, \dots, \zeta_j - 1, \\ \dot{\hat{e}}_{ij,\zeta_j}(t) = -\sum_{l=1}^{\zeta_j} (\kappa_{ij,l} [\hat{e}_{ij,l}]^{\gamma_{ij,l}} + \bar{\kappa}_{ij,l} [\hat{e}_{ij,l}]^{\bar{\gamma}_{ij,l}}). \end{cases} \quad (6.34)$$

It follows from Lemma 2.9 given in Appendix and (6.34) that $\hat{e}_{ij}$ converges to $\mathbf{0}_{\zeta_j}$ in fixed time, $i = 1, \dots, N$, $j = 1, \dots, p_i$, and the upper bound of the convergence time is given as

$$T_3^{ij} = \frac{\gamma_{ij,0} \lambda_{\max}(P_1^{ij})}{(1 - \gamma_{ij,0}) \lambda_{\min}(Q_1^{ij})} \lambda_{\max}^{\frac{1-\gamma_{ij,0}}{\gamma_{ij,0}}}(P_1^{ij}) + \frac{\bar{\gamma}_{ij,0} \lambda_{\max}(P_2^{ij})}{(\bar{\gamma}_{ij,0} - 1) \lambda_{\min}(Q_2^{ij})} \lambda_{\max}^{\frac{\bar{\gamma}_{ij,0}-1}{\bar{\gamma}_{ij,0}}}(P_2^{ij}), \quad (6.35)$$

where P_1^{ij} , P_2^{ij} , Q_1^{ij} , Q_2^{ij} are given as in Lemma 2.9, and $\lambda_{\max}(\cdot)$, $\lambda_{\min}(\cdot)$ denote the maximum and minimum eigenvalues of the matrix $(\cdot)$, respectively. Denote

$$T_3 = \max\{T_3^{ij}, i, j = 1, \dots, N\}. \quad (6.36)$$

Since $\hat{e}_i = \aleph_i e_i$ and $\aleph_i$ is invertible, it follows from (6.15) and Assumption 19 that $\mathbf{x}$ converges to ω in fixed time T_3 . In summary, the action $\mathbf{y}$ of the players of the heterogeneous linear dynamics (6.2) converges to the Nash equilibrium y^* in fixed time with the upper bound of the convergence time given by

$$T_L = T_{\text{vir}} + T_3. \quad (6.37)$$

It follows from the proof of Theorem 3.3 and (6.34) that the variables z_{ij} , ω_i , x_i and y_i are bounded for any $t \geq 0$, $i, j = 1, \dots, N$, that is, no finite-time escape occurs. The proof of this part is thus completed.

Part III: Proof that the actions of players with Euler-Lagrange dynamics (6.3) converge to x_i^* in fixed time.

It follows from (6.3) that

$$\ddot{x}_i = M_i^{-1}(x_i)[u_i + \bar{b}_i - C_i(x_i, \dot{x}_i)\dot{x}_i - G_i(x_i)]. \quad (6.38)$$

It further follows from (6.22), (6.23) and (6.38) that

$$\dot{s}_i = -p_i(\lceil s_i \rceil^{\mu_2} + \lceil s_i \rceil^{\nu_2} + \theta_i \text{sign}(s_i)) + M_i^{-1}(x_i)\bar{b}_i. \quad (6.39)$$

Consider the Lyapunov function candidate,

$$V_2 = \frac{1}{2} \sum_{i \in \mathcal{V}_E} \|s_i\|^2. \quad (6.40)$$

Taking the time derivative of V_4 along the trajectories of (6.39), one has

$$\begin{aligned} \dot{V}_2 &= - \sum_{i \in \mathcal{V}_E} p_i s_i^\top (\lceil s_i \rceil^{\mu_2} + \lceil s_i \rceil^{\nu_2}) + \sum_{i \in \mathcal{V}_E} (-\theta_i p_i s_i^T \text{sign}(s_i) + s_i^T M_i^{-1}(x_i)\bar{b}_i) \\ &\leq - \sum_{i \in \mathcal{V}_E} p_i (\|s_i\|_{\mu_2+1}^{\mu_2+1} + \|s_i\|_{\nu_2+1}^{\nu_2+1}). \end{aligned} \quad (6.41)$$

It follows from (6.40)-(6.41) and Lemma 2.2 given that

$$\dot{V}_4 \leq -H_1 V_4^{\frac{\mu_2+1}{2}} - H_2 V_4^{\frac{\nu_2+1}{2}}, \quad (6.42)$$

where $H_1 = 2^{\frac{\mu_1+1}{2}} p_{\min}$, $H_2 = 2^{\frac{\nu_1+1}{2}} p_{\min} n^{\frac{1-\nu_1}{2}}$, and $p_{\min} = \min\{p_i, i \in \mathcal{V}_E\}$.

Via Lemma 2.8, one thus has that s_i converges to $\mathbf{0}_{n_i}$ in fixed time. Moreover, the upper bound of the convergence time T_4 is given by

$$T_4 = \frac{2}{H_1(1 - \mu_2)} + \frac{2}{H_2(\nu_2 - 1)}. \quad (6.43)$$

After $t = T_{vir}$, $\omega_i = x_i^*$, where x_i^* is the i -th element of the Nash equilibrium $\mathbf{x}^*$. Then, it can be further obtained that after $t = T_{vir} + T_4$, $\omega_i = x_i^*$, $s_i = 0$, $i \in \mathcal{V}_E$. It follows from (6.22) that

$$\dot{x}_i = -q_i(\lceil x_i - x_i^* \rceil^{\mu_2} + \lceil x_i - x_i^* \rceil^{\nu_2}). \quad (6.44)$$

For $t \geq T_{vir} + T_4$, consider the Lyapunov function candidate,

$$V_3 = \frac{1}{2} \sum_{i \in \mathcal{V}_E} \|x_i - x_i^*\|^2. \quad (6.45)$$

Taking the time derivative of V_3 along the trajectories of (6.44), one has

$$\begin{aligned} \dot{V}_3 &= - \sum_{i \in \mathcal{V}_E} q_i (x_i - x_i^*)^\top (\lceil x_i - x_i^* \rceil^{\mu_2} + \lceil x_i - x_i^* \rceil^{\nu_2} \\ &= - \sum_{i \in \mathcal{V}_E} q_i (\|x_i - x_i^*\|_{\mu_2+1}^{\mu_2+1} + \|x_i - x_i^*\|_{\nu_2+1}^{\nu_2+1}). \end{aligned} \quad (6.46)$$

It follows from (6.45)-(6.46) and Lemma 2.2 given in Appendix that

$$\dot{V}_3 \leq -H_3 V_3^{\frac{\mu_2+1}{2}} - H_4 V_3^{\frac{\nu_2+1}{2}}, \quad (6.47)$$

where $H_3 = 2^{\frac{\mu_2+1}{2}} q_{min}$, $H_4 = 2^{\frac{\nu_2+1}{2}} q_{min} n^{\frac{1-\nu_2}{2}}$ and $q_{min} = \min\{q_i, i \in \mathcal{V}_E\}$.

Via Lemma 2.8, one thus has that x_i converges to x_i^* in fixed time. Moreover, the upper bound of the convergence time T_5 is given by

$$T_5 = \frac{2}{H_3(1 - \mu_2)} + \frac{2}{H_4(\nu_2 - 1)}. \quad (6.48)$$

In summary, the actions $\mathbf{x}$ of the players of Euler-Lagrange dynamics (6.3) converge to Nash equilibrium $\mathbf{x}^*$ in fixed time with the upper bound of the convergence time given by

$$T_E = T_{vir} + T_4 + T_5. \quad (6.49)$$

However, the proof of this part is not completed until the boundedness of $\|x_i\|$ is shown within the time interval $[0, T_N + T_4)$.

For $0 \leq t < T_{vir} + T_4$, consider the Lyapunov function candidate

$$W_1 = \frac{1}{2} \sum_{i \in \mathcal{V}_E}^N \|x_i - \omega_i\|^2. \quad (6.50)$$

Taking the time derivative of W_1 along the trajectories of (6.8), one has

$$\begin{aligned} W_1 &= \sum_{i \in \mathcal{V}_E}^N (x_i - \omega_i)^\top (s_i - q_i(\lceil x_i - \omega_i \rceil^{\mu_1} + \lceil x_i - \omega_i \rceil^{\nu_1} - \dot{\omega}_i)) \\ &\leq \sum_{i \in \mathcal{V}_E}^N -q_i \|x_i - \omega_i\|^2 + \|s_i - \dot{\omega}_i\| \|x_i - \omega_i\|. \end{aligned} \quad (6.51)$$

It follows from the proof of Theorem 3.3 that both ω_i and $\dot{\omega}_i$ are bounded within the time interval $[0, T_{vir} + T_4)$. Then there exists a positive constant D_2 such that

$$W_1 \leq \sum_{i \in \mathcal{V}_E}^N -q_i \|x_i - \omega_i\|^2 + D_2 \|x_i - \omega_i\|. \quad (6.52)$$

It further follows from Lemma 2.3 given that (6.52) can be rewritten as

$$W_1 \leq (2q_{min} - k_2)W_2 + \frac{1}{2k_2}D_2, \quad (6.53)$$

where $q_{min} = \min\{q_i, i = 1, \dots, N\}$ and k_2 is a positive constant. Choosing suitable parameter k_2 such that $2q_{min} - k_2 > 0$, then one has $\|x_i - \omega_i\|$ is bounded within the time interval $[0, T_{vir} + T_4)$. Since $\|\omega_i\|$ is bounded within the time interval $[0, T_{vir} + T_4)$, one thus has $\|x_i\|$ is also bounded within the time interval $[0, T_N + T_4)$, that is, no finite-time escape occurs. The proof is thus completed.

Remark 6.5. It should be pointed out that our result can include the cases of first- and second-order integrator dynamics as its special cases. If $r_i = 1$, then the players' dynamics given in (6.1) reduce to the first-order integrator dynamics, which was investigated in [li2019distributed, poveda2022fixed, meng2025distributed]. If $r_i = 2$, then the players' dynamics given in (6.1) reduce to the second-order integrator dynamics, which was investigated in [meng2025distributed].

6.4 Numerical Examples

6.5 Conclusions

This chapter has investigated the distributed fixed-time Nash equilibrium seeking problem for multi-agent systems governed by heterogeneous complex dynamics. To fundamentally bridge the gap between idealized Nash equilibrium seeking algorithms for players of first-order dynamics and practical physical executions, a novel virtual-physical two-layer design architecture has been established. This decoupled framework successfully isolates the game-theoretic optimization complexities from the severe mechanical nonlinearities inherent in physical systems, such as Euler-Lagrange equations. By synthesizing a novel fixed-time tracking protocol based on advanced sliding mode theory in the physical layer, it has been formally proven that the physical actions of all heterogeneous players accurately and robustly track the optimal virtual signals. Consequently, exact convergence to the Nash equilibrium is achieved within a globally bounded fixed time from any initial state. The proposed framework smoothly accommodates unmodeled dynamics and external disturbances, while naturally encompassing lower-order kinematics as special cases, thereby significantly advancing the physical model generality of distributed game-theoretic algorithms.

Chapter 7

Engineering Applications and Verification

7.1 Introduction

7.2 Application I: Distributed Energy Trading Game in Smart Microgrids

7.2.1 Background

7.2.2 Problem Formulation

7.2.3 Algorithm Deployment

7.2.4 Simulation Results and Discussion

7.3 Application II: Distributed Computation Offloading in Vehicular Edge Computing Systems

7.3.1 Background

To further demonstrate the practicability and superiority of the proposed static fixed-time Nash equilibrium seeking strategies, this section applies the theoretical framework

to a highly time-sensitive cyber-physical scenario: Vehicular Edge Computing (VEC) systems under Denial-of-Service (DoS) attacks.

In modern intelligent transportation systems, Connected and Autonomous Vehicles (CAVs) generate massive amounts of computation-intensive tasks (e.g., real-time environmental perception and autonomous decision-making). Due to the limited onboard computing resources, CAVs often compete to offload their tasks to a shared Roadside Unit (RSU). This inherent resource competition naturally constitutes a non-cooperative static game. Furthermore, the open wireless V2X communication environment exposes the system to severe malicious DoS jamming, making the fast and resilient fixed-time convergence property of the proposed algorithm particularly crucial.

7.3.2 Problem Formulation

Consider a VEC system consisting of one RSU and a set of N autonomous vehicles within its communication coverage, denoted by $\mathcal{V} = \{1, 2, \dots, N\}$. The vehicles communicate with each other via Vehicular Ad-hoc Networks (VANETs) to negotiate their offloading strategies.¹ Static Non-cooperative Game Formulation To alleviate the on-board computation burden, each vehicle $i \in \mathcal{V}$ decides its offloading data rate, denoted as the strategy variable $z_i \in \mathbb{R}^+$. Let $\mathbf{z} = [z_1, z_2, \dots, z_N]^T \in \mathbb{R}^N$ be the strategy profile of all vehicles. The total required computation task rate for vehicle i is assumed to be a constant D_i . The local cost function for each vehicle i , aiming to minimize a combination of local execution penalty, transmission energy consumption, and the shared RSU congestion delay, is formulated as a static quadratic function:

$$J_i(\mathbf{z}) = w_{i,1}(z_i - D_i)^2 + w_{i,2}z_i^2 + w_{i,3} \left(\frac{1}{N} \sum_{j=1}^N z_j - C_{RSU} \right)^2$$

where $w_{i,1}, w_{i,2}, w_{i,3} > 0$ are the weighting coefficients reflecting the preference of vehicle i . The first term penalizes the tasks left for the onboard CPU, the second term represents the energy consumption for wireless transmission, and the third term captures the coupled congestion penalty when the average offloaded data approaches the optimal processing capacity of the RSU, denoted by C_{RSU} . The objective of this non-cooperative game is to find the exact static Nash equilibrium $\mathbf{z}^* = [z_1^*, \dots, z_N^*]^T$, which satisfies:

$$J_i(z_i^*, z_{-i}^*) \leq J_i(z_i, z_{-i}^*), \quad \forall z_i \in \mathbb{R}^+, \forall i \in \mathcal{V}$$

where z_{-i} denotes the strategies of all vehicles except vehicle i .² Communication Network under DoS Attacks In practical VEC deployments, the information exchange among vehicles is vulnerable to intermittent DoS attacks (e.g., RF jamming). Let the time axis be partitioned into normal communication intervals and DoS attack intervals. The communication topology is modeled as a time-varying graph $\mathcal{G}(t)$, where the adjacency matrix $A(t) = [a_{ij}(t)]$ dynamically changes:

$$a_{ij}(t) = \begin{cases} a_{ij} > 0, & \text{during normal communication intervals} \\ 0, & \text{during DoS attack intervals} \end{cases}$$

During the attack intervals, the communication links are paralyzed, forcing the distributed gradient estimation process to suspend, which poses a severe threat to the convergence of the Nash equilibrium seeking.

The objective of this section is to design a resilient, fully distributed fixed-time algorithm for the VEC system that can drive the strategy of each vehicle $\omega_i(t)$ to the static Nash equilibrium point z_i^* in a fixed settling time T_{max} , despite the intermittent communication blockages caused by DoS attacks. Moreover, the upper bound of the convergence time T_{max} must be independent of the initial offloading strategies, thereby strictly satisfying the hard real-time requirements of fast-moving vehicles.

7.3.3 Algorithm Deployment

7.3.4 Simulation Results and Discussion

7.4 Application III: Target Protection of Air-Ground Heterogeneous Multi-Agent Systems with Conflicting Goals

7.4.1 Background

7.4.2 Problem Formulation

7.4.3 Algorithm Deployment

7.4.4 Simulation Results and Discussion

7.5 Conclusions

Chapter 8

Conclusions and Future Work

This thesis has presented a comprehensive investigation into [research domain], addressing fundamental challenges in [problem area]. The work contributes to both theoretical understanding and practical solutions in the field. This concluding chapter summarizes the key contributions, discusses their implications, and outlines promising directions for future research.

8.1 Conclusions

The primary objective of this thesis was to address the critical challenges in [research domain], particularly focusing on [main problem]. Through a systematic approach, we have made significant contributions across three main areas:

8.2 Future Work

Our theoretical contributions include:

- **Novel algorithmic framework:** We introduced a new algorithmic framework that addresses [specific problem] with improved complexity bounds. The framework provides [theoretical advantage] over existing approaches.
- **Mathematical analysis:** We provided rigorous mathematical analysis of [system/algorithm], establishing theoretical foundations for [key result]. This analysis revealed important insights about [theoretical insight].

- **Complexity characterization:** We characterized the computational complexity of [problem type], showing that [complexity result] and providing optimal algorithms for specific problem instances.

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